Name: _____ Class: _____

N5 Practical Electronics

Past paper questions and marking instructions

Every current past paper question, by topic, easy to hard



How to use this booklet

1. Pick a part (below). Start at the first question of each topic: they get harder as you go. Write every answer in your jotter, with its number, for example 1.4.

2. Mark first, then look. The marking instructions come straight after each question. Tap Answers beside it in the list, or turn the page.

3. Mark in green pen, one mark at a time. Only the points in the marking instructions get marks. Write your score beside the question, for example 4/6.

4. Some questions say "use the data sheet". Open the data sheet here. Every year's data sheet is the same.

5. Stuck? Read the Electronics notes for that topic here, then try again.

Contents

Part 1 · Components and circuits

Electronic components: questions 1.1 to 1.8, 51 marks
Circuits and calculations: questions 1.9 to 1.15, 72 marks
Analogue circuits: questions 1.16 to 1.22, 50 marks

Part 2 · Logic and systems

Logic: questions 2.1 to 2.7, 41 marks
Systems and block diagrams: questions 2.8 to 2.14, 42 marks
Simulation: questions 2.15 to 2.21, 35 marks

Part 3 · Construction and testing

Construction: questions 3.1 to 3.17, 81 marks
Testing and fault-finding: questions 3.18 to 3.28, 44 marks
Safety: questions 3.29 to 3.30, 4 marks

Tap a part, a question or Answers to jump to it. If links don't work, use the page numbers in each list.

Part 1

Components and circuits

Start with

the first two questions of each topic below. Mark them in green pen with their marking instructions (tap Answers). Then carry on down the list.

			Question	Marking	
Electronic components					
1.1	2025 paper, Question 2	1 mark	page 3	Answers p. 4	
1.2	2026 paper, Question 1	8 marks	page 5	Answers p. 8	
1.3	2018 paper, Question 1	7 marks	page 10	Answers p. 12	
1.4	2021 paper, Question 1	8 marks	page 13	Answers p. 16	
1.5	Specimen paper, Question 1	7 marks	page 17	Answers p. 19	
1.6	2025 paper, Question 1	7 marks	page 20	Answers p. 22	
1.7	2019 paper, Question 1	6 marks	page 23	Answers p. 25	
1.8	2024 paper, Question 1	7 marks	page 26	Answers p. 28	
Circuits and calculations					
1.9	2026 paper, Question 3	11 marks	page 29	Answers p. 31	
1.10	2018 paper, Question 3	11 marks	page 34	Answers p. 36	
1.11	2019 paper, Question 3	10 marks	page 37	Answers p. 39	
1.12	Specimen paper, Question 3	8 marks	page 40	Answers p. 41	
1.13	2021 paper, Question 3	11 marks	page 42	Answers p. 44	
1.14	2025 paper, Question 3	11 marks	page 46	Answers p. 48	
1.15	2024 paper, Question 3	10 marks	page 50	Answers p. 52	*Challenge*
Analogue circuits					
1.16	2021 paper, Question 9	7 marks	page 55	Answers p. 57	
1.17	2019 paper, Question 7	7 marks	page 59	Answers p. 60	
1.18	2025 paper, Question 9	8 marks	page 61	Answers p. 64	
1.19	Specimen paper, Question 7	8 marks	page 65	Answers p. 67	
1.20	2018 paper, Question 7	6 marks	page 68	Answers p. 71	
1.21	2026 paper, Question 7	7 marks	page 72	Answers p. 73	
1.22	2024 paper, Question 9	7 marks	page 75	Answers p. 76	*Challenge*

2. The table below shows information about the resistance of a thermistor at different temperatures.

MARKS DO NOT WRITE IN THIS MARGIN

Temperature (°C)	Resistance (kΩ)
20	50.0
40	30.0
60	
80	10.0

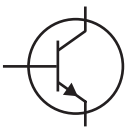
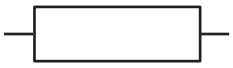
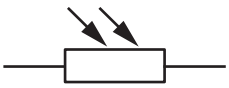
Complete the table above to predict the resistance of the thermistor at 60 °C. **1**

2.			Any value greater than 10.0kΩ and less than 30.0kΩ **(1 mark)**	**1**	Accept a value only. Award correct mark if answer is out with table but must have correct unit.

1. The table below gives information about some circuit components.

Some of the boxes have been left blank.

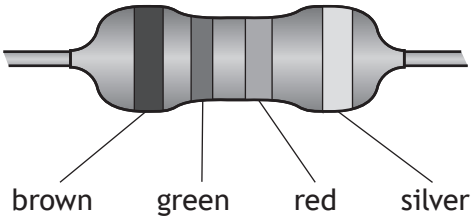
(a) Complete the table for the missing entries. **3**

Component name	Symbol	Function
Bipolar transistor		electronic switch
		controls the current in a circuit
Ammeter		measures current in a circuit
Light dependent resistor (LDR)		

MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(b) The diagram below shows the colour coding for a resistor.



Using information from the data sheet, answer the questions below.

(i) Determine the resistance of the resistor. **1**

(ii) State the percentage tolerance of this resistor. **1**

(iii) Determine the maximum and minimum resistance of this resistor in ohms. **2**

[Turn over

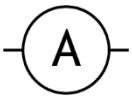
MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(c) The table below shows possible values for the resistance of a resistor in different notations.

220 R	22
2.2 mΩ	2M2

Identify which of the values shown is the **largest** value of resistance. **1**

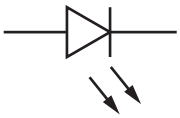
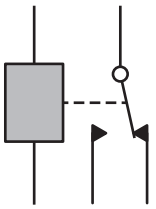
1.	(a)		Resistor **(1 mark)**  **(1 mark)** Changes resistance with changes in light intensity. **(1 mark)**	**3**	LURD on its own not accepted, needs further explanation.
	(b)	(i)	1500 Ω **OR** 1.5 kΩ **OR** 1K5	**1**	Accept: 1500R, 1.5×10^3 Ω Answer must have correct ohms unit or correct R notation.
		(ii)	(±) 10 (%)	**1**	Accept correct answer without percentage symbol. If –10 or +10 given on their own, don't award marks.
		(iii)	(Maximum=) 1650 (Ω) **(1 mark)** (Minimum=) 1350 (Ω) **(1 mark)**	**2**	Accept answer consistent with 1b(i) & 1b(ii). Accept in any order. Accept without max/min labelling, but if given must be correct responses. No penalty for omitting units as "in ohms" is stated in question.
	(c)		2M2	**1**	Accept correct answer converted in different format. Accept a tick or circle of the final answer.

Question	Expected response	Max mark	Additional guidance

1. (a) The table gives information about some circuit components.

Some of the boxes have been left blank.

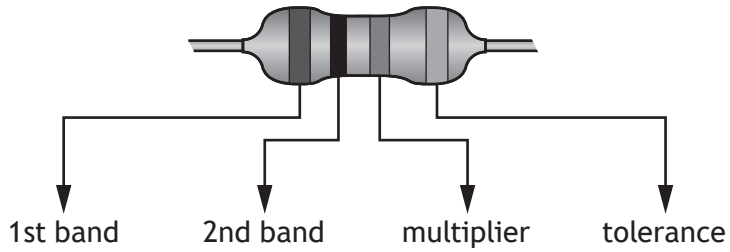
Complete the table for the missing entries. **3**

Name	*Symbol*	*Function*
electrolytic capacitor		stores charge
		emits light
relay		
741 operational amplifier (op-amp)		comparator

MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(b) The following diagram shows the colour coding for a resistor.



A student is comparing two resistors R_1 and R_2. The colour code for each resistor is given in the table below.

Resistor	*1st band*	*2nd band*	*Multiplier*	*Tolerance*
R_1	brown	red	orange	silver
R_2	brown	red	orange	brown

Using information from the data sheet:

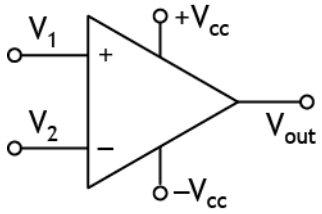
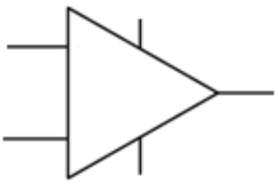
(i) determine the resistance of resistor R_1; **1**

(ii) state the percentage tolerance value of resistor R_2; **1**

(iii) determine the colour codes for a resistor of value 6K8 $\pm$ 5%. **2**

1st band	*2nd band*	*Multiplier*	*Tolerance*
			gold

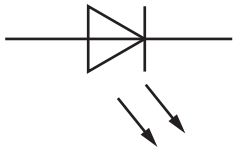
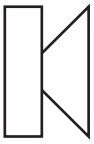
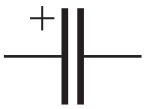
[Turn over

1.	(a)		Light emitting diode/LED **(1 mark)** Electromagnetic switch **(1 mark)**  **(1 mark)**	**3**	Relay- explains control of high power/current/voltage circuit via a low power/current/voltage circuit. Minimum acceptable 741 diagram 
	(b)	(i)	12 000 Ω/ 12 kΩ / 12K	**1**	
		(ii)	(±) 1 (%)	**1**	
		(iii)	Blue and grey **(1 mark)** Red **(1 mark)**	**2**	1 mark for first and second bands correct. 1 mark for multiplier.

1. The table below gives information about some circuit components.

Some of the boxes have been left blank.

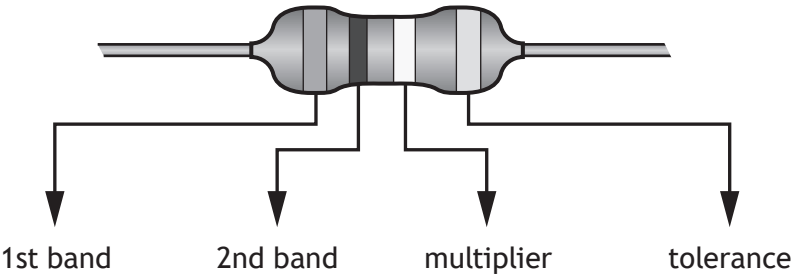
(a) Complete the table for the missing entries. **3**

Component name	Symbol	Function
Light emitting diode		Emits light indicating a current flow
Speaker		
		Stores charge
Variable resistor		A resistor whose resistance value can be adjusted

MARKS | DO NOT WRITE IN THIS MARGIN

1. **(continued)**

(b) The diagram below shows the colour coding for a resistor.



Use the information in the data sheet to answer the question below.

Determine the colour codes for a resistor of value 4K7 with a ±1% tolerance. **2**

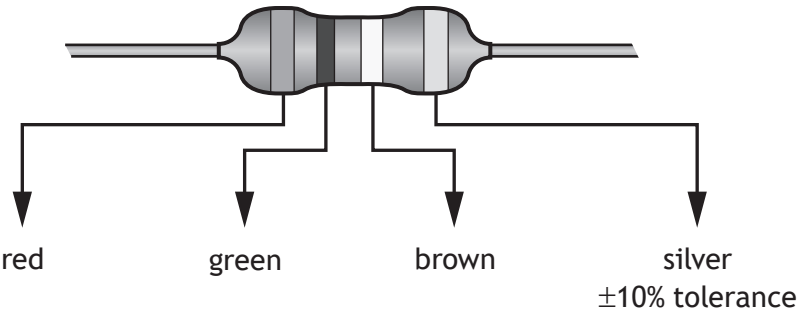
	1st band	2nd band	Multiplier	Tolerance
Colour code				brown

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(c) The diagram below shows the colour coding for a second resistor.



Use the information in the data sheet to answer the questions below.

(i) Determine the resistance of the second resistor. **1**

Space for working and answer

(ii) Determine the minimum and maximum resistance of the second resistor. **2**

Space for working and answer

<table>
<tr><td>1.</td><td>(a)</td><td></td><td>Changes electrical energy to sound (energy) (1 mark)

Electrolytic capacitor (1 mark)

  (1 mark)</td><td>3</td><td>Do not accept changes electricity to sound

Must state electrolytic</td></tr>
<tr><td></td><td>(b)</td><td></td><td><table><tr><th>1st band</th><th>2nd band</th><th>multiplier</th></tr><tr><td>yellow</td><td>violet</td><td>red</td></tr></table></td><td>2</td><td>1 mark for both colour bands
1 mark for multiplier colour band</td></tr>
<tr><td></td><td>(c)</td><td>(i)</td><td>250 Ω

OR

250R</td><td>1</td><td>Answer must have ohms unit or R notation</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Min = 225 Ω OR 225R (1 mark)
Max = 275 Ω OR 275R (1 mark)</td><td>2</td><td>Accept answer consistent with 1(c)(i)</td></tr>
</table>

1. (a) The table gives information about some circuit components. Some of the boxes have been left blank.

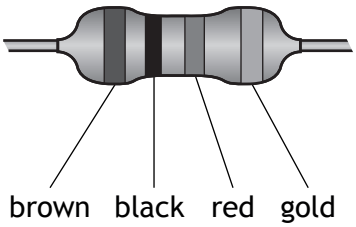
Complete the table for the missing entries. **3**

Name	*Symbol*	*Function*
resistor		limits current, drops voltage
		stores charge
bipolar transistor		electronic switch
diode		

MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(b) The diagram below shows the colour coding for a resistor.



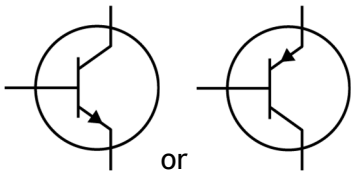
Using information in the data sheet:

(i) determine the resistance of the resistor; **1**

(ii) state the tolerance in the resistance of the resistor; **1**

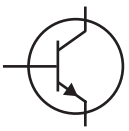
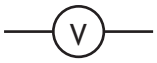
(iii) determine the maximum and minimum resistance of the resistor. **2**

[Turn over

1.	(a)		capacitor 1  or 1 allows current to flow in only one direction or similar 1	**3**	Accept symbol for npn or pnp transistor (with or without circle).
	(b)	(i)	1000 Ω or 1 kΩ or 1k or 1k0	**1**	
		(ii)	5%	**1**	Percentage sign is required.
		(iii)	950 Ω or 950R 1 1050 Ω or 1k05 1	**2**	If unit omitted award a maximum of 1 mark.

1. The table below gives information about some circuit components.
Some of the boxes have been left blank.

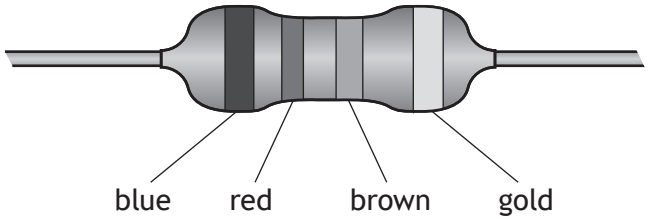
(a) Complete the table for the missing entries. **3**

Component name	Symbol	Function
Bipolar transistor		electronic switch
Cell		
		measures voltage across a circuit
Lamp		converts electrical energy into light energy

MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(b) A pupil selects a resistor with the colour coding as shown below.



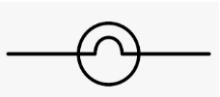
Use the information in the data sheet to answer the questions below.

(i) Determine the resistance of the resistor. **1**

(ii) State the percentage tolerance of this resistor. **1**

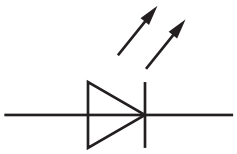
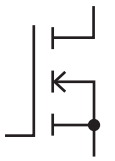
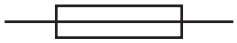
(iii) Determine the maximum and minimum resistance of this resistor in ohms. **2**

[Turn over

1.	(a)		Supplies electrical (potential) energy/charge **(1 mark)** voltmeter **(1 mark)**  **(1 mark)**	**3**	Do not accept: - Supplies power Power supply Stores power Also accept: - Converts chemical energy to electrical Do not accept old bulb symbol 
	(b)	(i)	620Ω **(1 mark)** **OR** 620R **OR** 0.62kΩ	**1**	If given in ohms must have unit, otherwise 0 marks. Also accept scientific notation with unit. 6.2×10^2 Ω
		(ii)	(±)5(%)	**1**	Accept answer without tolerance and/or percentage. Do not accept + or – on its own.
		(iii)	(Maximum) = 651Ω OR 651R **(1 mark)** (Minimum) = 589Ω OR 589R **(1 mark)**	**2**	Accept answer consistent with 1(b) (i) & (ii). Accept in any order. Accept value only. If maximum and minimum is incorrectly labelled award 0 marks.

1. The table below gives information about some circuit components.
Some of the boxes have been left blank.

(a) Complete the table for the missing entries. **3**

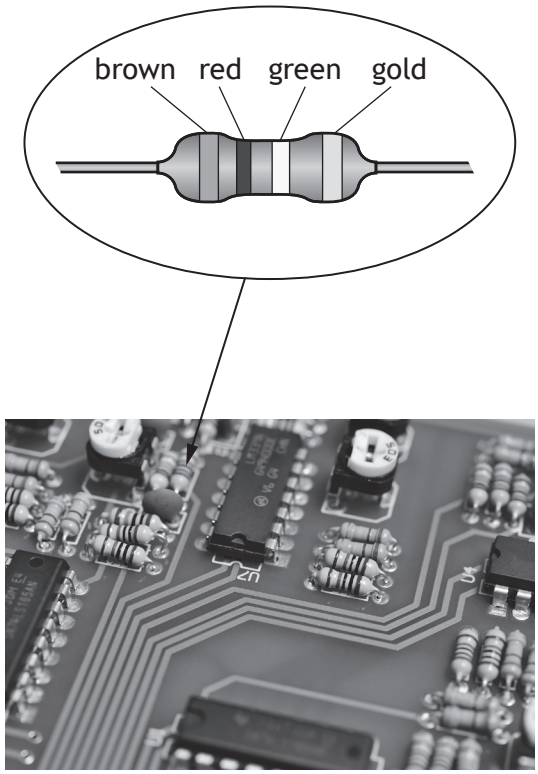
Component name	Symbol	Function
Light emitting diode		Emits light indicating a current flow
MOSFET transistor		
Motor		Converts electrical energy to kinetic energy
		To protect wiring from too much current

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1. (continued)

(b) A technician is examining the circuit board shown below when a fault is discovered.

A resistor with an incorrect value has been used.



Use the information in the data sheet to answer the questions below.

(i) Determine the resistance of the resistor. **1**

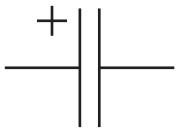
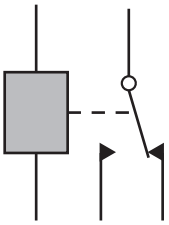
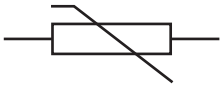
(ii) The technician replaces the incorrect component with a resistor of value 600R which has a 2% tolerance.

Determine the minimum and maximum resistance of the resistor. **2**

1.	(a)		Switch electrical **(1 mark)**  **(1 mark)** FUSE **(1 mark)**	**3**	
	(b)	(i)	1M2 **OR** 1·2MΩ **OR** 1,200,000Ω	**1**	Also accept scientific notation with unit
		(ii)	Minimum 588Ω **OR** 588R **(1 mark)** Maximum 612Ω **OR** 612R **(1 mark)**	**2**	

1. The table below gives information about some circuit components.
Some of the boxes have been left blank.

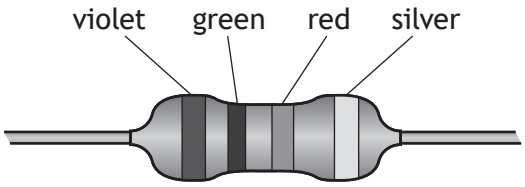
(a) Complete the table for the missing entries. **3**

Component name	Symbol	Function
Electrolytic capacitor		stores charge
		electromagnetic switch
Diode		allows current to flow in one direction only
Thermistor		

MARKS | DO NOT WRITE IN THIS MARGIN

1. (continued)

(b) A technician selects a resistor with the colour coding as shown below.

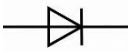


Use the information in the data sheet to answer the questions below.

(i) Determine the resistance of this resistor. **1**

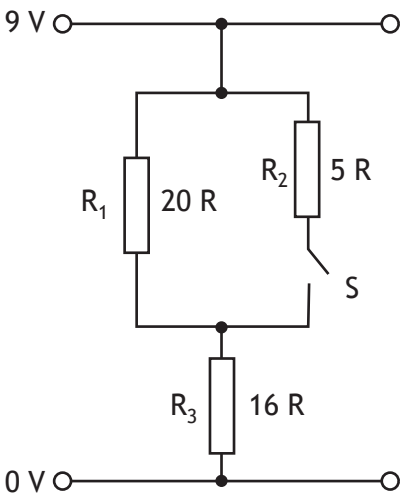
(ii) State the percentage tolerance of this resistor. **1**

(iii) Determine the maximum and minimum resistance of this resistor in **ohms**. **2**

1.	(a)		Relay **(1 mark)** **(1 mark)**  Changes resistance with (changes in) temperature **(1 mark)**	**3**	Do not accept switch Accept alternatives:  Also accept -  Must have input & output lines. Accept heat Accept expressions involving TURD (Temperature goes Up, Resistance goes Down)
	(b)	(i)	7.5 $k\Omega$ **OR** 7K5 **OR** 7500 Ω	**1**	If given in ohms must have unit. Also accept scientific notation with unit Accept 7500R Don not accept 7.5KR
		(ii)	(±)10%	**1**	Accept answer without percentage If + or – is given on its own, do not award marks
		(iii)	Maximum = 8250 Ω **(1 mark)** Minimum = 6750 Ω **(1 mark)**	**2**	Accept answer consistent with 1(b)(i). Accept in any order. Accept without Max & Min name, but if given must be correct response.No penalty for omitting units, as “in ohms” is stated in the question.

MARGIN

3. A circuit diagram is shown below.



(a) Switch S is **open**.

(i) Calculate the total resistance of this circuit. **1**

Space for working and answer

(ii) Calculate the current in this circuit. **3**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

3. (a) (continued)

(iii) Calculate the power dissipated in resistor R_3. **3**

Space for working and answer

(b) Switch S is now **closed**.

(i) Calculate the effective resistance of the two resistors in parallel. **3**

Space for working and answer

(ii) Calculate the overall circuit resistance. **1**

Space for working and answer

[Turn over

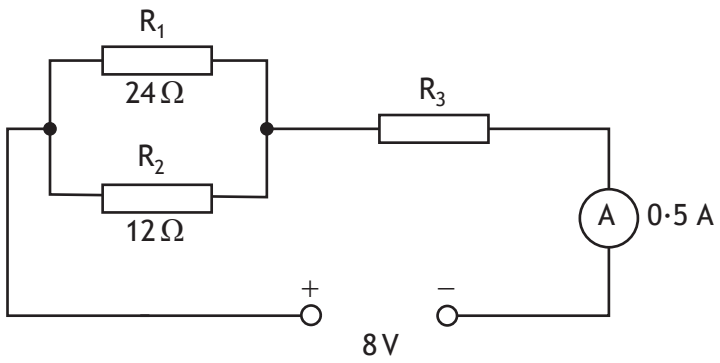
3.	(a)	(i)	Rt = 20 + 16 = 36 Ω	**1**	Accept $36R$ Must have correct unit.
		(ii)	V = I R 9 = I × 36 I = 0.25 A	**3**	Accept answer consistent with 3(a)(i).
		(iii)	$P = I^2 R$ $P = 0.25^2 \times 16$ P = 1 W	**3**	Accept answer consistent with 3(a)(ii) with correct substitutions. **1 mark** for both relationships. **1 mark** for substitutions. **1 mark** for final answer including unit. Alternative method V = IR& $P = V^2R$ **OR** Voltage divider equation method.

Question			Expected response	Max mark	Additional guidance
3.	(b)	(i)	1/Rt = 1/R1 + 1/R2 1/Rt = 1/20 + 1/5 Rt = 4 Ω	**3**	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ Is an acceptable alternative method. If wrong equation used, eg $RT = \frac{1}{R1} + \frac{1}{R2}$ then **(0) marks**. Accept imprecise working towards a final answer. $\frac{1}{R_T} = 4\Omega$ ↑ accept Accept = 4 R Accept any subscript on component resistors.
		(ii)	Rt = 4 + 16 = 20 Ω	**1**	Must have correct unit. Accept $20R$ Accept answer consistent with 3(b) (i).

Question	Expected response	Max mark	Additional guidance

MARGIN

3. A student sets up the circuit shown.



(a) Calculate the total resistance of the circuit. **3**

Space for working and answer

(b) Calculate the effective resistance of R_1 and R_2 in parallel. **3**

Space for working and answer

(c) Determine the resistance of resistor R_3. **1**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

3. (continued)

(d) Determine the voltage across R_3. **1**

Space for working and answer

(e) Calculate the power dissipated in resistor R_3. **3**

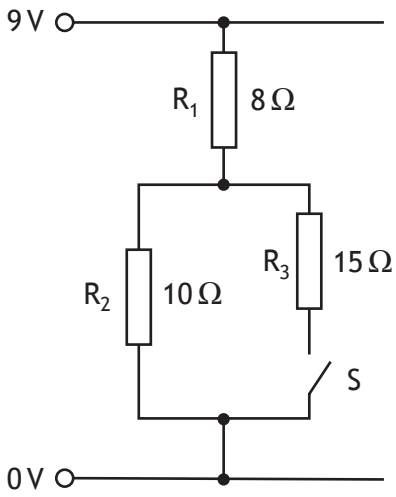
Space for working and answer

[Turn over

3.	(a)		$V = IR$ **(1 mark)** $8 = 0{\cdot}5 \times R$ **(1 mark)** $R = 16\ \Omega$ **(1 mark)**	**3**	Accept 16R
	(b)		$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ **(1 mark)** $\frac{1}{R_T} = \frac{1}{24} + \frac{1}{12}$ **(1 mark)** $R_T = 8\ \Omega$ **(1 mark)**	**3**	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ is an acceptable alternative method. If wrong equation used, eg $R_T = \frac{1}{R_1} + \frac{1}{R_2}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{12} + \frac{1}{24} = 8\ \Omega$ ↑ accept Accept 8R
	(c)		$R_3 = 8\ \Omega$	**1**	Or consistent with (a) and (b). Accept 8R
	(d)		$V_3 = 4\ \text{V}$	**1**	Or consistent with (a), (b), and (c).
	(e)		$P = \frac{V^2}{R}$ **(1 mark)** $P = \frac{4^2}{8}$ **(1 mark)** $P = 2\ \text{W}$ **(1 mark)**	**3**	Or consistent with (a), (b), (c), and (d). $P = IV$ **1** $P = 0{\cdot}5 \times 4$ **1** $P = 2\ \text{W}$ **1** Or $P = I^2 R$ **1** $P = 0{\cdot}5^2 \times 8$ **1** $P = 2\ \text{W}$ **1**

MARGIN

3. A circuit diagram is shown below.



Switch S is **open.**

(a) (i) Calculate the total resistance of the circuit. **1**

Space for working and answer

(ii) Calculate the current in the circuit. **3**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

3. (a) (continued)

(iii) Calculate the power dissipated in resistor R_1. **3**

Space for working and answer

(b) Switch S is now **closed**.

Calculate the effective resistance of R_2 and R_3 in parallel. **3**

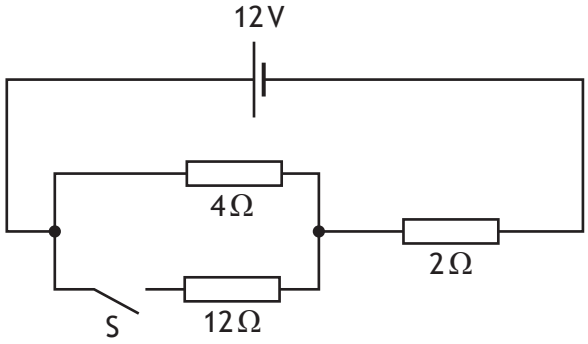
Space for working and answer

[Turn over

3.	(a)	(i)	18Ω	**1**	Accept $18R$
		(ii)	$V = IR$ **(1 mark)** $9 = I \times 18$ **(1 mark)** $I = 0{\cdot}5\ A$ **(1 mark)**	**3**	Accept answer consistent with 3(a)(i) Answer must have unit
		(iii)	$P = I^2 R$ **(1 mark)** $P = 0{\cdot}5^2 \times 8$ **(1 mark)** $P = 2W$ **(1 mark)**	**3**	Accept answer consistent with 3(a)(ii) Answer must have unit Accept P=IV =v2/R if full working given
	(b)		$\frac{1}{R_T} = \frac{1}{R_2} + \frac{1}{R_3}$ **(1 mark)** $\frac{1}{R_T} = \frac{1}{10} + \frac{1}{15}$ **(1 mark)** $R_T = 6\Omega$ **(1 mark)**	**3**	$R_T = \frac{R_2 R_3}{R_2 + R_3}$ Is an acceptable alternative method If wrong equation used, eg $R_T = \frac{1}{R_2} + \frac{1}{R_3}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{15} + \frac{1}{10} = 6\Omega$ accept Accept = 6 R Accept any subscript on component resistors

WRITE IN THIS MARGIN

3. A student sets up a circuit as shown.



(a) Switch S is open.

(i) Calculate the total resistance of the circuit. **1**

(ii) Calculate the current in the circuit. **3**

(b) Switch S is now closed.

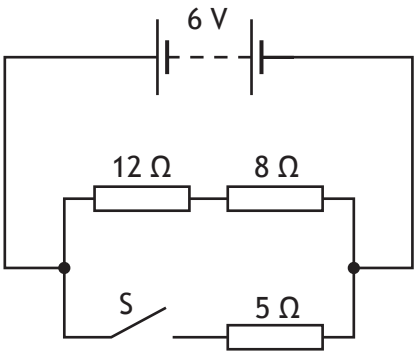
(i) Calculate the effective resistance of the two resistors in parallel. **3**

(ii) Calculate the overall circuit resistance. **1**

3.	(a)	(i)	6 Ω	**1**	Unit required.
		(ii)	$V = IR$ 1 $12 = I \times 6$ 1 $I = 2$ A 1	**3**	Or consistent with (a)(i)
	(b)	(i)	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ 1 $\frac{1}{R} = \frac{1}{4} + \frac{1}{12}$ 1 $R_T = 3\ \Omega$ 1	**3**	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ is an acceptable alternative method. If wrong equation used, eg $R_T = \frac{1}{R_1} + \frac{1}{R_2}$ then (0) marks Accept imprecise working towards a final answer. $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{12} = 3\ \Omega$ ↑ accept
		(ii)	5 Ω	**1**	Unit required. Or consistent with (b)(i)

MARGIN

3. A circuit diagram is shown below.



(a) Switch S is **open**.

(i) Calculate the total resistance of the circuit. **1**

Space for working and answer

(ii) Calculate the current in the circuit. **3**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

3. (continued)

(b) Switch S is now **closed**.

(i) Calculate the total resistance of the circuit. **3**

Space for working and answer

(ii) State the voltage across the 5 Ω resistor. **1**

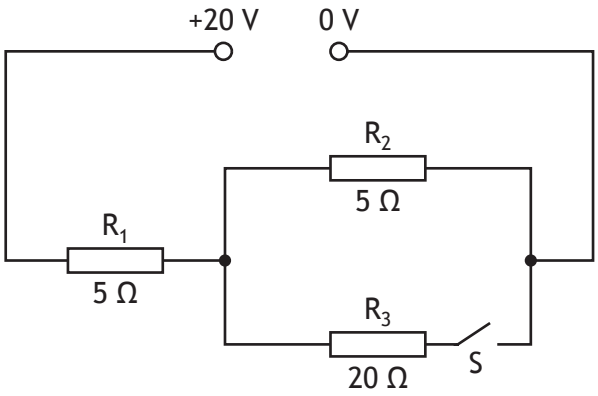
(iii) Calculate the power dissipated in the 5 Ω resistor. **3**

Space for working and answer

3.	(a)	(i)	20 Ω **OR** 20 R	**1**	Answer must have ohms unit or R notation
		(ii)	$V = IR$ **(1 mark)** $6 = I \times 20$ **(1 mark)** $I = 0{\cdot}3\ A$ **(1 mark)**	**3**	Accept answer consistent with 3(a)(i)
	(b)	(i)	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ **(1 mark)** $\frac{1}{R_T} = \frac{1}{20} + \frac{1}{5}$ **(1 mark)** $R_T = 4\Omega$ **(1 mark)**	**3**	$R_T = \frac{R_2 R_3}{R_2 + R_3}$ Is an acceptable alternative method If wrong equation used, eg $R_T = \frac{1}{R_2} + \frac{1}{R_3}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = 4\Omega$ Accept Accept = 4 R Accept any subscript on component resistors
		(ii)	6V **(1 mark)**	**1**	Answer must have unit

Question			Expected response	Max mark	Additional guidance
		(iii)	$P = \frac{V^2}{R}$ **(1 mark)** $P = \frac{6^2}{5}$ **(1 mark)** $P = 7 \cdot 2 \text{ W}$ **(1 mark)**	**3**	Accept answer consistent with 3(b)(ii) Both equations 1 mark Both substitutions 1 mark Answer and unit 1 mark $V = IR$ $6 = I \times 5$ $I = 1 \cdot 2 (A)$ $P = IV$ $P = 1 \cdot 2 \times 6$ $P = 7 \cdot 2 W$ **OR** $P = I^2 R$ $R = 1 \cdot 2^2 \times 5$ $P = 7 \cdot 2 W$

3. A circuit diagram is shown below.



(a) Switch S is **open**.

(i) Calculate the total resistance of this circuit. **1**

Space for working and answer

(ii) Calculate the current in this circuit. **3**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

3. (a) (continued)

(iii) Calculate the power dissipated in resistor R_1. **3**

Space for working and answer

(b) Switch S is now **closed**.

(i) Calculate the effective resistance of R_2 and R_3 in parallel. **3**

Space for working and answer

(ii) Calculate the overall circuit resistance. **1**

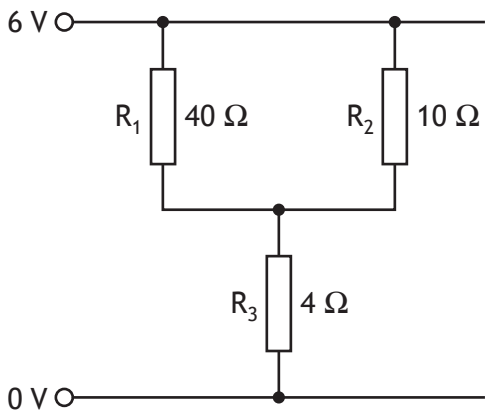
Space for working and answer

[Turn over

3.	(a)	(i)	(R_T = 5 + 5) **(1 mark)** R_T = 10 Ω	**1**	Accept 10R Must have correct unit.
		(ii)	V = IR **(1 mark)** 20 = I × 10 **(1 mark)** I = 2 A **(1 mark)**	**3**	Accept answer consistent with 3(a)(i) 1 mark for both relationships. 1 mark for substitutions. 1 mark for final answer including unit. Also accept alternative methods. Do not accept the term "Amps" as units.
		(iii)	$P = I^2R$ **(1 mark)** $P = \cdot 2^2 \times 5$ **(1 mark)** $P = 20W$ **(1 mark)**	**3**	Accept answer consistent with 3(a)(ii) 1 mark for relationship. 1 mark for substitutions. 1 mark for final answer including unit. Also accept alternative formulas but with correct substitution.
	(b)	(i)	$\frac{1}{R_T} = \frac{1}{R_2} + \frac{1}{R_3}$ **(1 mark)** $\frac{1}{R_T} = \frac{1}{5} + \frac{1}{20}$ **(1 mark)** $R_T = 4\Omega$ **(1 mark)**	**3**	$R_T = \frac{R_1 R_2}{R_1 + R_2}$ Is an acceptable alternative method. If wrong equation used, eg, $R_T = \frac{1}{R_1} + \frac{1}{R_2}$ then (0) marks. Accept imprecise working towards a final answer. $\frac{1}{R_T} = 4\Omega$ ↑ accept Accept = 4 R Accept any subscript on component resistors.

Question			Expected response	Max mark	Additional guidance
3.	(b)	(ii)	(R_T = 5 + 4) **(1 mark)** R_T = 9 Ω	**1**	Accept answer consistent with 3(b)(i) Accept 9R Must have correct unit.

3. A circuit diagram is shown below.



(a) (i) Calculate the effective resistance of R_1 and R_2 in parallel. **3**

Space for working and answer

(ii) Calculate the total resistance of this circuit. **1**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

3. (continued)

(b) Calculate the current in resistor R_3. **3**

Space for working and answer

(c) Calculate the power dissipated in resistor R_3. **3**

Space for working and answer

[Turn over

3.	(a)	(i)	$\frac{1}{R_T}=\frac{1}{R_1}+\frac{1}{R_2}$ **(1 mark)** $\frac{1}{R_T}=\frac{1}{40}+\frac{1}{10}$ **(1 mark)** $R_T=8\Omega$ **(1 mark)**	**3**	$R_T=\frac{R_1R_2}{R_1+R_2}$ Is an acceptable alternative method If wrong equation used, eg $RT=\frac{1}{R1}+\frac{1}{R2}$ then (0) marks Accept imprecise working towards a final answer $\frac{1}{R_T}=8\Omega$ ↑ accept Accept = 8 R Accept any subscript on component resistors
		(ii)	(R_T = 8 + 4) R_T = 12 Ω	**1**	Accept answer consistent with 3(a)(i) Accept 12R Must have correct unit
	(b)		$V = IR$ **(1 mark)** 6 = I × 12 **(1 mark)** I = 0·5 A **(1 mark)**	**3**	Accept answer consistent with 3(a)(ii) 1 mark for both relationships 1 mark for substitutions 1 mark for final answer including unit Accept: $V_{R_3}=\left(\frac{R_3}{R_p+R_3}\right)V_s$ $V_{R_3}=\left(\frac{4}{8+4}\right)6$ = 2 V And $V_{R_3}=I\times R_3$ $2=I\times4$ I = 2/4 =0.5A

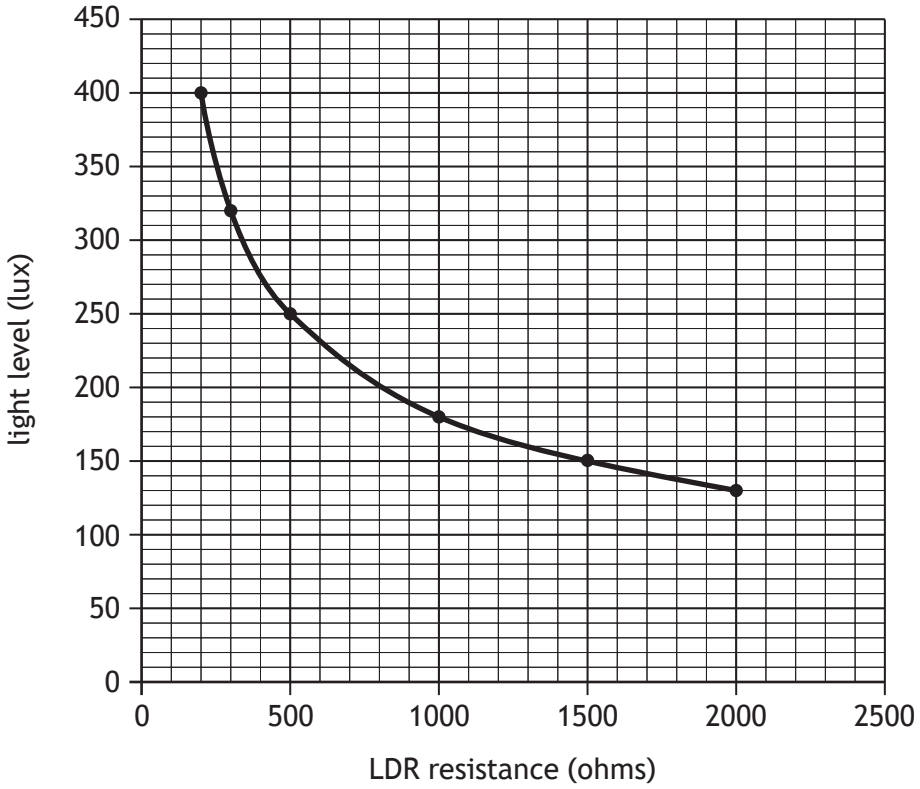
Question			Expected response	Max mark	Additional guidance
3.	(c)		$P = I^2R$ **(1 mark)** $P = 0 \cdot 5^2 \times 4$ **(1 mark)** $P = 1W$ **(1 mark)**	3	Accept answer consistent with 3(b) with correct substitutions 1 mark for both relationships 1 mark for substitutions 1 mark for final answer including unit Accept $V = I\ R_3$ $0.5 \times 4 = 2\ V$ $P = \frac{V^2}{R}$ $P = IV$ $P = \frac{2^2}{4}$ $P = 0.5 \times 2$ $P = 1W$ $P = 1W$

Question	Expected response	Max mark	Additional guidance

MARGIN

9. A technician needs to work in a well lit room. For safety reasons when the light level decreases to 250 lux an alarm sounds.

The alarm circuit uses a light dependent resistor as a sensor. The graph below shows how the resistance of the LDR varies with changing light levels.

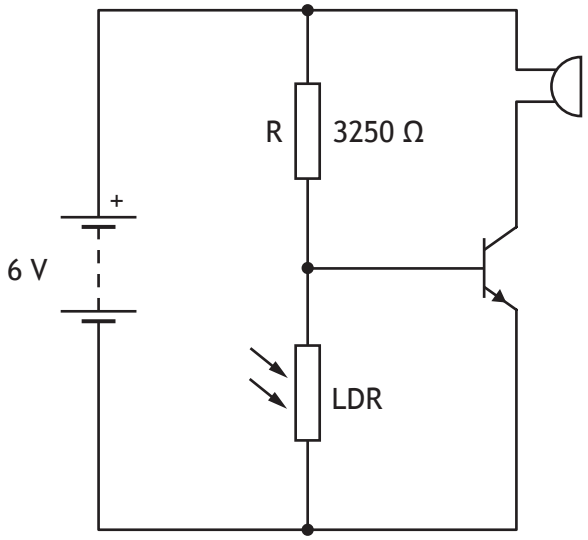


(a) State the resistance of the LDR when the alarm sounds. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

9. (continued)

(b) The alarm circuit is shown below.



Calculate the voltage across the LDR when the alarm (buzzer) sounds. **3**

Space for working and answer

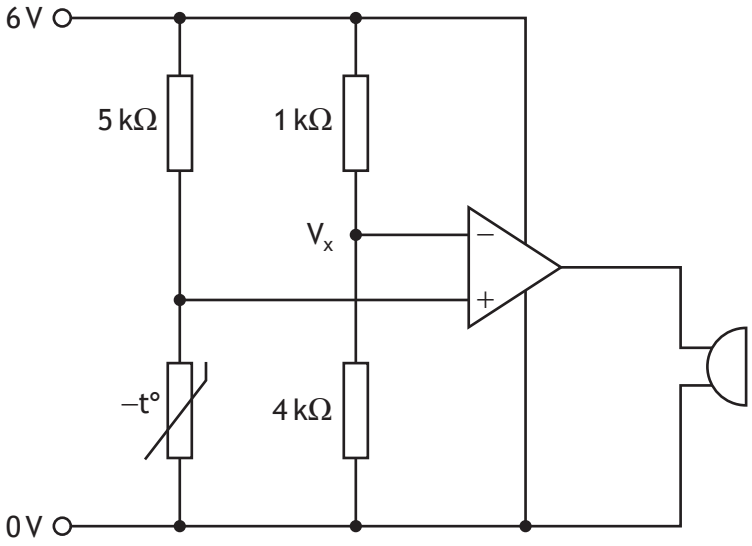
(c) Describe how this circuit works. **3**

9.	(a)		500 Ω **OR** 500R	**1**	
	(b)		$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$ **(1 mark)** $V_2 = \frac{500}{(3250 + 500)} \times 6$ **(1 mark)** $V_2 = 0 \cdot 8V$ **(1 mark)**	**3**	Accept answer consistent with 9(a) $V = IR$ $6 = I \times 3750$ $I = 0.0016$ $I = 1 \cdot 6 \times I0^{-3}(A)$ $V = IR$ $V = 1 \cdot 6 \times I0^{-3} \times 500$ $V = 0 \cdot 8V$ 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit. Or by voltage ratio method. If correct answer given with no working award 3 marks.
	(c)		As light level reduces LDR resistance increases **(1 mark)** As LDR resistance increases the voltage across LDR increases. **(1 mark)** When voltage across the LDR reaches 0·8V transistor switches on **(1 mark)**	**3**	

Question	Expected response	Max mark	Additional guidance

THIS MARGIN

7. A low temperature warning system is controlled using a LM741 comparator circuit as shown in the simulation below.



(a) Calculate the reference voltage V_x. **3**

Space for working and answer

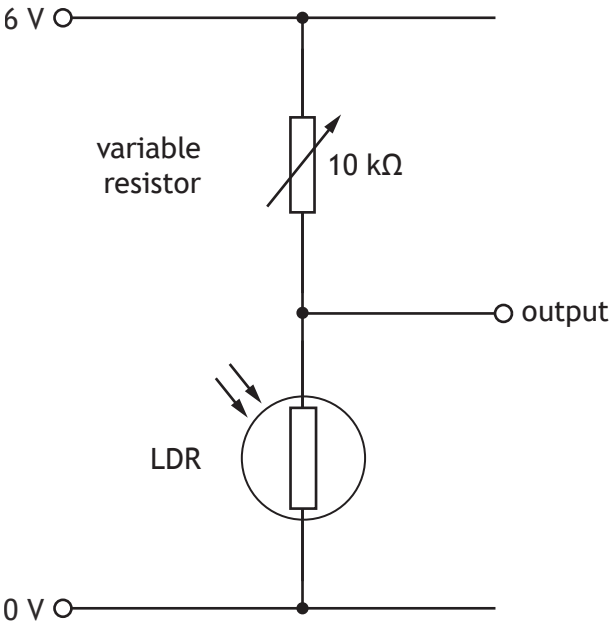
(b) State the resistance of the thermistor when the voltage across it is equal to the reference voltage V_x. **1**

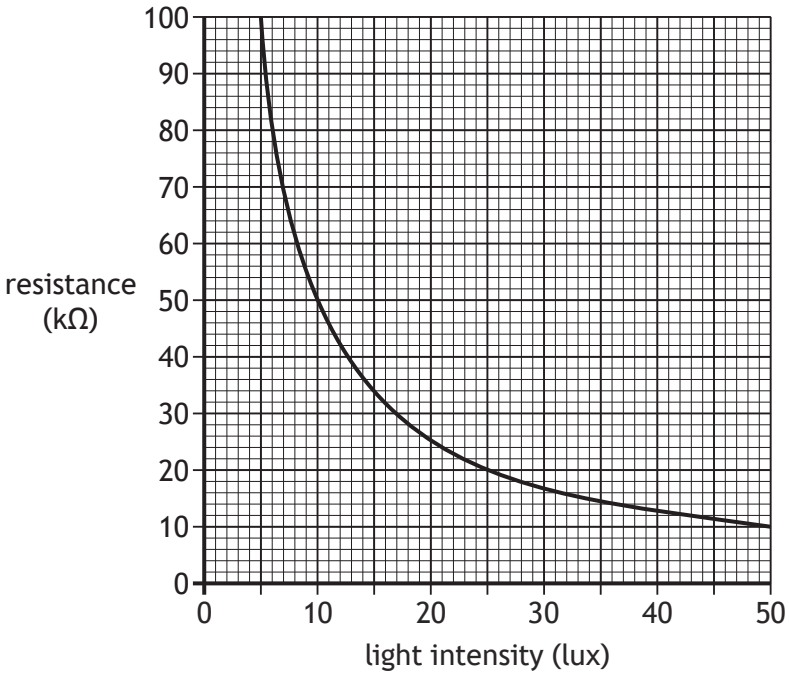
(c) Describe how this circuit works. **3**

7.	(a)		$V_x = \frac{R_2}{R_1 + R_2} \times V_S$ **(1 mark)** $V_x = \frac{4000}{5000} \times 6$ **(1 mark)** $V_x = 4.8V$ **(1 mark)**	**3**	Accept (4/5) x 6. 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit. Or by voltage ratio method. If correct answer given with no working award 3 marks.
	(b)		20K or 20kΩ or 20,000Ω	**1**	Must have unit or be in R notation
	(c)		As temperature decreases thermistor resistance increases. **(1 mark)** As thermistor resistance increases the voltage across the thermistor increase. **(1 mark)** When voltage across thermistor = Vx then LM741/comparator/IC/op amp switches on **(1 mark)**	**3**	Accept V ref = Vx = trigger voltage = switch on voltage = threshold (voltage) Accept buzzer/alarm sounds in place of LM741 switching on but not bell

9. An LDR is used as a light sensor in a circuit to monitor and control the automatic mode setting of a car's headlights. Part of the circuit is shown below.

MARKS DO NOT WRITE IN THIS MARGIN





(a) (i) The graph above shows how the resistance of the LDR varies with the light intensity, measured in lux.

Use the graph to determine the resistance of the LDR when the light intensity is 25 lux. **1**

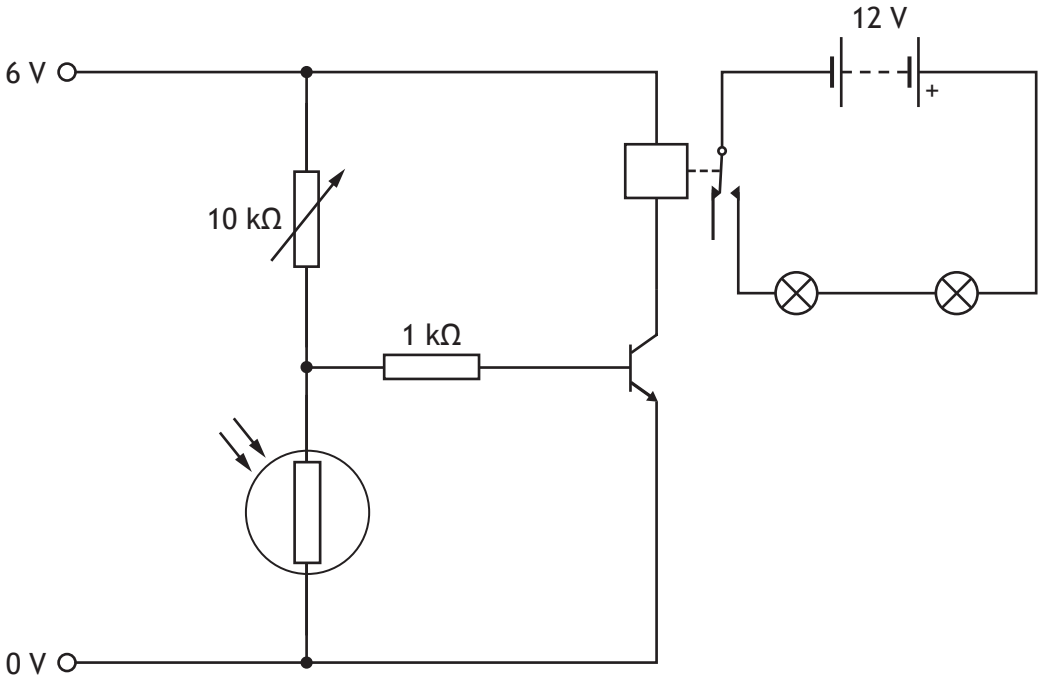
9. (a) (continued) **MARKS** DO NOT WRITE IN THIS MARGIN

(ii) The variable resistor is now set at a resistance of 10 kΩ.

Calculate the voltage across the LDR when the light intensity is kept at 25 lux. **3**

Space for working and answer

(b) The potential divider is now connected to a switching circuit via the car battery to operate the car lights, as shown below.



(i) Explain how the circuit operates in order to switch on the car lights when the light intensity falls below a certain value. **3**

MARKS | DO NOT WRITE IN THIS MARGIN

9. (b) (continued)

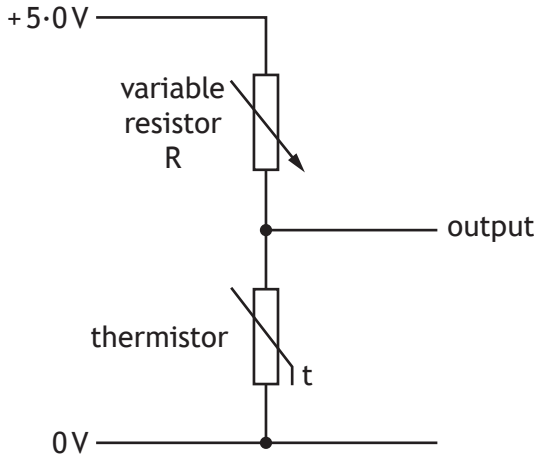
(ii) The operation of the circuit can be improved by the addition of one component connected in parallel with the relay.

Name this component. **1**

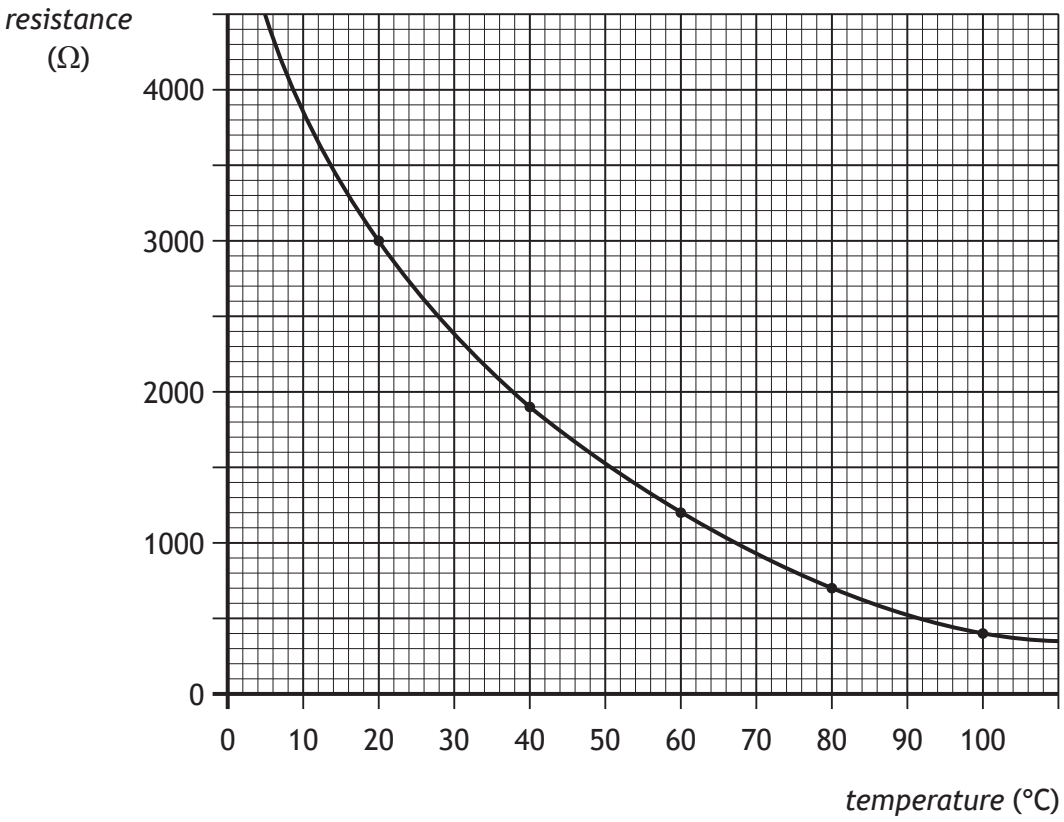
9.	(a)	(i)	20kΩ **(1 mark)** **OR** 20000Ω **OR** 20K	**1**	Must have unit or be in R notation. Allow a tolerance of + or – half a box.
		(ii)	$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$ **(1 mark)** $V_2 = \frac{20}{(20+10)} \times 6$ **(1 mark)** $V_2 = 4V$ **(1 mark)**	**3**	Accept (20/30) × 6 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit. Or by voltage ratio method. $V_2/V_s=R_2/R_{total}$ $V_2/6=20/30$ $V_2=4V$ If correct answer with unit and no working award 3 marks. Accept follow through rule from Q9 a(i).
	(b)	(i)	LDR resistance increases and voltage across it increases. **(1 mark)** Transistor switches on at a certain voltage/0.7v **(1 mark)** Relay activates/turns on/closes **(1 mark)**	**3**	Independent marks Accept Transistor switches on/switch on voltage at 0.7v Answer must include the following three key points: - • LDR resistance increases • voltage increases to turn on Transistor at certain voltage • turns on RELAY (external circuit switches on) If transistor switching voltage is given as 2v award a maximum of 2 marks.
		(ii)	Diode **(1 mark)**	**1**	Do not accept the circuit symbol, the name must be given.

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7. A thermistor is used as a temperature sensor in a circuit to monitor and control the temperature of water in a tank. Part of the circuit is shown.



(a) (i) The graph shows how the resistance of the thermistor varies with temperature.



Use the graph to determine the resistance of the thermistor when the temperature is 60 °C. **1**

[Turn over

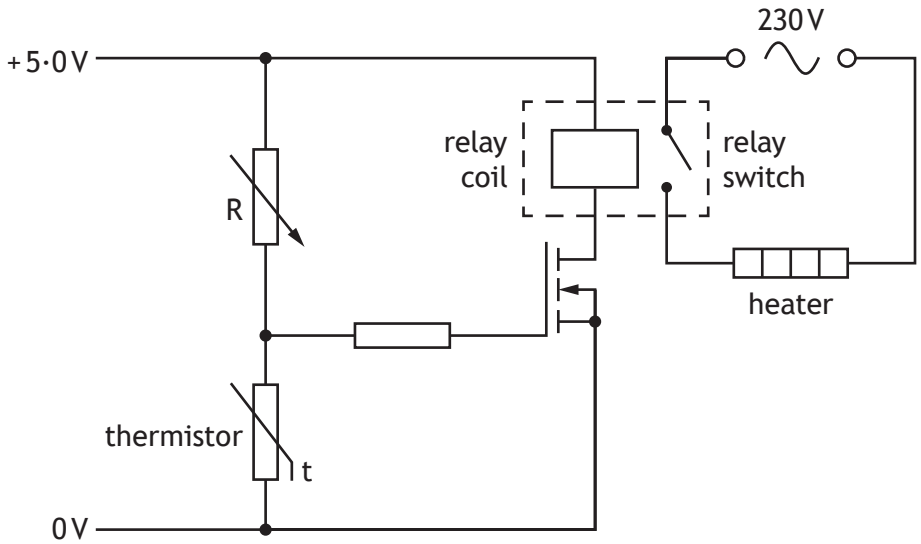
MARKS | DO NOT WRITE IN THIS MARGIN

7. (a) (continued)

(ii) The variable resistor R is now set at a resistance of 800 Ω.

Calculate the voltage across the thermistor when the temperature is 60 °C. **3**

(b) The potential divider is now connected to a switching circuit to operate a heater.



(i) Explain how the circuit operates to switch on the heater when the temperature falls below a certain value. **3**

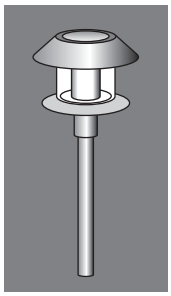
(ii) The operation of this circuit can be improved by the addition of one component connected across the relay.

Name this component. **1**

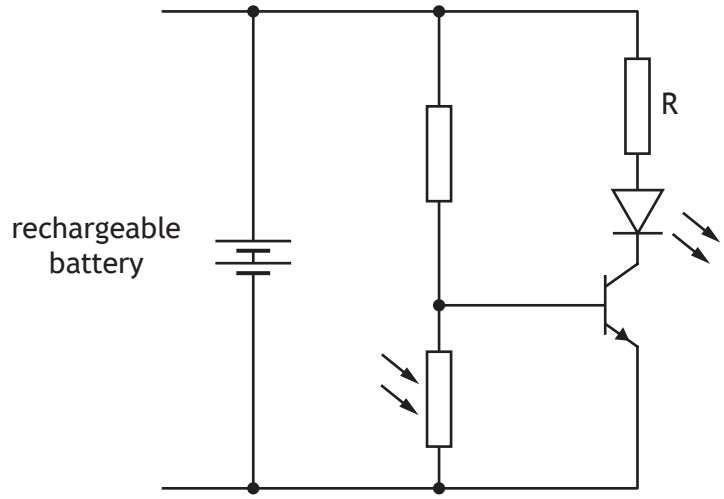
7.	(a)	(i)	1200 Ω or 1k2	**1**	
		(ii)	$V_1 = \frac{R_1}{R_1 + R_2} \times V_S$ 1 $V_1 = \frac{1200}{1200 + 800} \times 5 \cdot 0$ 1 $V_1 = 3$ V 1	**3**	Or consistent with (a)(i) Using Ohm's law: $R_T\,(= 1200 + 800) = 2000$ $I = \frac{V}{R}$ $I = \frac{5 \cdot 0}{2000}$ $I = 2 \cdot 5 \times 10^{-3}$ A $V_1 = IR$ $V_1 = 2 \cdot 5 \times 10^{-3} \times 1200$ $V_1 = 3$ V 1 mark for Ohm's Law relationship 1 mark for all substitutions 1 mark for final answer including unit
	(b)	(i)	Resistance of thermistor increases so voltage across it increases. 1 Transistor switches on at a certain voltage/2 V. 1 Relay activates and closes contacts in heater circuit. 1	**3**	If transistor switching voltage is given as 0·7 V award a maximum of 2 marks.
		(ii)	diode	**1**	

MARGIN

7. A high intensity LED is used as a garden light. The light turns on automatically when it becomes dark.



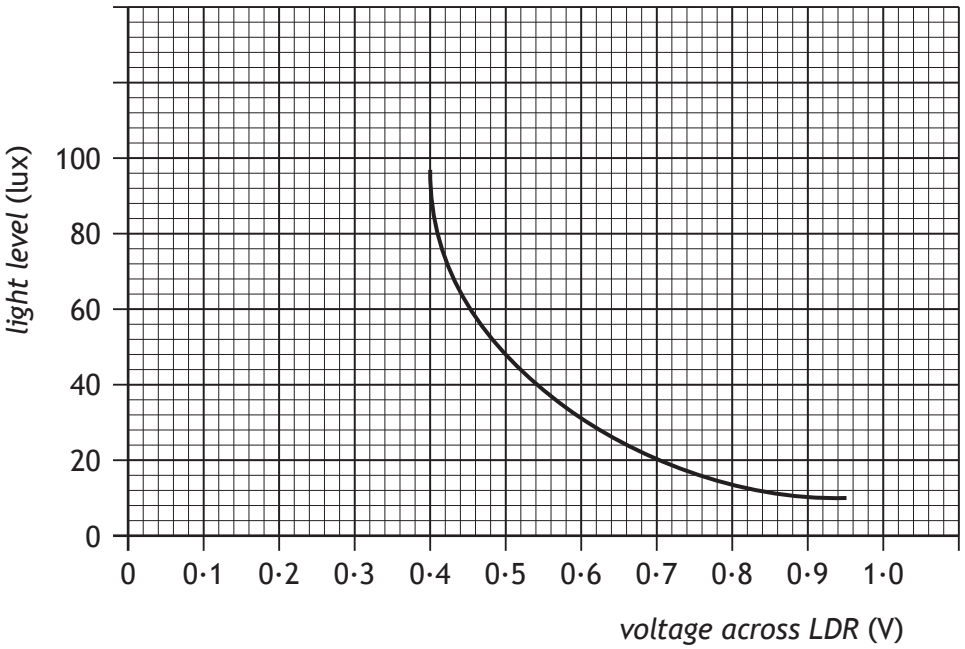
(a) The LED is switched on using the following circuit.



MARKS | DO NOT WRITE IN THIS MARGIN

7. (a) (continued)

The graph shows the voltage across the LDR in this circuit for different light levels.



(i) For the LED to switch on, the voltage across the LDR must be at least 0·7 V.

Determine the light level at which the LED switches on. **1**

(ii) Explain the purpose of resistor R. **1**

(iii) The manufacturer has used a bipolar transistor, as a MOSFET cannot be used in this circuit.

Explain why a MOSFET cannot be used in this circuit. **1**

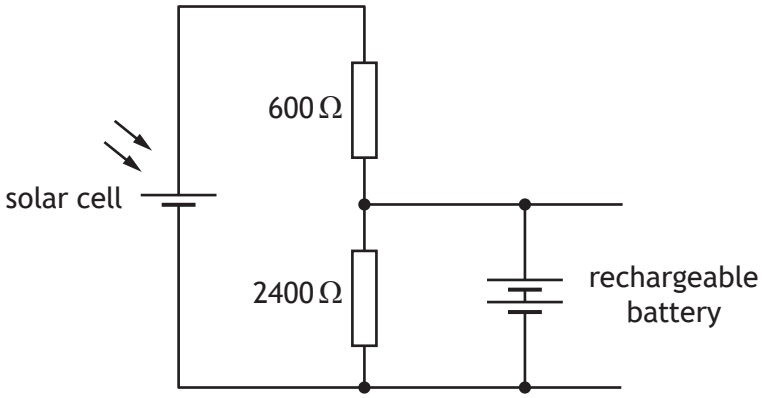
[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

7. (continued)

(b) The light also contains a solar cell which charges the rechargeable battery during daylight hours.

Part of the circuit is shown.



At a particular light level, the voltage generated by the solar cell is 1·5 V.

Calculate the voltage across the rechargeable battery at this light level. **3**

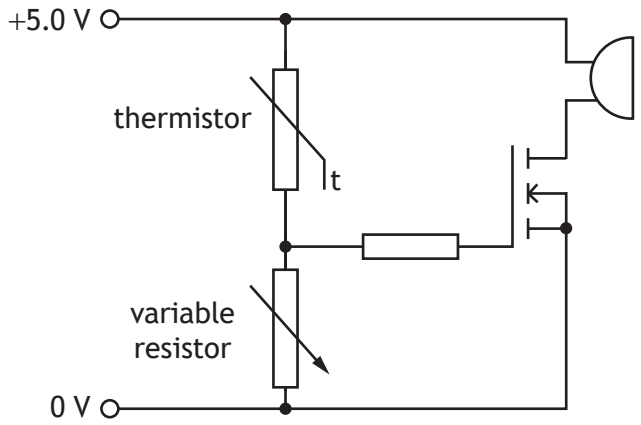
Space for working and answer

7.	(a)	(i)	20 lux	**1**	
		(ii)	Drop voltage/reduce current/protect LED/limit power.	**1**	Do not accept: voltage 'through' or current 'across' on their own.
		(iii)	Switch on voltage too high/2 V.	**1**	
	(b)		$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$ **(1 mark)** $V_2 = \frac{2400}{600 + 2400} \times 1{\cdot}5$ **(1 mark)** $V_2 = 1{\cdot}2$ V **(1 mark)**	**3**	$V = IR$ $1{\cdot}5 = I \times 3000$ $I = 5 \times 10^{-4}$ (A) $V = IR$ $V = 5 \times 10^{-4} \times 2400$ $V = 1{\cdot}2$ V 1 mark for Ohm's Law anywhere. 1 mark for **both** substitutions. 1 mark for final answer including unit. Or by voltage ratio method. If correct answer given with no working award 3 marks.

MARGIN

7. An alarm circuit has a buzzer that sounds when the temperature is too high.

The circuit diagram for the buzzer system is shown below.



(a) At a certain temperature, the thermistor has a resistance of 1000 Ω.

The variable resistor is set at a resistance of 1500 Ω.

Calculate the voltage across the variable resistor. **3**

Space for working and answer

(b) Explain how the circuit operates to sound the buzzer. **3**

(c) State the purpose of the variable resistor in this circuit. **1**

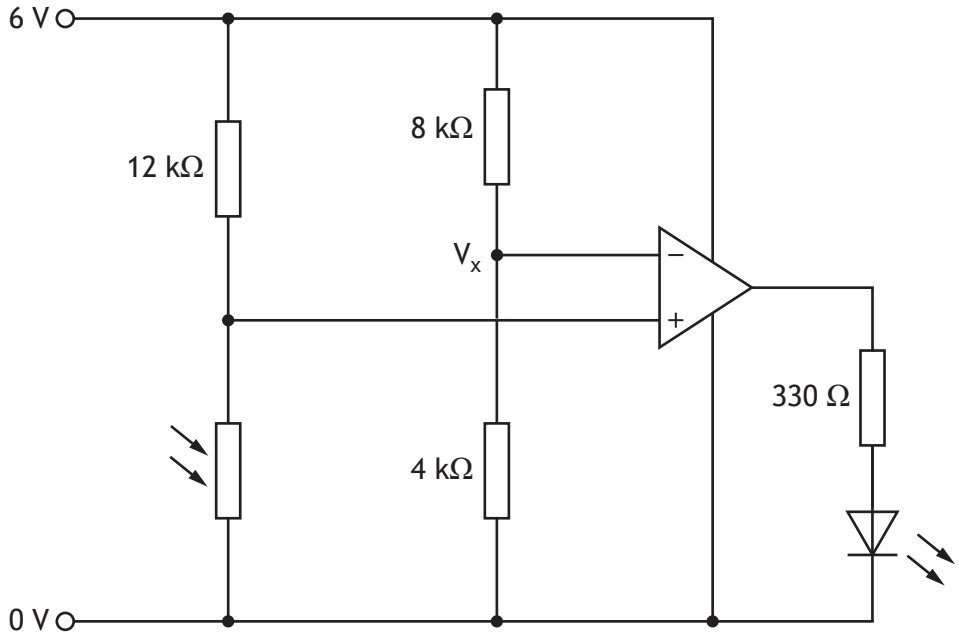
[Turn over

7.	(a)		$V_2 = \frac{R2}{R1+R2} \times v_s$ **(1 mark)** $V_2 = \frac{1500}{1500+1000} \times 5$ **(1 mark)** V_2 = 3.0 V **(1 mark)**	**3**	Accept (1.5/2.5) x 5 Accept 3 V V = IR 5.0 = I × 2500 I = 0.002 (A) **OR** 2×10^{-3} (A) V = IR V = 0.002 × 1500 **OR** V = 2×10^{-3} × 1500 V = 3.0 V **OR** V = IR 5.0 = I × 2.5 from 2.5 (kΩ) I = 2 (mA) V = IR V = 2 × 1.5 from 2(mA) x 1.5(kΩ) V = 3.0 V **1 mark** for Ohm's Law anywhere. **1 mark** for both substitutions. **1 mark** for final answer including unit. **OR** by voltage ratio method: $V_2/V_s = R_2/R_{total}$ V_2/5.0 = 1500/2500 V_2 = 3.0 V **OR** V_2/5 = 1.5/2.5 If correct answer given with no working award **3 marks**.

Question			Expected response	Max mark	Additional guidance
7.	(b)		(As the temperature increases) the resistance of thermistor decreases so **voltage across thermistor decreases.** **(1 mark)** Voltage across variable resistor increases. **(1 mark)** Transistor/MOSFET switches **on** and buzzer sounds. **(1 mark)**	**3**	Independent marks. Accept: 'transistor' in place of MOSFET. Do not accept: npn transistor or pnp transistor in place of MOSFET. Accept: MOSFET conducts. Ignore any value for switching voltage stated.
	(c)		To vary the temperature at which the buzzer sounds.	**1**	Accept change, alter and set.

9. A child's LED night light is controlled using a LM741 comparator circuit as shown below.

MARKS DO NOT WRITE IN THIS MARGIN



(a) Calculate the reference voltage V_x. **3**

Space for working and answer

(b) State the resistance of the light dependent resistor when the voltage across it is equal to the reference voltage V_x. **1**

(c) Describe how this circuit works. **3**

9.	(a)		$V_x = \frac{R_2}{R_1 + R_2} \times V_S$ **(1 mark)** $V_x = \frac{4000}{(8000 + 4000)} \times 6$ **(1 mark)** $V_x = 2V$ **(1 mark)**	**3**	Accept (4/12) × 6 1 mark for Ohm's Law anywhere. 1 mark for both substitutions. 1 mark for final answer including unit Or by voltage ratio method V_x / V_s = R_2 / R_{total} V_x / 6 = 4000 / 12000 V_x = $2V$ If correct answer with unit and no working award 3 marks
	(b)		6 $k\Omega$ **OR** 6000 Ω **OR** 6K	**1**	Must have unit or be in R notation Accept answer consistent with Vx calculated in 9(a)
	(c)		As light level reduces LDR resistance increases **(1 mark)** As LDR resistance increases the voltage across LDR increases **(1 mark)** When voltage across the LDR = V_x then LM741/comparator/IC/op amp output switches on, powering the LED **(1 mark)**	**3**	Accept V ref = V_x = trigger voltage = switch on voltage Accept marks independently Accept LED lights up Accept alternatives to = Do not accept arrow symbols as alternatives to increase and decrease

Question	Expected response	Max mark	Additional guidance

Part 2

Logic and systems

Start with

the first two questions of each topic below. Mark them in green pen with their marking instructions (tap Answers). Then carry on down the list.

			Question	Marking	
Logic					
2.1	2024 paper, Question 4	5 marks	page 79	Answers p. 81	
2.2	2026 paper, Question 4	7 marks	page 83	Answers p. 85	
2.3	2018 paper, Question 4	6 marks	page 87	Answers p. 89	
2.4	2019 paper, Question 4	5 marks	page 90	Answers p. 92	
2.5	2021 paper, Question 4	6 marks	page 93	Answers p. 95	
2.6	2025 paper, Question 4	6 marks	page 98	Answers p. 100	
2.7	Specimen paper, Question 4	6 marks	page 101	Answers p. 103	*Challenge*
Systems and block diagrams					
2.8	2024 paper, Question 10	6 marks	page 104	Answers p. 105	
2.9	2026 paper, Question 9	6 marks	page 107	Answers p. 108	
2.10	2018 paper, Question 10	6 marks	page 110	Answers p. 112	
2.11	2019 paper, Question 9	6 marks	page 113	Answers p. 114	
2.12	2021 paper, Question 10	6 marks	page 115	Answers p. 116	
2.13	Specimen paper, Question 8	6 marks	page 117	Answers p. 118	
2.14	2025 paper, Question 10	6 marks	page 119	Answers p. 120	*Challenge*
Simulation					
2.15	2026 paper, Question 6	3 marks	page 121	Answers p. 122	
2.16	2018 paper, Question 8	4 marks	page 124	Answers p. 125	
2.17	2021 paper, Question 6	4 marks	page 126	Answers p. 127	
2.18	Specimen paper, Question 6	8 marks	page 128	Answers p. 130	
2.19	2019 paper, Question 8	7 marks	page 131	Answers p. 133	
2.20	2024 paper, Question 5	5 marks	page 134	Answers p. 135	
2.21	2025 paper, Question 5	4 marks	page 136	Answers p. 137	*Challenge*

MARGIN

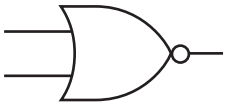
4. There are many different types of logic gate used in electronics.

(a) Complete the truth table for an AND gate. **1**

A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 16*.)

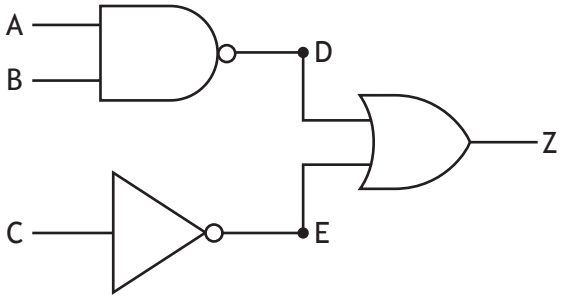
(b) Name the logic gate shown below. **1**



MARKS DO NOT WRITE IN THIS MARGIN

4. (continued)

(c) Complete the truth table for the logic circuit shown below. **3**



A	B	C	D	E	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(An additional truth table, if required, can be found on *page 16*.)

[Turn over

4. (a)

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

1

(b) NOR (gate) **1**

(c)

A	B	C	D	E	Z
0	0	0	1	1	1
0	0	1	1	0	1
0	1	0	1	1	1
0	1	1	1	0	1
1	0	0	1	1	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

3

1 mark per column

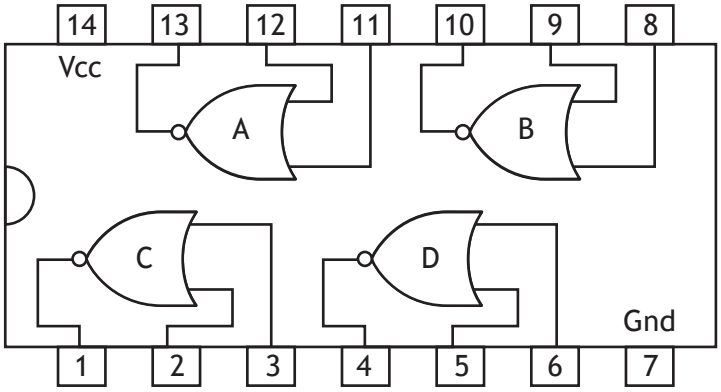
Apply follow through between columns

Question	Expected response	Max mark	Additional guidance

MARGIN

4. Logic gates are widely used in electronic circuits.

(a) A pinout for a 74 series logic chip is shown below.



(i) Using information from the data sheet, identify the **integrated circuit** (IC) shown. **1**

(ii) Identify an **output** pin on the integrated layout diagram above. **1**

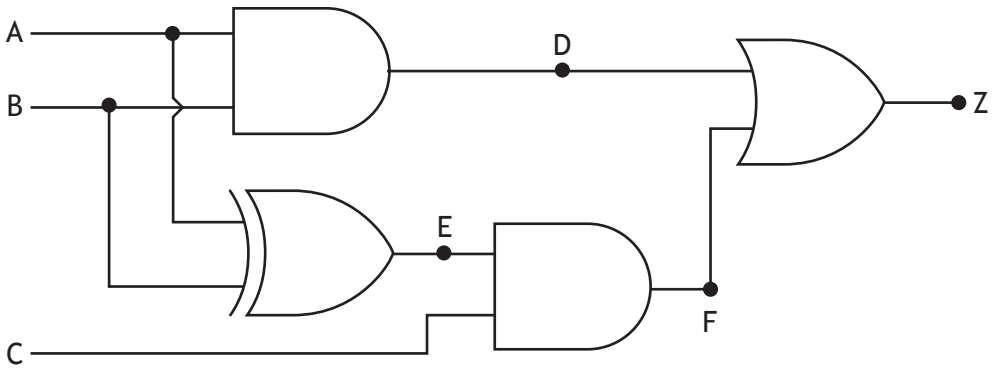
(iii) With reference to logic gate **A**, on the above pinout diagram, pins 12 and 13 have a logic state of 0.

State the logic state at pin 11. **1**

MARKS DO NOT WRITE IN THIS MARGIN

4. (continued)

(b) Complete the truth table for the logic circuit shown below. **4**



A	B	C	D	E	F	Z
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

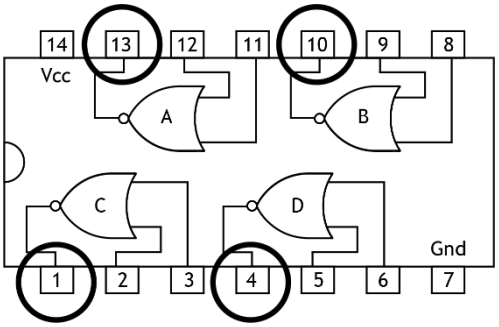
(An additional truth table, if required, can be found on *page 20*.)

[Turn over

<table>
<tr><td>4.</td><td>(a)</td><td>(i)</td><td>7402 (quad 2 input NOR gates)

OR

Quad 2 input NOR (gates)</td><td>1</td><td>Do not accept NOR on its own.</td></tr>
<tr><td></td><td></td><td>(ii)</td><td>Any one from 1, 4, 10 or 13.</td><td>1</td><td>Accept circles or arrows.

Award 1 mark if candidate circles more than one correct output pin.

Award 0 marks if candidate identifies any one of the pins: 2,3,5,6,7,8,9,11,12,14.

The candidate has provided more than the required number of responses then each incorrect response negates a correct response.</td></tr>
<tr><td></td><td></td><td>(iii)</td><td>(logic state) 1</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>F</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
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</table>
</td><td>4</td><td>1 mark for each column.

Apply follow through between columns.</td></tr>
</table>

Question	Expected response	Max mark	Additional guidance

MARGIN

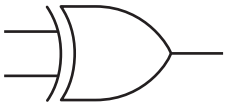
4. Logic gates are widely used in electronic circuits.

(a) Complete the truth table for an OR gate. **1**

A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 24*.)

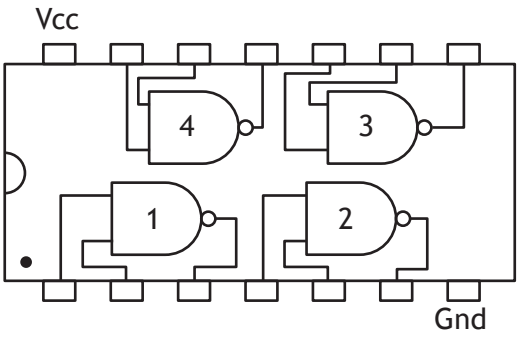
(b) Name the logic gate shown below. **1**



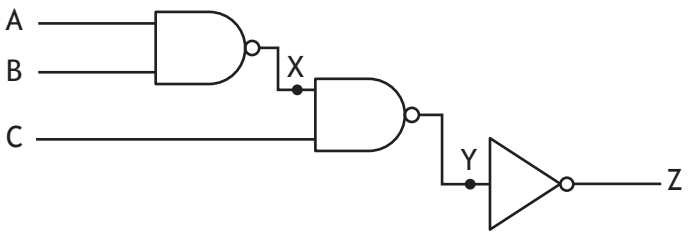
MARKS | DO NOT WRITE IN THIS MARGIN

4. (continued)

(c) Using information from the data sheet, identify the **integrated circuit** (IC) shown. **1**



(d) Complete the truth table for the logic circuit shown. **3**



A	B	C	X	Y	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(An additional truth table, if required, can be found on *page 24*.)

<table>
<tr>
<td>4.</td>
<td>(a)</td>
<td></td>
<td>
<table>
<tr><th>A</th><th>B</th><th>Output</th></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td></tr>
</table>
</td>
<td>1</td>
<td></td>
</tr>
<tr>
<td></td>
<td>(b)</td>
<td></td>
<td>Exclusive OR / XOR</td>
<td>1</td>
<td></td>
</tr>
<tr>
<td></td>
<td>(c)</td>
<td></td>
<td>7400 (quad 2 input NAND gates)

OR

Quad 2 input NAND (gates)</td>
<td>1</td>
<td>Do not accept NAND on its own.</td>
</tr>
<tr>
<td></td>
<td>(d)</td>
<td></td>
<td>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>X</th><th>Y</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td></tr>
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<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td></tr>
</table>
</td>
<td>3</td>
<td>1 mark for each column.

Apply follow through between columns.</td>
</tr>
</table>

MARGIN

4. There are many different types of logic gates used in electronic circuits.

(a) Draw the circuit symbol for a NAND gate. **1**

(b) The truth table for a logic gate is shown below.

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

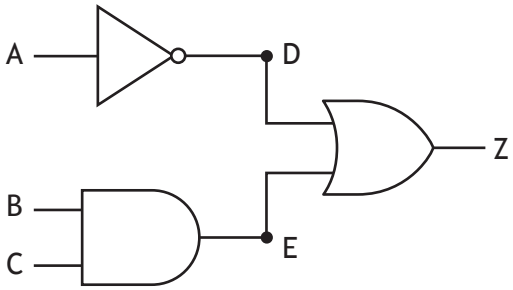
Name the logic gate that produces this truth table. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

4. (continued)

Many electronic devices use several logic gates connected to one another.

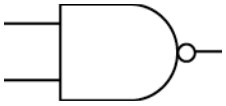
(c) Complete the truth table for the logic circuit shown below. **3**



A	B	C	D	E	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(An additional truth table, if required, can be found on *page 20*)

[Turn over

<table>
<tr><td>4.</td><td>(a)</td><td></td><td>  </td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>NOR (gate)</td><td>1</td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
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<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr>
</table>
</td><td>3</td><td>1 mark per column

Apply follow through between columns</td></tr>
</table>

MARGIN

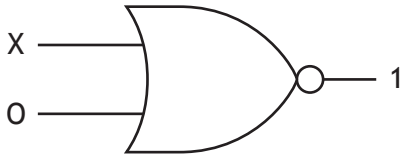
4. Logic gates are widely used in electronic circuits.

(a) Complete the truth table for a NAND gate. **1**

A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 22*)

(b) A logic gate is shown below.

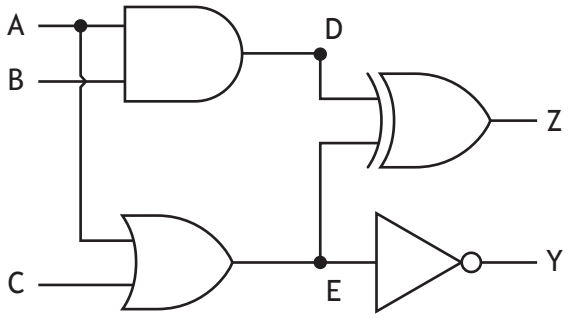


Determine the logic state at X that would produce the output shown. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

4. (continued)

(c) Many electronic devices use a combination of interconnected logic gates. Complete the truth table for the logic circuit shown below. **4**



A	B	C	D	E	Y	Z
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

(An additional truth table, if required, can be found on *page 22*)

[Turn over

4.	(a)		<table><tr><th>A</th><th>B</th><th>Output</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	**1**	

<table>
<tr><th colspan="3">Question</th><th>Expected response</th><th>Max mark</th><th>Additional guidance</th></tr>
<tr><td></td><td>(b)</td><td></td><td>0</td><td>1</td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td>
<table>
<tr><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th><th>Y</th><th>Z</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>4</td><td>1 mark per column

Apply follow through between columns</td></tr>
</table>

Question	Expected response	Max mark	Additional guidance

MARGIN

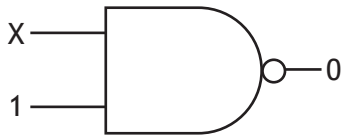
4. There are many different types of logic gate used in electronics.

(a) Complete the truth table for an **NOR** gate. **1**

A	B	Output
0	0	
0	1	
1	0	
1	1	

(An additional truth table, if required, can be found on *page 20*.)

(b) A logic gate is shown below, with given logic states.

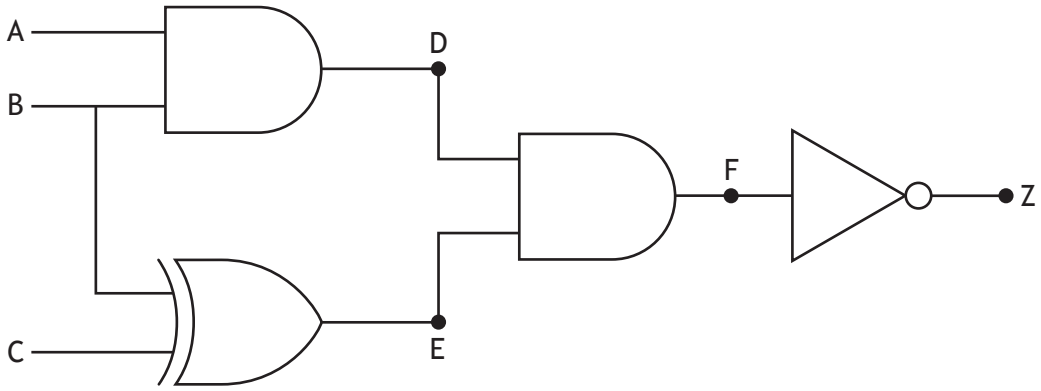


Determine the logic state of X that would produce the output shown. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

4. (continued)

(c) Complete the truth table for the logic circuit shown below. **4**



A	B	C	D	E	F	Z
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

(An additional truth table, if required, can be found on *page 20*.)

[Turn over

4. (a)

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

1

(b) (Logic State) 1 **(1 mark)**

1 Accept 1, High, ON

(c)

A	B	C	D	E	F	Z
0	0	0	0	0	0	1
0	0	1	0	1	0	1
0	1	0	0	1	0	1
0	1	1	0	0	0	1
1	0	0	0	0	0	1
1	0	1	0	1	0	1
1	1	0	1	1	1	0
1	1	1	1	0	0	1

4 1 mark per column.

Apply follow through between columns.

MARGIN

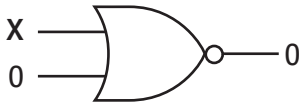
4. Logic gates are widely used in electronic circuits.

(a) The truth table for a logic gate is shown.

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

Name the gate that produces this truth table. **1**

(b) A logic gate is shown below.



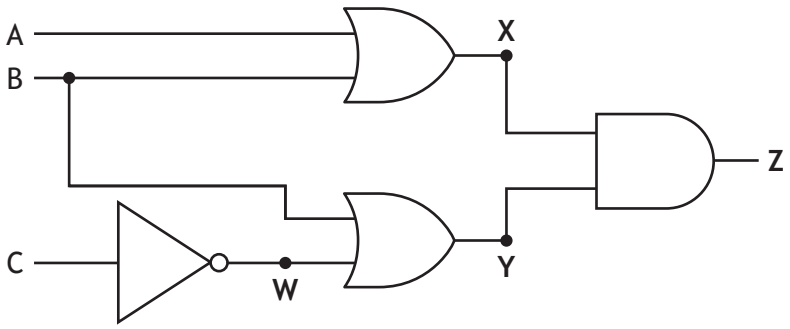
Determine the logic state at **X** that would produce the output shown. **1**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

4. (continued)

(c)



Complete the truth table for the circuit. **4**

A	B	C	W	X	Y	Z
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0				
1	0	1				
1	1	0				
1	1	1				

(An additional truth table, if required, can be found on *page 18*.)

<table>
<tr><td>4.</td><td>(a)</td><td></td><td>XOR or exclusive OR</td><td>1</td><td></td></tr>
<tr><td></td><td>(b)</td><td></td><td>1</td><td>1</td><td></td></tr>
<tr><td></td><td>(c)</td><td></td><td>
<table>
<tr><td>W</td><td>X</td><td>Y</td><td>Z</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td></tr>
</table>

1 mark for each correct column.
Apply follow through between columns.
</td><td>4</td><td></td></tr>
</table>

MARGIN

10. A security system for an art gallery is required to protect an expensive vase.

The system must include a sensor that can detect if the vase has been lifted.

When the vase is lifted, this will turn on an alarm and a motor which closes the shutter on the gallery door.

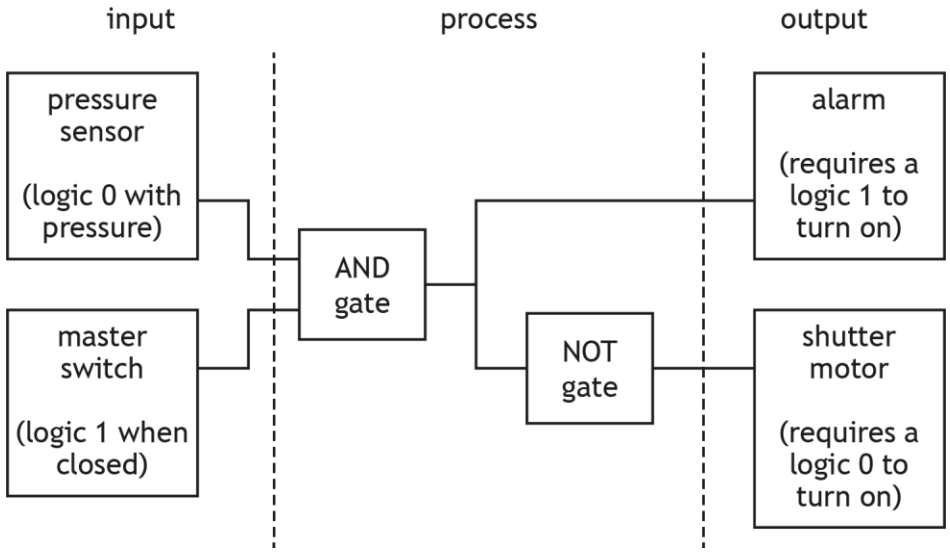
There is also a master switch to turn the whole system on and off.

Selecting from the elements given below, draw a **block diagram** of an electronic solution for this system.

On your diagram, clearly indicate the input, process and output sections of your solution. **6**

pressure sensor (logic 0 with pressure)	NOT gate	temperature sensor (logic 1 when hot)	XOR gate	master switch (logic 1 when closed)
NOR gate	alarm (requires a logic 1 to turn on)	AND gate	LED (requires a logic 1 to turn on)	shutter motor (requires a logic 0 to turn on)

10.			**1 mark** for selecting both input devices **1 mark** for selecting correct gates **1 mark** for selecting both outputs **1 mark** for correctly linking the inputs and gates **1 mark** for linking **both** AND gate and NOT gate to outputs **1 mark** for correctly identifying the three sections of the system	**6**	All selected elements must be shown as part of a system/block diagram (arrows are NOT required on the connections) Accept switch, alarm, motor and pressure sensor as block descriptors Accept solid sepreation lines Do not accept a list/table of elements



Question	Expected response	Max mark	Additional guidance

MARGIN

9. An engineer designs a system to control the temperature of the water in a tropical fish tank.

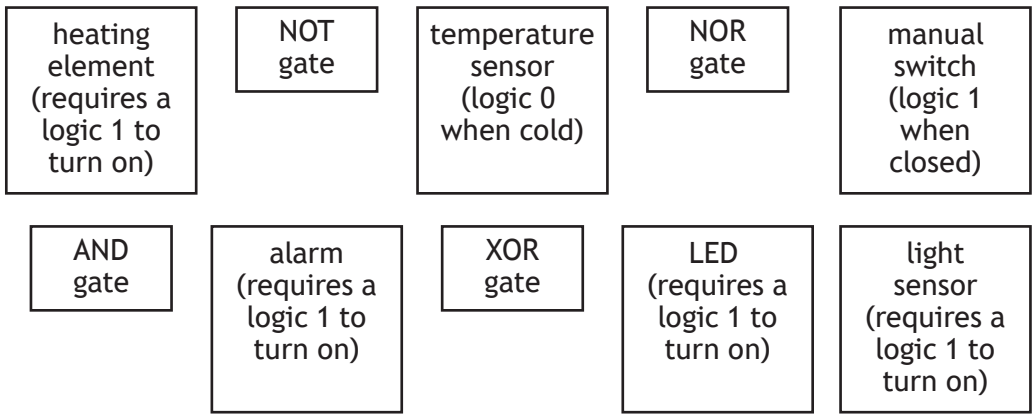
The system requires an electric heater to turn on when the temperature of the water becomes too cold.

An LED should switch on when the heater is on.

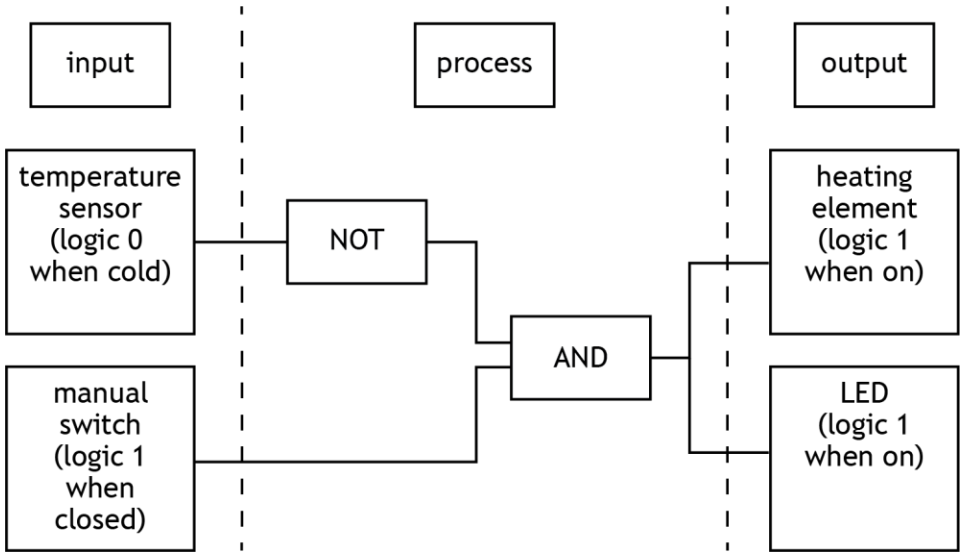
There is also a manual switch to turn the whole system on and off.

Selecting from the elements given, draw a block diagram of an electronic solution for this system.

On your diagram, clearly indicate the input, process and output sections of your solution. **6**



9.			**1 mark** for correctly identifying the three sections of the system with dashed lines (- - - -). **1 mark** for selecting both input devices. **1 mark** for selecting both outputs. **1 mark** for selecting both gates correctly. **1 mark** for linking both inputs to gates correctly. **1 mark** for linking the outputs to gates correctly.	**6**	All selected elements must be shown as part of a system/block diagram. (arrows are NOT required on the connections). Logic gate symbols not accepted. Accept switch and temp sensor as block descriptors.



Question	Expected response	Max mark	Additional guidance

MARGIN

10. An engineer designs a system to control the temperature within a greenhouse.

The system includes a sensor which measures the temperature within the greenhouse.

If the temperature exceeds 24 °C this turns on a motor to open a window.

An LED indicator lights when the window is open.

There is also a manual switch to turn the whole system on and off.

Selecting from the elements given, draw a block diagram of an electronic solution for this system.

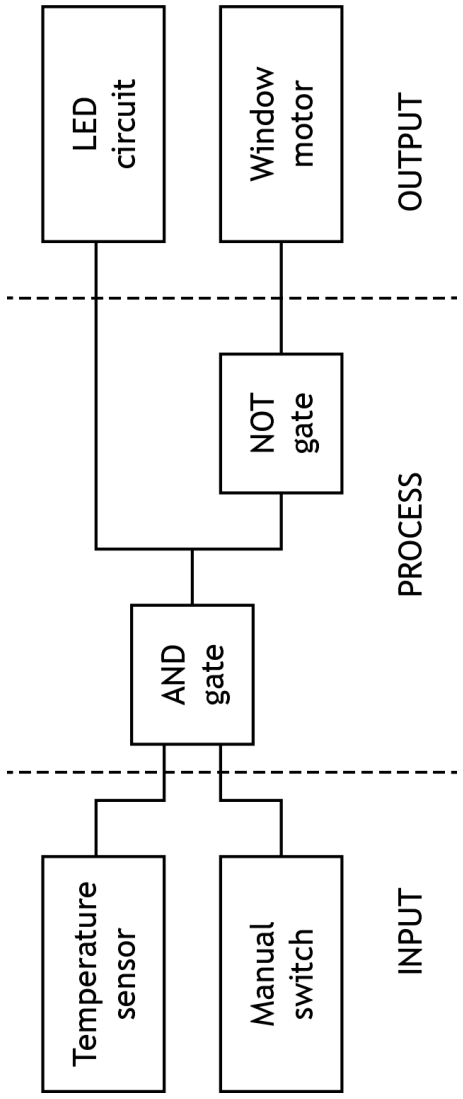
On your diagram, clearly indicate the input, process and output sections of your solution. **6**

AND gate	NOR gate	NOT gate	Window motor (requires a logic 0 to turn the motor on)	Manual switch circuit (logic 1 when closed)
OR gate	XOR gate	LED circuit (requires a logic 1 to turn the LED on)	Temperature sensor (logic 1 when temperature is greater than 24 °C)	
Light sensor (logic 1 when light)				

MARKS | DO NOT WRITE IN THIS MARGIN

10. (continued)

[Turn over for next question

10.			**1 mark** for selecting both input devices. **1 mark** for selecting correct gates. **1 mark** for selecting both output devices. **1 mark** for correctly linking the inputs to gates. **1 mark** for correctly linking the outputs to the gates. **1 mark** for correctly identifying the three sections of the system diagram.  	**6**	All selected elements must be shown as part of a system/block diagram. (arrows are NOT required on the connections).

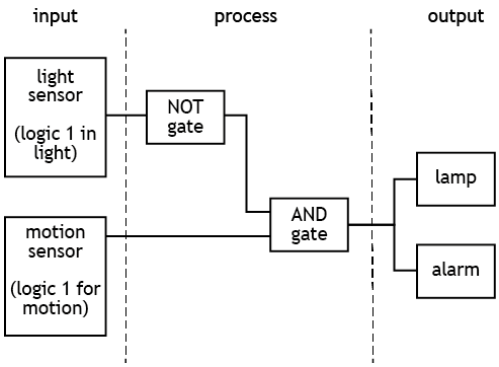
MARGIN

9. A security system for a house requires a lamp and an alarm to turn on when movement is detected outside the house. The security system only needs to operate at night.

Selecting from the elements given below, draw a block diagram of an electronic solution for the security system.

On your diagram, clearly indicate the input, process and output sections of your solution. **6**

light sensor (logic 1 in light)	alarm	temperature sensor (logic 1 for hot)	lamp	motion sensor (logic 1 for motion)
	video camera		NOT gate	
	AND gate	XOR gate	dvd recorder	

9.			**1 mark** for selecting both inputs **1 mark** for selecting correct gates **1 mark** for selecting both outputs **1 mark** for correctly linking inputs to gates **1 mark** for linking AND gate to outputs **1 mark** correctly identifying the three sections of the system diagram	**6**	 All selected elements must be shown as part of a system/block diagram. (arrows are NOT required on the connections).

MARGIN

10. A zoo uses an incubator to keep eggs warm.

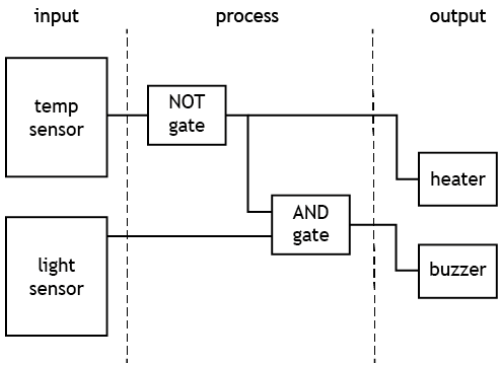
A safety system for the incubator requires an electric heater to turn on when the temperature in the incubator becomes too cold.

When the heater is turned on during daylight hours an alarm should sound.

Selecting from the elements given below, draw a **block diagram** of an electronic solution for this system.

On your diagram, clearly indicate the input, process and output sections of your solution. **6**

light sensor (logic 1 for light)	motion sensor (logic 1 for motion)	temperature sensor (logic 1 for hot)	buzzer	XOR gate	AND gate
			heating element	NOT gate	lamp

10.			**1 mark** for selecting both input devices **1 mark** for selecting correct gates **1 mark** for selecting both output devices **1 mark** for correctly linking the inputs and gates **1 mark** for correctly linking outputs to the gates **1 mark** for correctly identifying the three sections of the system	**6**	 All selected elements must be shown as part of a system/block diagram. (arrows are NOT required on the connections).

MARGIN

8. An engineer designs a system to control the moisture of the soil in a greenhouse.

The system must include a sensor which will measure the moisture content of the soil and turn on a water sprinkler system if the soil is too dry. An LED indicator should be included to show when the system is operating. There should also be a manual shut down switch to turn the system off when it is not required.

Selecting from the elements given below, draw a block diagram of an electronic solution that meets the engineer's criteria.

On your diagram, clearly indicate the input, process and output sections of your solution. **6**

light sensor (logic 1 in light)	moisture sensor (logic 0 when dry)	buzzer	XOR gate	NOT gate	LED
		AND gate	switch	water sprinkler	

[Turn over

8.			1 mark for selecting both input devices 1 mark for selecting correct gates 1 mark for selecting both output devices 1 mark for correctly linking the inputs and gates 1 mark for correctly linking the outputs to the gates (arrows are NOT required on the connections) 1 mark for correctly identifying the three sections of the system	**6**	

MARGIN

10. A gas pipeline mechanism is required to detect gas leaks.

When a leak is detected, the gas should be switched off by a motor.

The system must include a gas sensor to monitor for any leaks.

The gas can be switched off when either an engineer presses an emergency switch or when the gas sensor detects gas.

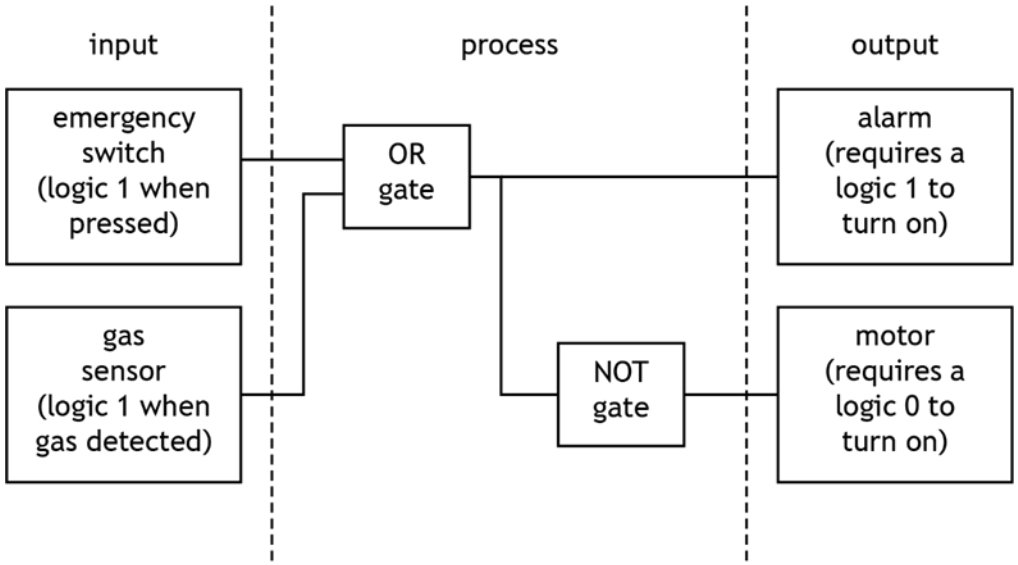
If the gas is switched off, an audible alert should also be sounded.

Selecting from the elements given below, draw a block diagram of an electronic solution for this system.

On your diagram, clearly indicate the input, process and output sections of your solution. **6**

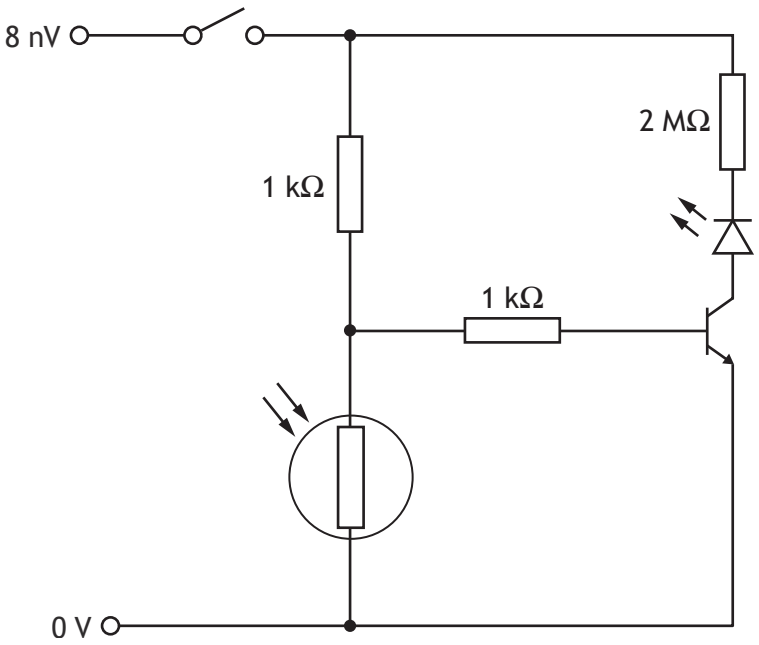
gas sensor (logic 1 when gas detected)	NOT gate	temperature sensor (logic 1 when cold)	NAND gate	emergency switch (logic 1 when pressed)
AND gate	alarm (requires a logic 1 to turn on)	OR gate	LED (requires a logic 1 to turn on)	motor (requires a logic 0 to turn on)

10.			**1 mark** for correctly identifying the three separate sections of the system. **1 mark** for selecting both input devices. **1 mark** for selecting correct gates. **1 mark** for selecting both outputs. **1 mark** for correctly linking the inputs to the gates. **1 mark** for correctly linking the outputs to the gates.	**6**	All selected elements must be shown as part of a system/block diagram. (Arrows are NOT required on the connections). Accept switch, alarm, motor and gas sensor as block descriptors. Max 5 marks if not block diagram. Do not accept logic gate symbols.



MARGIN

6. A student simulated the circuit shown.



Identify three errors in this simulation. **3**

Error 1:

Error 2:

Error 3:

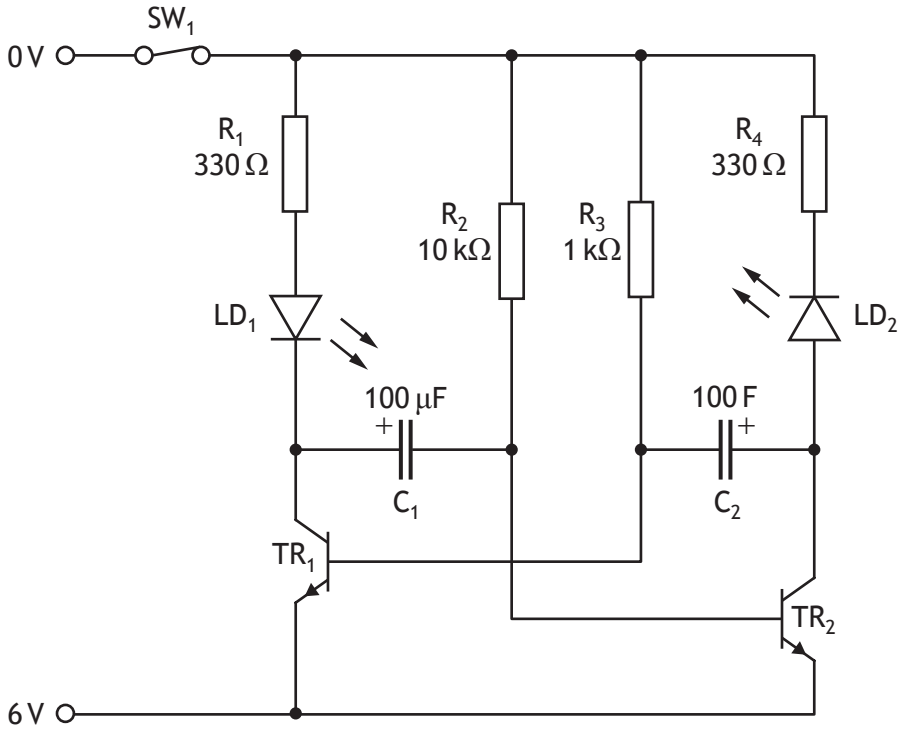
6.			• voltage supply value <u>too small</u> • <u>protective</u> resistor for LED too large a value • LED incorrect polarity/orientation	**3**	Do not accept too large a resistor.

Question	Expected response	Max mark	Additional guidance

MARGIN

8. When switched on, the circuit shown should have the LEDs alternately **flashing at the same rate.** However the student has made four errors.

Identify the **four** errors. **4**



Error 1:

Error 2:

Error 3:

Error 4:

[Turn over

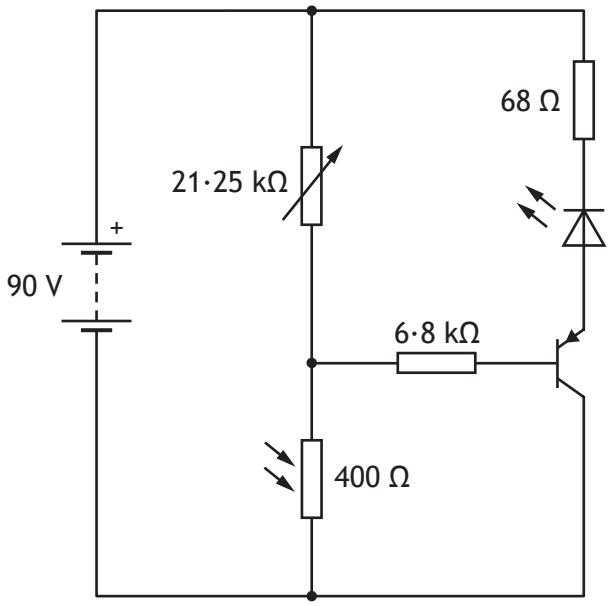
8.			Power supply reversed/ wrong way round. **(1 mark)** LD_2 wrong polarity/wrong way round. **(1 mark)** C_2 value incorrect/too big/ different. **(1 mark)** R_2 / R_3 value incorrect/ different. **(1 mark)**	**4**	Errors can be listed in any order. Battery not accepted

MARGIN

6. A student produces a simulation of a circuit that will turn on a LED when it gets dark.

However, the simulation does not work as specified.

Identify four errors in the simulation below. **4**



Error 1:

Error 2:

Error 3:

Error 4:

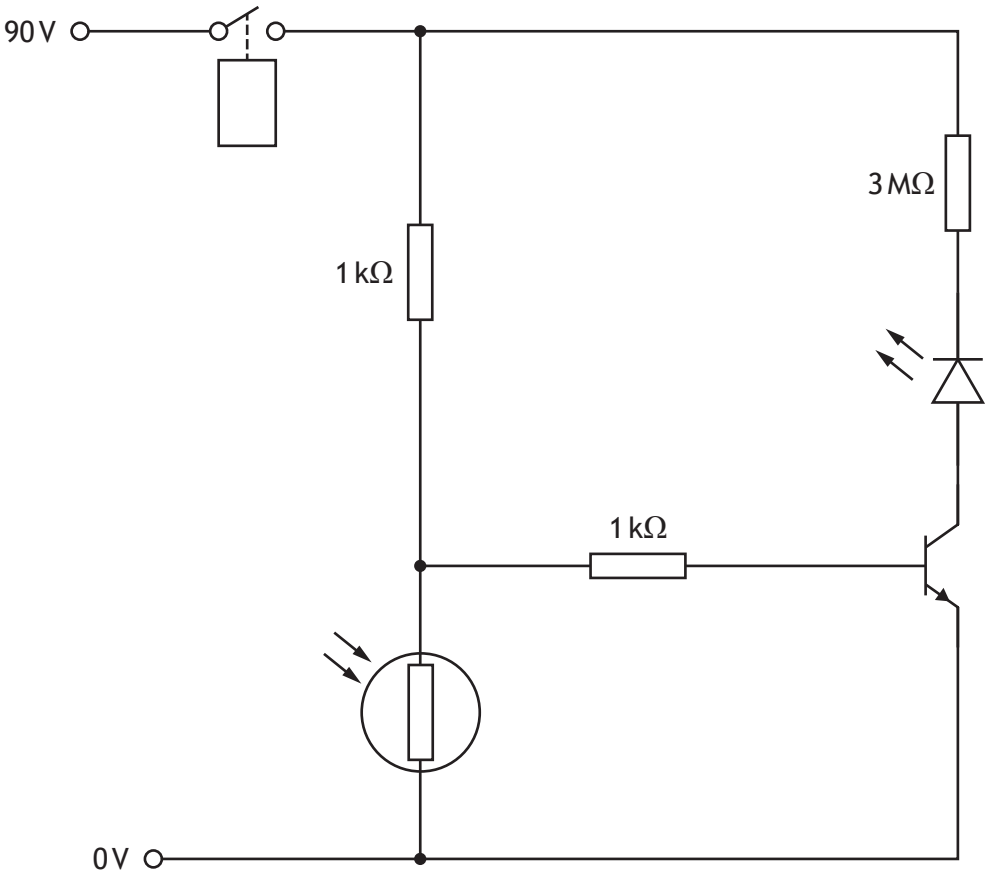
6.			Supply voltage too high **(1 mark)** LED wrong way round **(1 mark)** Protective resistor for LED has too low a resistance **(1 mark)** Transistor should be a npn/mosfet **(1 mark)** **OR** LDR and variable resistor should be changed over position in circuit **(1 mark)**	**4**	Errors can be listed in any order

WRITE IN THIS MARGIN

6. Simulation is used extensively in circuit design.

(a) Give two reasons for simulating a circuit before it is constructed. **2**

(b) A student simulated the circuit shown.



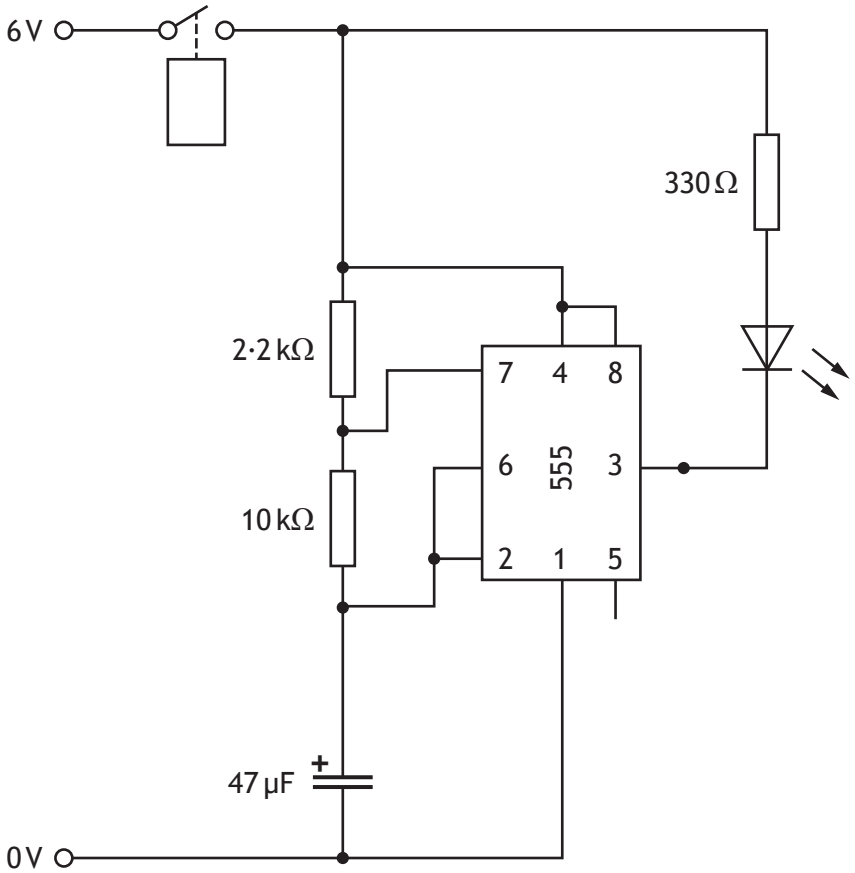
Identify the **three** errors in the circuit. **3**

[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

6. (continued)

(c) The student also simulated the 555 timer circuit shown below.



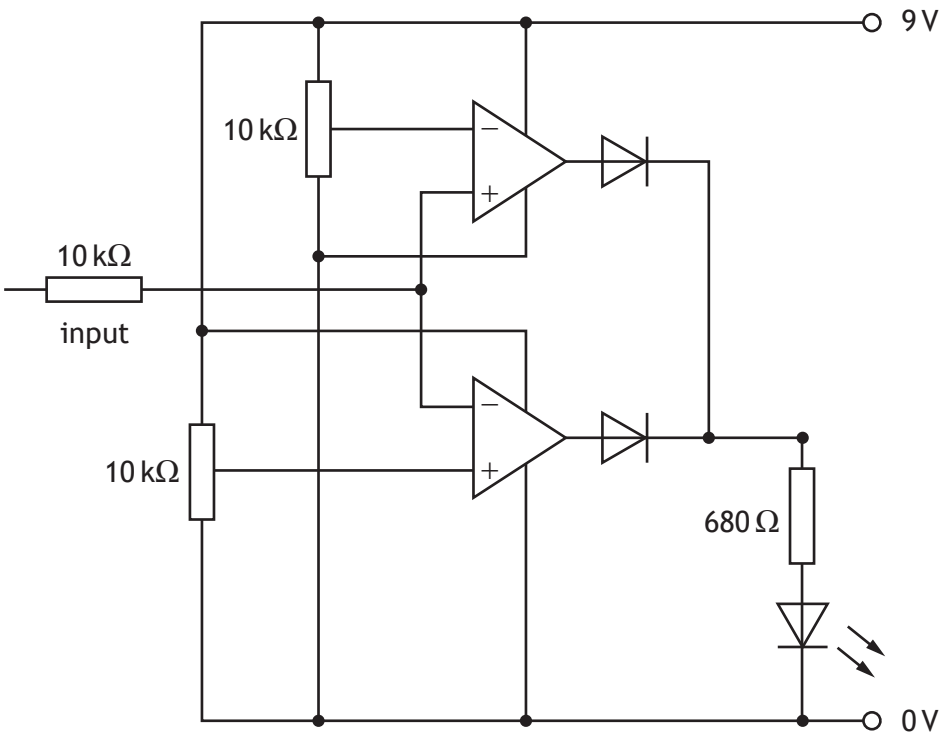
Complete the following table by giving **three** pre-power up checks for this circuit. **3**

Pre-power up checklist
Check IC orientation

6.	(a)		to establish if the circuit works to improve the circuit design easily edited to try different combinations allows testing of sub-systems Any other suitable response. Any two for 1 mark each.	**2**	Not: allows you to cost the circuit.
	(b)		incorrect/too large supply voltage 1 protective resistance for LED is too large 1 LED connected wrong way round 1	**3**	
	(c)		orientation of capacitor orientation of LED supply voltage resistance values Any three for 1 mark each.	**3**	

THIS MARGIN

8. A student simulated the circuit shown below.



(a) Complete the following table by giving **three** pre-power up checks for this circuit. **3**

Pre-power up checklist
Check both IC's orientation

[Turn over

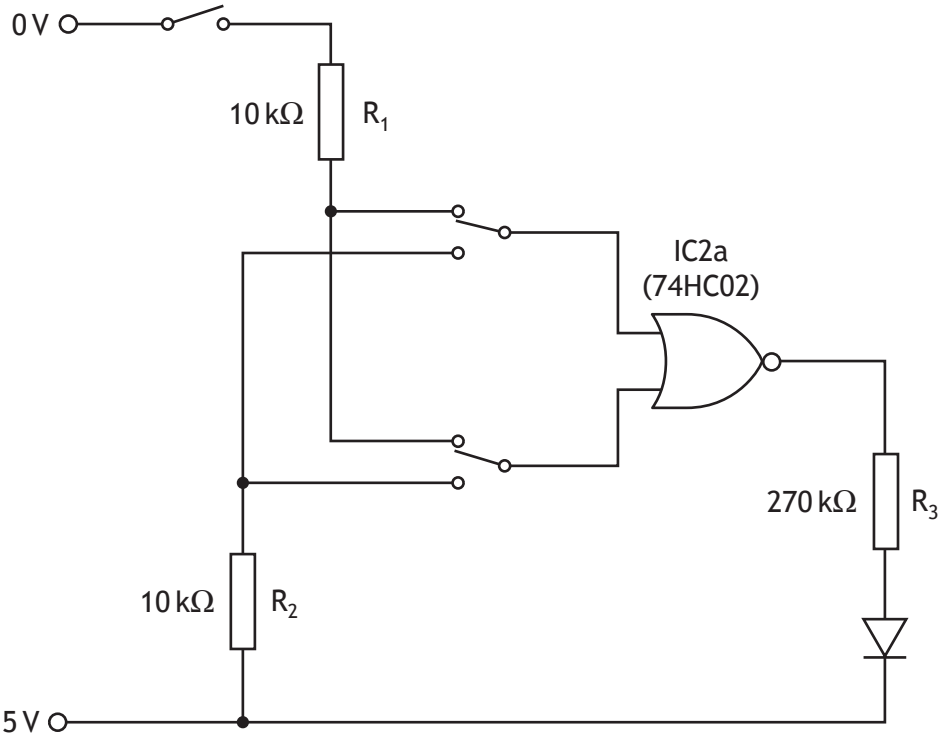
MARKS | DO NOT WRITE IN THIS MARGIN

8. (continued)

(b) A student also produced a simulation of a circuit to logic test a single AND gate using a LED as the output device.

However the simulation does not work as specified.

Identify **the four** errors in the simulation below. **4**



Error 1:

Error 2:

Error 3:

Error 4:

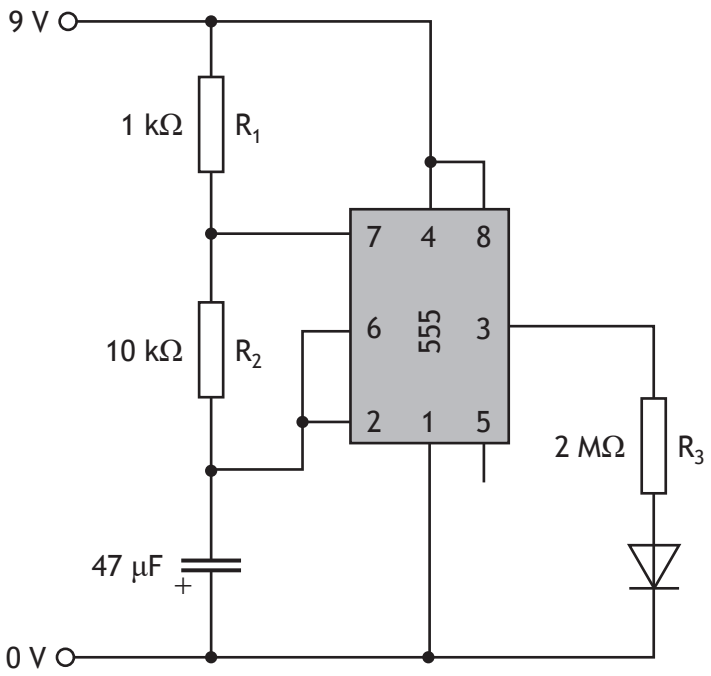
8.	(a)		Any three from • check diode orientation • check LED orientation • correct value of resistors • both ICs are the same type • both potentiometers are of the same type (linear or log) • both potentiometers have same maximum resistance • check power supply connected correctly/right way round	**3**	Do not accept variable resistors.
	(b)		Voltage supply rails are reversed/ wrong way around **(1 mark)** Incorrect logic gate used **(1 mark)** R_3 resistance too large **(1 mark)** Using diode not LED **(1 mark)**	**4**	Accept using a NOR gate Accept not using a LED Not resistor too big/large If power not identified then accept diode/led wrong way round

5. Simulation is used in circuit design.

MARKS | DO NOT WRITE IN THIS MARGIN

(a) State **two** reasons for simulating a circuit before it is constructed. **2**

(b) A student simulated the flashing LED circuit shown below.



Identify **three** errors in the student's circuit. **3**

Error 1:

Error 2:

Error 3:

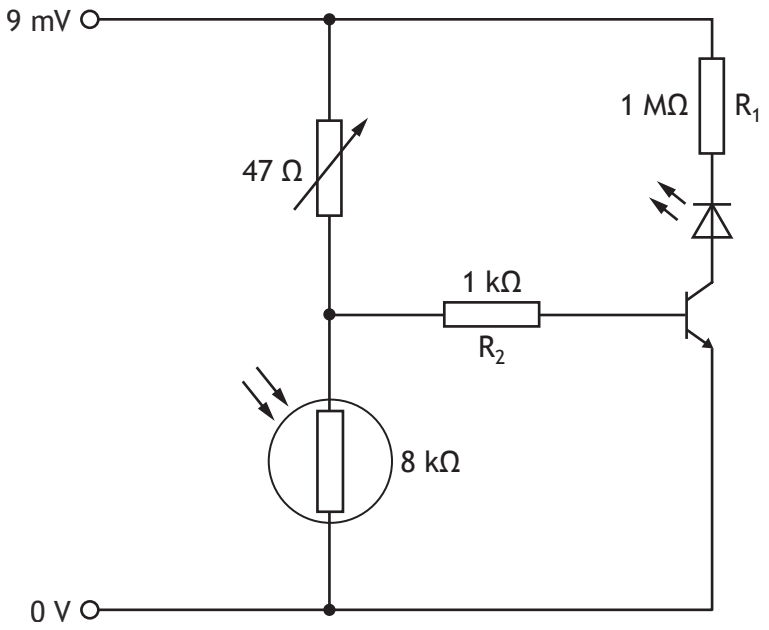
5.	**(a)**		Any two from • To establish if the circuit works • To improve the circuit design • Easily edited to try different combinations • Allows testing of sub-systems	**2**	Any other suitable response Do not accept: allows you to cost the circuit
	(b)		Orientation of (electrolytic) capacitor or (electrolytic) capacitor wrong way round **(1 mark)** Diode used, not LED **(1 mark)** (Protective resistor) R3 (for LED) has too high a resistance/value **(1 mark)**	**3**	Errors can be listed in any order Accept not using a LED rather than a diode Do not accept resistor too big/large Do not accept pin 5 is not connected/floating

MARGIN

5. A student produces a simulation of a circuit that will turn on an LED when it gets too cold in a greenhouse.

However, the simulation does not work as specified.

Identify **four** errors in the student's simulation shown below. **4**



Error 1:

Error 2:

Error 3:

Error 4:

5.			Any four from • voltage supply value too small. • LED orientation/polarity wrong way round. • protective resistor (R_1) for LED has too high a resistance value. • incorrect component symbol used to measure temperature (Thermistor not LDR).	**4**	Errors can be listed in any order. Do not accept resistor too big/large.

Part 3

Construction and testing

Start with

the first two questions of each topic below. Mark them in green pen with their marking instructions (tap Answers). Then carry on down the list.

Construction

			Question	Marking	
3.1	2026 paper, Question 8	5 marks	page 139	Answers p. 141	
3.2	2024 paper, Question 6	3 marks	page 143	Answers p. 144	
3.3	2018 paper, Question 9	5 marks	page 146	Answers p. 148	
3.4	2021 paper, Question 8	5 marks	page 149	Answers p. 151	
3.5	Specimen paper, Question 9	7 marks	page 153	Answers p. 160	
3.6	2026 paper, Question 5	6 marks	page 162	Answers p. 164	
3.7	2024 paper, Question 11	6 marks	page 165	Answers p. 167	
3.8	2018 paper, Question 2	2 marks	page 168	Answers p. 169	
3.9	2021 paper, Question 2	2 marks	page 170	Answers p. 171	
3.10	Specimen paper, Question 2	5 marks	page 173	Answers p. 175	
3.11	2019 paper, Question 2	4 marks	page 176	Answers p. 178	
3.12	2018 paper, Question 11	6 marks	page 179	Answers p. 184	
3.13	2019 paper, Question 10	6 marks	page 185	Answers p. 190	
3.14	2021 paper, Question 11	6 marks	page 191	Answers p. 196	
3.15	2026 paper, Question 10	5 marks	page 197	Answers p. 199	
3.16	2025 paper, Question 11	6 marks	page 200	Answers p. 202	
3.17	2025 paper, Question 6	2 marks	page 203	Answers p. 204	*Challenge*

Testing and fault-finding

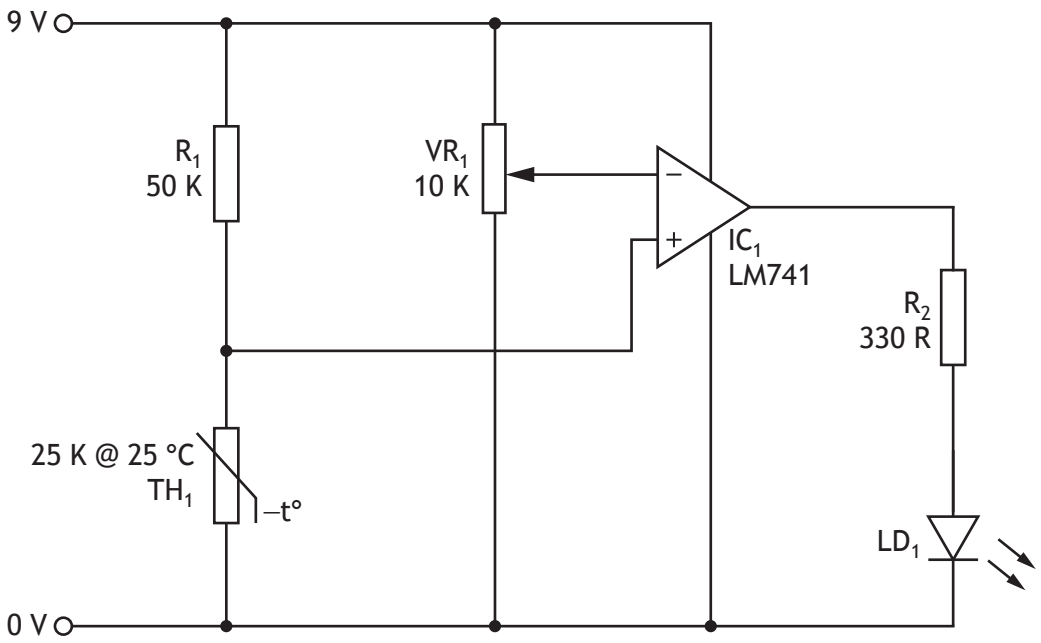
3.18	2024 paper, Question 8	5 marks	page 205	Answers p. 206	
3.19	Specimen paper, Question 5	5 marks	page 208	Answers p. 210	
3.20	2019 paper, Question 6	4 marks	page 211	Answers p. 213	
3.21	2019 paper, Question 5	5 marks	page 214	Answers p. 216	
3.22	2021 paper, Question 5	3 marks	page 217	Answers p. 218	
3.23	2018 paper, Question 6	4 marks	page 219	Answers p. 221	
3.24	2021 paper, Question 7	2 marks	page 222	Answers p. 223	
3.25	2018 paper, Question 5	3 marks	page 224	Answers p. 225	
3.26	2024 paper, Question 7	4 marks	page 226	Answers p. 227	
3.27	2025 paper, Question 8	4 marks	page 228	Answers p. 229	
3.28	2025 paper, Question 7	5 marks	page 230	Answers p. 232	*Challenge*

Safety

3.29	2026 paper, Question 2	2 marks	page 233	Answers p. 234	
3.30	2024 paper, Question 2	2 marks	page 235	Answers p. 236	

MARGIN

8. A student builds the circuit shown.



Using the information from the two suppliers' catalogues shown on the opposite page, complete the costings sheet below to produce the **lowest** cost for the circuit. **5**

Supplier	Component	Product code	Cost (p)
JIMSON	LM741CN	TC 133	56
SWIFT	8 way DIL	SK-0080	10
	LED 5 mm std		
	330 R		
	50 K		
	10 K potentiometer		
	thermistor		

(An additional costings sheet, if required, can be found on *page 20*.)

8. (continued)

DO NOT WRITE IN THIS MARGIN

Supplier — SWIFT				
	Component	**Description**	**Product code**	**Cost**
Integrated circuits	LM741CN	op-amp	IC-0258	75 p
	UA741CP	op-amp	IC-0267	41 p
	OP070CP	op-amp	IC-0234	110 p
	8 way DIL	IC socket	SK-0080	10 p
Semi-conductors	LED	5 mm std red	SC-0155	4 p
	LED	10 mm std red	SC-0177	10 p
Thermistor	thermistor	5 K @ 25 °C	BB-0545	65 p
	thermistor	10 K @ 25 °C	BB-0556	50 p
	thermistor	25 K @ 25 °C	BB-0568	28 p
Resistors	50 K	0.25 W carbon film 5%	FR-0855	2.8 p
	4K7	0.25 W carbon film 5%	FR-0856	2 p
	330 R	0.25 W carbon film 5%	FR-0857	1.5 p
Potentiometer	25 K	0.5 W trimmer	PT-0765	95 p
	10 K	0.5 W trimmer	PT-0744	55 p
	1 M	0.5 W trimmer	PT-0732	35 p
Supplier — JIMSON				
	Component	**Description**	**Product code**	**Cost**
Integrated circuits	LM741CN	op-amp	TC 133	56 p
	UA741CP	op-amp	TC 134	36 p
	OP070CP	op-amp	TC 135	105 p
	8 way DIL	IC socket	SK 090	30 p
Semi-conductors	LED	5 mm std red	LD 345	12 p
	LED	10 mm std red	LD 346	25 p
Thermistor	thermistor	5 K @ 25 °C	TH 114	32 p
	thermistor	10 K @ 25 °C	TH 115	45 p
	thermistor	25 K @ 25 °C	TH 116	22 p
Resistors	50 K	0.25 W carbon film 5%	CR 654	2.5 p
	4K7	0.25 W carbon film 5%	CR 655	1.7 p
	330 R	0.25 W carbon film 5%	CR 656	1.2 p
Potentiometer	25 K	0.5 W trimmer	VT 250	92 p
	10 K	0.5 W trimmer	VT 251	65 p
	1 M	0.5 W trimmer	VT 252	40 p

8.

Supplier	Component	Product code	Cost (p)
JIMSON	LM741CN	IC-0258	56
SWIFT	8-way DIL	SK-0080	10
SWIFT	LED 5mm std red	SC-0155	4
JIMSON	330R	CR 656	1•2
JIMSON	50 K	CR 654	2•5
SWIFT	10K Potentiometer	PT-0744	55
JIMSON	Thermistor	TH 116	22

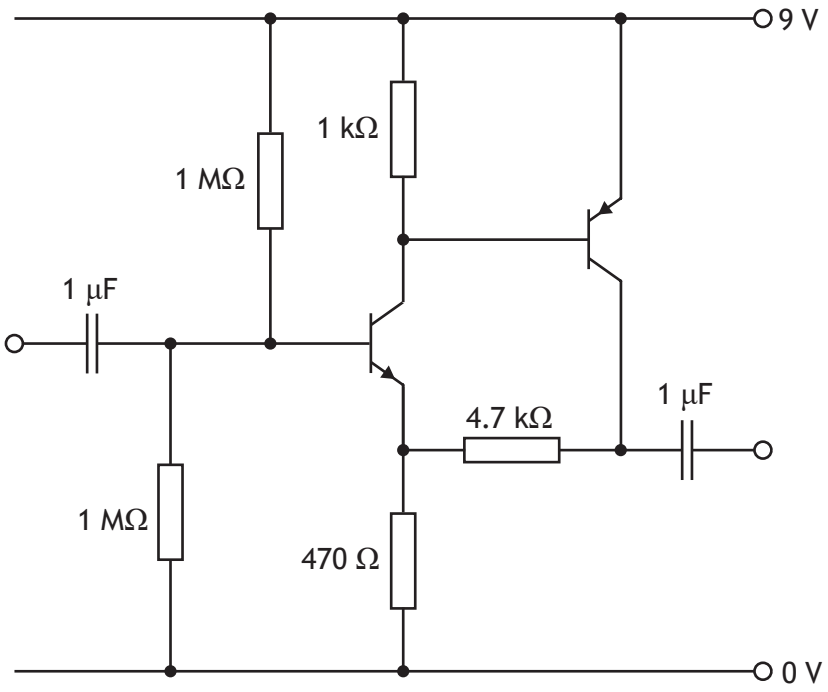
5

1 mark for each correct row.

Question	Expected response	Max mark	Additional guidance

MARGIN

6. A technician constructed a circuit from the schematic diagram shown below.



Complete the following table by giving **three** pre-power up checks for this circuit. **3**

Pre-power up checklist
Correct power supply voltage

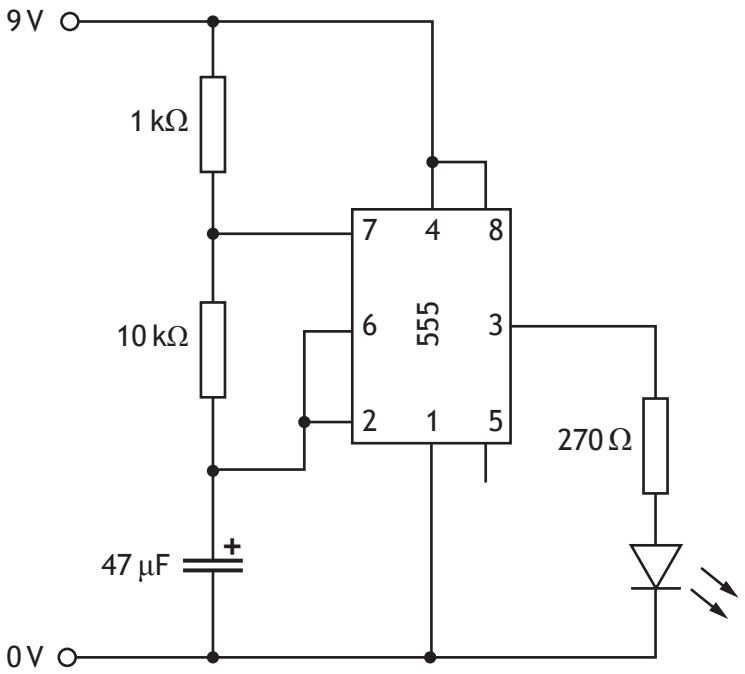
[Turn over

6.			Any three from • Check that there is no short circuit across the circuit supply terminals • Power supply correct way round (0V lower rail) • Transistors in correct position or location • Ttransistor orientations correct • Resistor values • Capacitor values correct • Resistors in correct location • Capacitors in correct location • Check wiring is secure • Check for solder bridges	**3**	Any other suitable response

Question	Expected response	Max mark	Additional guidance

MARGIN

9. A student builds the circuit shown.



Using the information from two suppliers' catalogues shown on the opposite page, complete the costings sheet to produce the **lowest** cost for the circuit. **5**

Supplier	*Component*	*Product code*	*Cost (p)*
JIMSON	NE555	TC124	20
SWIFT	8 way DIL socket	SK-0080	10
	LED 5 mm std		
	47 µF electrolytic capacitor		
	270R		
	1K		
	10K		

DO NOT WRITE IN THIS MARGIN

9. (continued)

Supplier	SWIFT			
	Component	Description	Product code	Cost
Integrated circuits	LM555CM	timers	IC-0283	45p
	NE555	timers	IC-0254	32p
	NE556	timers	IC-0216	25p
	8 way dil	ic socket	SK-0080	10p
Semi-conductors	LED	5 mm std red	SC-0155	4p
	LED	10 mm std red	SC-0177	10p
Electrolytic capacitors	10 µF	16 V	CP-0555	10p
	47 µF	16 V	CP-0566	18p
	47 µF	6 V	CP-0588	10p
	47 µF	5 V	CP-0599	8p
Resistors	220R	0·25 W carbon film 5%	EC-0159	0·5p
	270R	0·25 W carbon film 5%	EC-0161	0·5p
	1K	0·25 W carbon film 5%	EC-0175	0·5p
	10K	0·25 W carbon film 5%	EC-0182	1·5p
	100K	0·25 W carbon film 5%	EC-0198	3p
Supplier	**JIMSON**			
	Component	**Description**	**Product code**	**Cost**
Integrated circuits	LM555CM	timers	TC 123	90p
	NE555	timers	TC 124	20p
	NE556	timers	TC 125	80p
	8 way dil	ic socket	SK 099	50p
Semi-conductors	LED	5 mm std red	LD345	12p
	LED	10 mm std red	LD346	20p
Electrolytic capacitors	10 µF	16 V	EC 798	10p
	47 µF	16 V	EC 799	14p
	47 µF	6 V	EC 800	10p
	47 µF	5 V	EC 801	8p
Resistors	220R	0·25 W carbon film 5%	FR 921	0·25p
	270R	0·25 W carbon film 5%	FR 922	0·25p
	1K	0·25 W carbon film 5%	FR 923	0·25p
	10K	0·25 W carbon film 5%	FR 924	2p
	100K	0·25 W carbon film 5%	FR 925	5p

9.

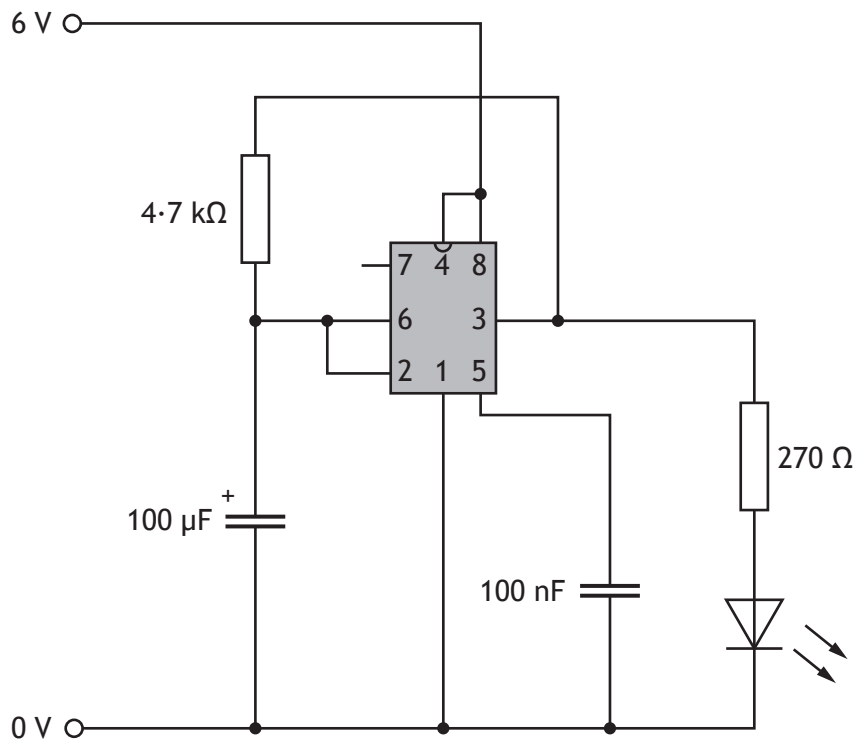
Supplier	*Component*	*Product code*	*Cost (p)*
JIMSON	NE555	TC124	20
SWIFT	8 way DIL socket	SK-0080	10
SWIFT	LED 5mm std	SC-0155	4
JIMSON	47µF electrolytic capacitor	EC799	14
JIMSON	270R	FR922	0·25
JIMSON	1K	FR923	0·25
SWIFT	10K	EC-0182	1·5

5

1 mark for each correct row.

MARGIN

8. A student builds the circuit shown below.



Using the information from the two suppliers catalogues shown on the opposite page, complete the costings sheet below to produce the **lowest** cost for the circuit. **5**

Supplier	Component	Product code	Cost (p)
JIMSON	NE555	TC124	20
SWIFT	8 way DIL socket	SK-0080	10
	LED 5 mm std		
	100 µF electrolytic capacitor		
	100 nF capacitor		
	270R		
	4K7		

(An additional costings sheet, if required, can be found on *page 23*)

8. (continued)

DO NOT WRITE IN THIS MARGIN

Supplier	SWIFT			
	Component	Description	Product code	Cost
Integrated circuits	LM555CM	timers	IC-0283	45p
	NE555	timers	IC-0254	32p
	NE556	timers	IC-0216	25p
	8 way dil	ic socket	SK-0080	10p
Semi-conductors	LED	5 mm std red	SC-0155	4p
	LED	10 mm std red	SC-0177	10p
Capacitors	10 nF	16 V	CP-2020	30p
	100 nF	16 V	CP-2030	35p
	100 nF	5 V	CP-2040	40p
Electrolytic capacitors	10 µF	16 V	CP-0555	10p
	100 µF	16 V	CP-0566	18p
	100 µF	5 V	CP-0599	8p
Resistors	270R	0·25 W carbon film 5%	EC-0161	0·5p
	4K7	0·25 W carbon film 5%	EC-0175	0·5p
	47K	0·25 W carbon film 5%	EC-0182	1·5p
Supplier	**JIMSON**			
	Component	**Description**	**Product code**	**Cost**
Integrated circuits	LM555CM	timers	TC 123	90p
	NE555	timers	TC 124	20p
	NE556	timers	TC 125	80p
	8 way dil	ic socket	SK 099	50p
Semi-conductors	LED	5 mm std red	LD345	12p
	LED	10 mm std red	LD346	20p
Capacitors	10 nF	16 V	CP 120	12p
	100 nF	16 V	CP 135	45p
	100 nF	5 V	CP 140	25p
Electrolytic capacitors	10 µF	16 V	EC 799	14p
	100 µF	16 V	EC 800	10p
	100 µF	5 V	EC 801	8p
Resistors	270R	0·25 W carbon film 5%	FR 922	0·25p
	4K7	0·25 W carbon film 5%	FR 923	0·25p
	47K	0·25 W carbon film 5%	FR 924	2p

8.

Supplier	Component	Product code	Cost (p)
JIMSON	NE555	TC124	20
SWIFT	8 way DIL socket	SK-0080	10
SWIFT	LED 5mm std	SC-0155	4
JIMSON	100µF electrolytic capacitor	EC 800	10
SWIFT	100nF capacitor	CP-2030	35
JIMSON	270R	FR 922	0·25
JIMSON	4K7	FR 923	0·25

5

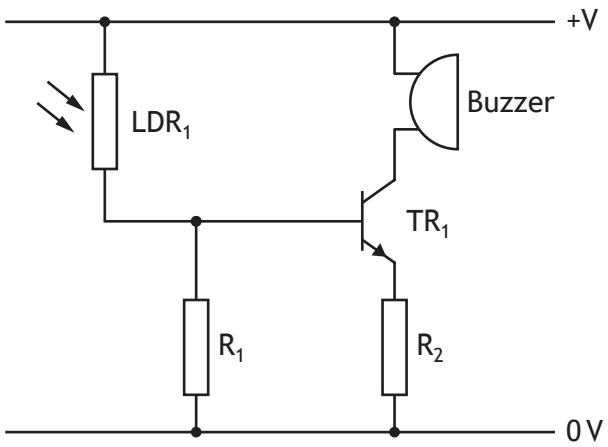
1 mark for each row

Question	Expected response	Max mark	Additional guidance

MARGIN

9. (a) State two precautions that should be taken when soldering. **2**

(b) For the circuit diagram shown, complete the layout diagram. **5**



Component data

R_1 – carbon film 10 kΩ 0·25 W

R_2 – carbon film 1 kΩ 0·25 W

LDR_1 – ORP12

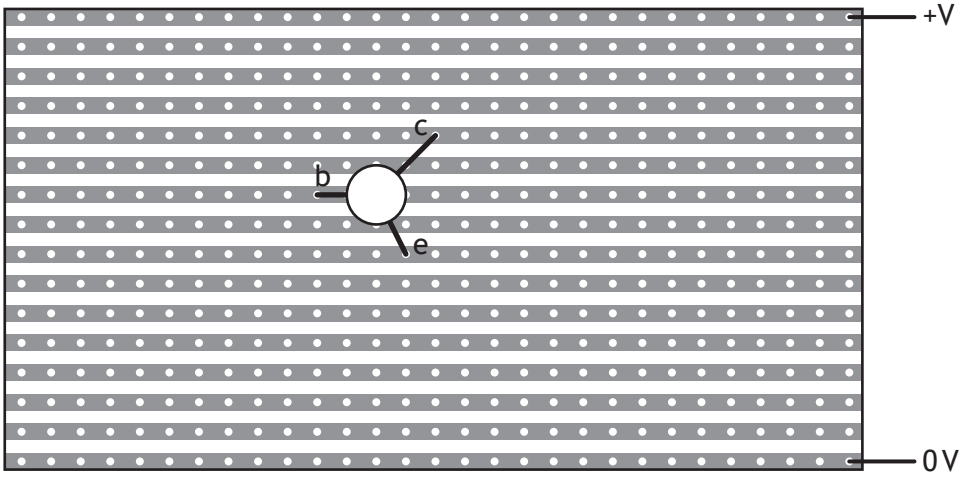
TR_1 – BC108 npn bipolar

Buzzer – 6 V dc

MARKS | DO NOT WRITE IN THIS MARGIN

9. (b) (continued)

Layout diagram



(Additional layout diagrams, if required, can be found on *page 19*.)

[END OF SPECIMEN QUESTION PAPER]

N5

National
Qualifications
SPECIMEN ONLY

S844/75/11

Electronics Data Sheet

Date — Not applicable

Duration — 1 hour

SQA ©

Relationships required for National 5 Electronics

$$V = IR$$

$$R_T = R_1 + R_2 + \ldots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$$

$$P = IV$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

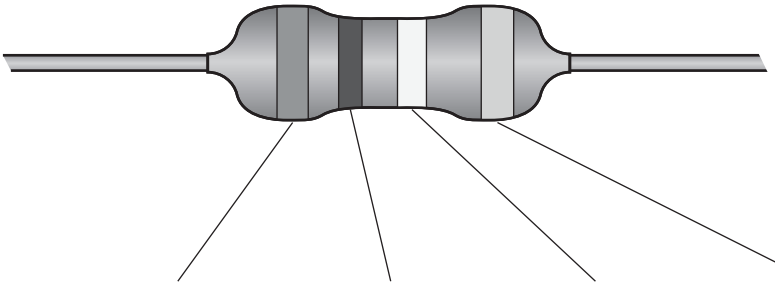
$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$$

$$f = \frac{1}{T}$$

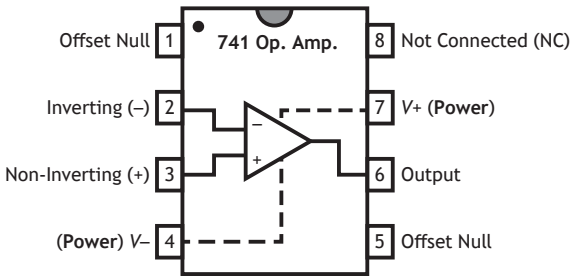
Resistor Colour Codes

4-band Resistor

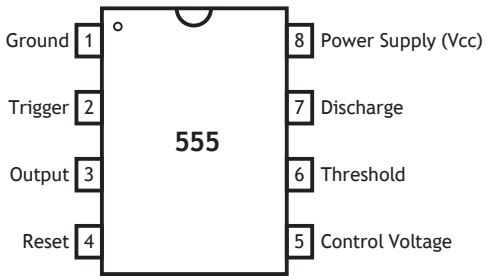


Colour	*1st band value*	*2nd band value*	*Multiplier*	*Tolerances*
Black	0	0	× 1	
Brown	1	1	× 10	±1%
Red	2	2	× 100	±2%
Orange	3	3	× 1 000	±3%
Yellow	4	4	× 10 000	±4%
Green	5	5	× 100 000	±0·5%
Blue	6	6	× 1 000 000	±0·25%
Violet	7	7	× 10 000 000	±0·10%
Grey	8	8	× 100 000 000	±0·05%
White	9	9	× 1 000 000 000	
Gold			× 0·1	±5%
Silver			× 0·01	±10%
No band				±20%

IC Pinout diagrams

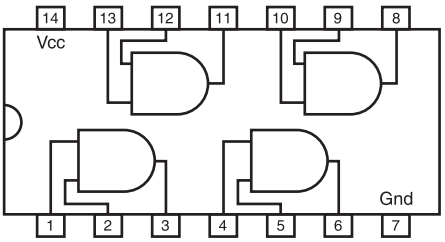


741 Op-amp

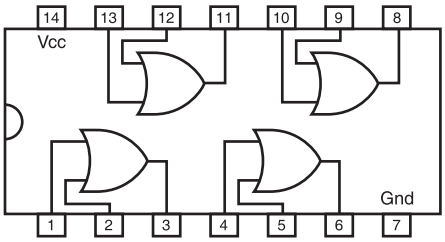


555 timer

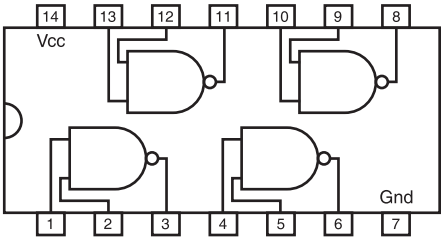
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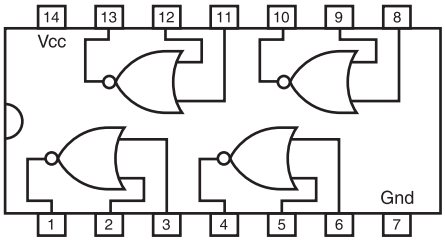
7408 Quad 2 input
AND Gates



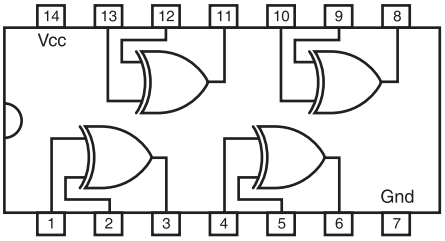
7432 Quad 2 input
OR Gates



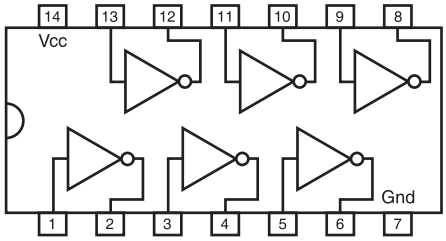
7400 Quad 2 input
NAND Gates



7402 Quad 2 input
NOR Gates



7486 Quad 2 input
XOR Gates



7404 Hex NOT Gates
(Inverters)

National Qualifications SPECIMEN ONLY

S844/75/01 **Electronics**

Marking Instructions

These marking instructions have been provided to show how SQA would mark this specimen question paper.

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9.	(a)		wear eye protection suitable ventilation avoid breathing in fumes Any other sensible suggestion. Any two for 1 mark each.	**2**	
	(b)		1 mark for each of the correctly positioned components. 1 mark for labelling all of the components.	**5**	'footprint' of components should be of an appropriate size

[END OF SPECIMEN MARKING INSTRUCTIONS]

Published: January 2026

Change since last published:

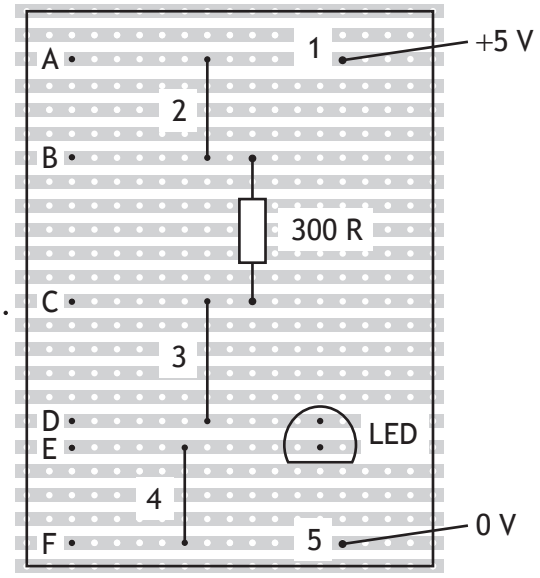
Change to question paper name and code.

MARGIN

5. (a) State an advantage of colour coding wires. **1**

(b) A student plans a stripboard layout of an LED in series with a resistor.

The layout, shown below, includes wires and test points. The wires are labelled one to five. The test points are marked A to F.



(i) The layout includes wires of different insulation colours: red, blue and black.

Identify the wire number that should have blue insulation. **1**

MARKS | DO NOT WRITE IN THIS MARGIN

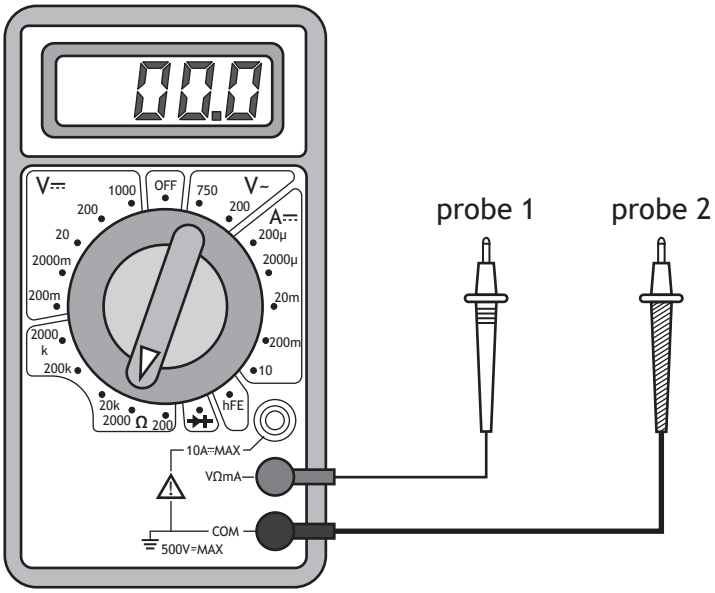
5. (b) (continued)

(ii) Explain the purpose of the resistor in the circuit. **1**

(iii) The student now constructs the circuit from part (b).

Before connecting the stripboard to the supply, the student tests for continuity using a multimeter.

The multimeter used is shown below and will sound a 'beep' to indicate continuity.



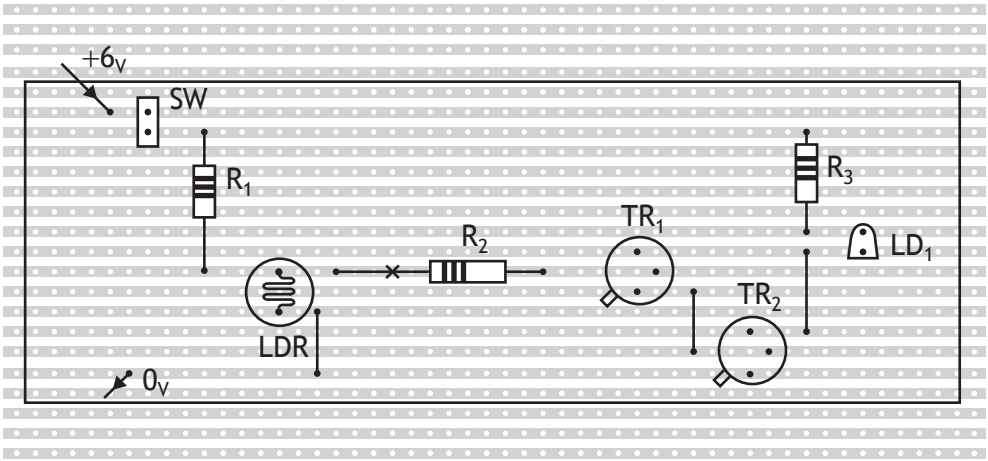
Describe how the above multimeter can be used to test for continuity between the LED and 0 V rail line on the student's stripboard. **3**

[Turn over

5.	(a)		**Accept one of the following:** • help to identify wires between subsystems • identify part(s) of a circuit that has the same potential/voltage/ signal • follow the same signal/logic state/potential between components in a circuit • test for continuity of the same wire between subsystems/ components	**1**	
	(b)	(i)	(wire) 3	**1**	
		(ii)	**Any one of the following:** • **reduce** voltage across LED • **reduce** (limit) current in LED • protect the LED • limit the power (dissipated in/ developed in/supplied to) the LED	**1**	Answers must reference LED. Do not accept: • voltage 'through' or current 'across' on their own • limit/reduce current across the LED • reduce voltage in LED • prevent LED explode
		(iii)	• (connect probe) at test point E • (connect other probe) at test point F • turn dial to 	**3**	**1 mark** for each correct test point stated. **1 mark** for the correct dial setting. Must refer to test points. Do not accept wire number(s) on their own eg 4 only or 4 and 5. If wire number used, must refer to top and bottom of wire. Must use the symbol used on the given multimeter for setting.

MARGIN

11. The stripboard plan below shows a **component (top) view** of a circuit with the following layout.



Component data

TR_1 and TR_2 — BFY51 npn transistors

LD_1 — 5 mm standard led (red)

LDR — ORP12 light dependent resistor

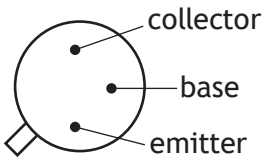
R_1 — carbon film 10 K 0.25 w

R_2 — carbon film 6K8 0.25 w

R_3 — carbon film 270 R 0.25 w

SW — SPST switch (on/off switch)

TR_1 and TR_2 pin connections



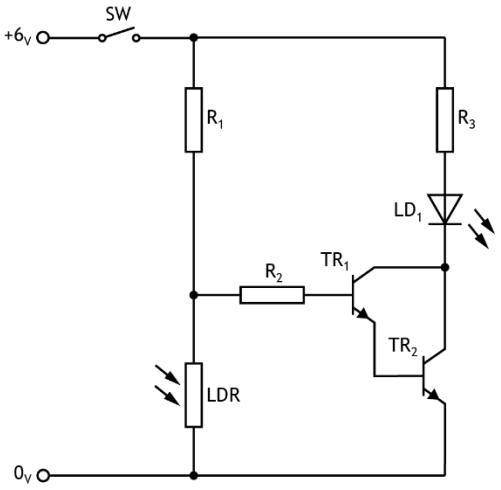
Draw a circuit diagram for this circuit.

Each component must be labelled. **6**

MARKS | DO NOT WRITE IN THIS MARGIN

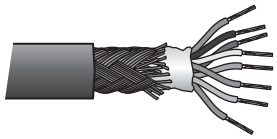
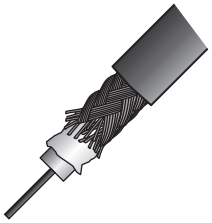
11. (continued)

[END OF QUESTION PAPER]

11.			Both power rails labelled with nodes **(1 mark)** Switch in correct position **(1 mark)** R_1, LDR voltage divider to R_2 **(1 mark)** Transistor positions/connections in relation to each other and 0v rail. **(1 mark)** Collectors to + 6V via 270R resistor and LED in series with correct orientation. **(1 mark)** Component labelling (either values or identifiers) **(1 mark)**	**6**	 Any errors in circuit symbols max 5 marks, only if candidate has achieved maximum marks for the question Accept LED$_1$ as the label Accept SPST as a descriptor for SW

[END OF MARKING INSTRUCTIONS]

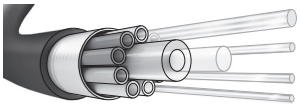
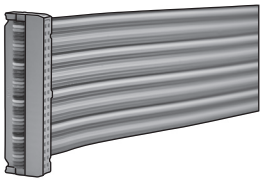
2. Complete the table by stating a typical use for each cable type shown. **2**

	Cable type	*Typical use*
	multi-strand	
	coaxial	

[Turn over

2.			Multi-strand: phone/telecom **(1 mark)** Coaxial: aerial/TV signal/radio signals/connecting CRO/audio **(1 mark)**	**2**	Accept any sensible suggestions.

2. Complete the table below by stating a typical use for each cable type shown. **2**

	Cable type	Typical use
	fibre optic	
	ribbon cable	

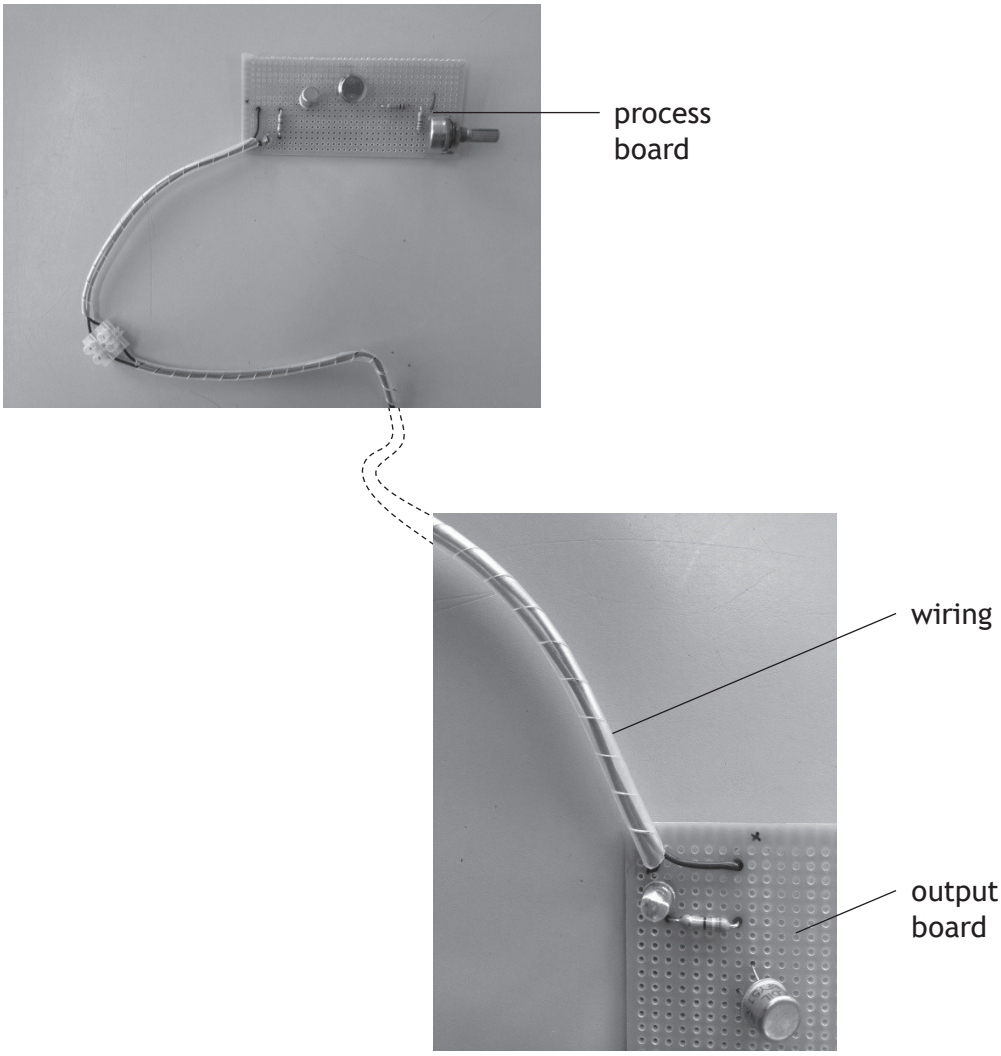
[Turn over

2.			Cable tv, telephone cabling, secure landline communications cabling, broadband **(1 mark)** Computer cabling, internal connections inside mobile phones Computer peripheral cabling **(1 mark)**	**2**	Any two suitable answers.

Question	Expected response	Max mark	Additional guidance

MARGIN

2. The photographs show the wiring connections between a process board and an output board for a circuit.



(a) Name the type of wiring shown which connects the two boards. **1**

(b) Describe how a continuity tester could be used to ensure good connections between the boards. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

2. (continued)

(c) State two methods that could be used to identify the correct wiring between the boards. **2**

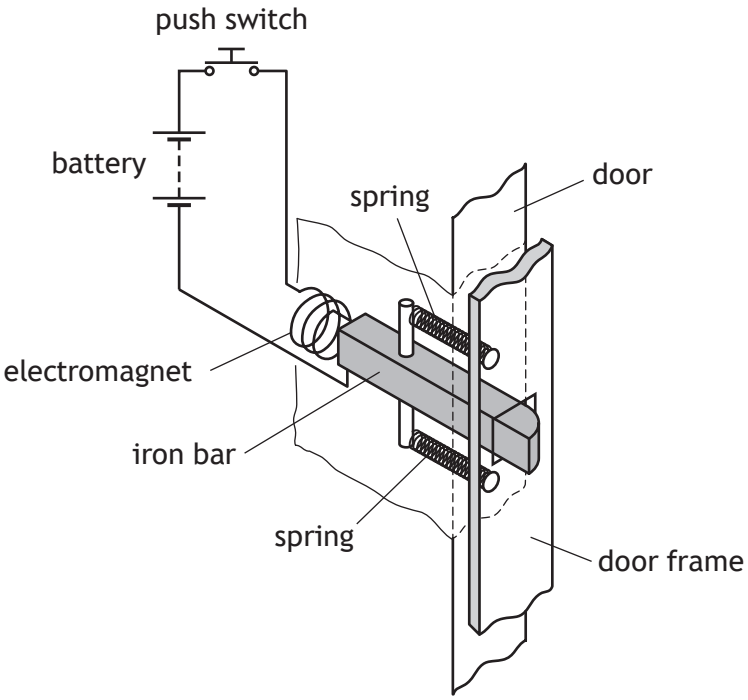
[Turn over

2.	(a)		spiral wrap(ping)		**1**	
	(b)		connect to both ends of cable lamp/LED lights	1 1	**2**	
	(c)		colour coding numbering	1 1	**2**	

MARGIN

2. A remote entry system for a block of flats allows a resident to unlock the outside door from inside their flat.

Part of this system is shown in the diagram below.

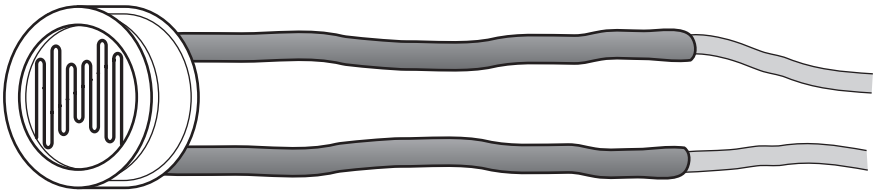


(a) Explain how this part of the system operates to unlock the door. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

2. (continued)

(b) A light dependent resistor and sleeving is connected to a process board by two wires as shown in the picture below.



(i) Name the wiring technique shown. **1**

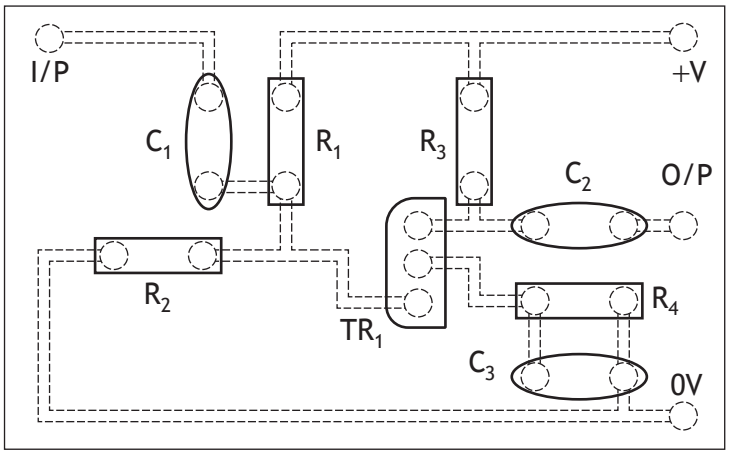
(ii) State why this technique is used. **1**

[Turn over

2.	(a)		(When switch closed) current flows through electromagnet/coil **(1 mark)** The iron bar is attracted by magnetism to the electromagnet/ coil **(1 mark)**	**2**	
	(b)	(i)	Heat shrink (tubing)	**1**	Not sleeving Accept flying leads
		(ii)	To avoid short circuiting the LDR/ stop the bare wires touching	**1**	If flying leads then for remote connection to other parts of circuitry

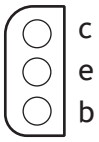
MARGIN

11. The following PCB layout shows a circuit with the following component data.



Component data

R_1 — carbon film 82K 0·25 W
R_2 — carbon film 10K 0·25 W
R_3 — carbon film 8K2 0·25 W
R_4 — carbon film 1K 0·25 W
C_1 — 0·1 μF
C_2 — 0·1 μF
C_3 — 10 μF
TR_1 — BC182 npn bipolar



TR_1 pin connections

Draw a circuit diagram for this circuit. **6**

Each component must be labelled.

DO NOT WRITE IN THIS MARGIN

11. (continued)

[END OF QUESTION PAPER]

National Qualifications 2018

X860/75/11

Practical Electronics Data Sheet

WEDNESDAY, 30 MAY
9:00 AM – 10:00 AM

SQA ©

Relationships required for National 5 Practical Electronics

$$V = IR$$

$$R_T = R_1 + R_2 + \ldots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$$

$$P = IV$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

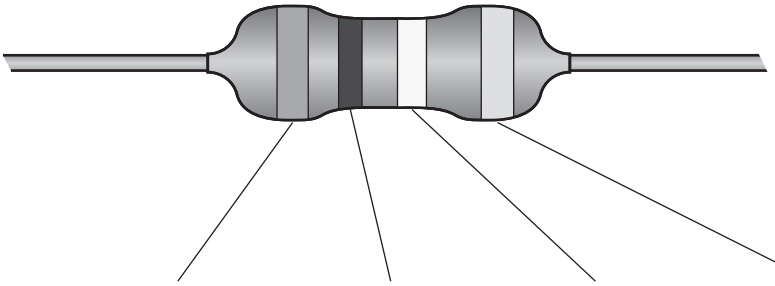
$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$$

$$f = \frac{1}{T}$$

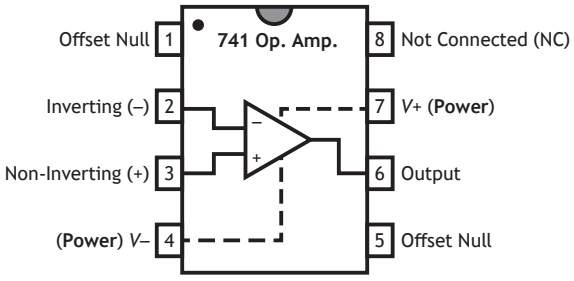
Resistor Colour Codes

4-band Resistor

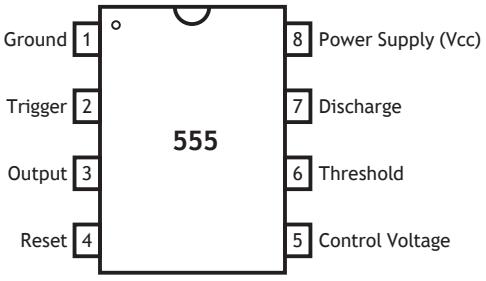


Colour	*1st band value*	*2nd band value*	*Multiplier*	*Tolerances*
Black	0	0	× 1	
Brown	1	1	× 10	±1%
Red	2	2	× 100	±2%
Orange	3	3	× 1 000	±3%
Yellow	4	4	× 10 000	±4%
Green	5	5	× 100 000	±0·5%
Blue	6	6	× 1 000 000	±0·25%
Violet	7	7	× 10 000 000	±0·10%
Grey	8	8	× 100 000 000	±0·05%
White	9	9	× 1 000 000 000	
Gold			× 0·1	±5%
Silver			× 0·01	±10%
No band				±20%

IC Pinout diagrams

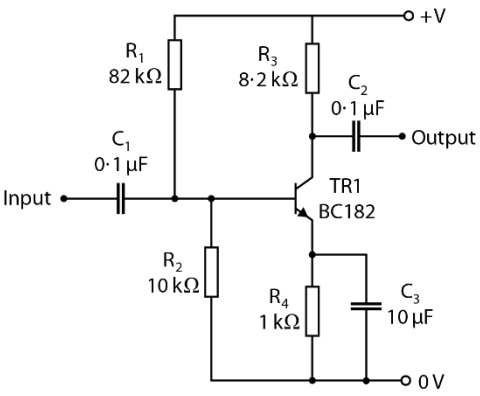


741 Op-amp



555 timer

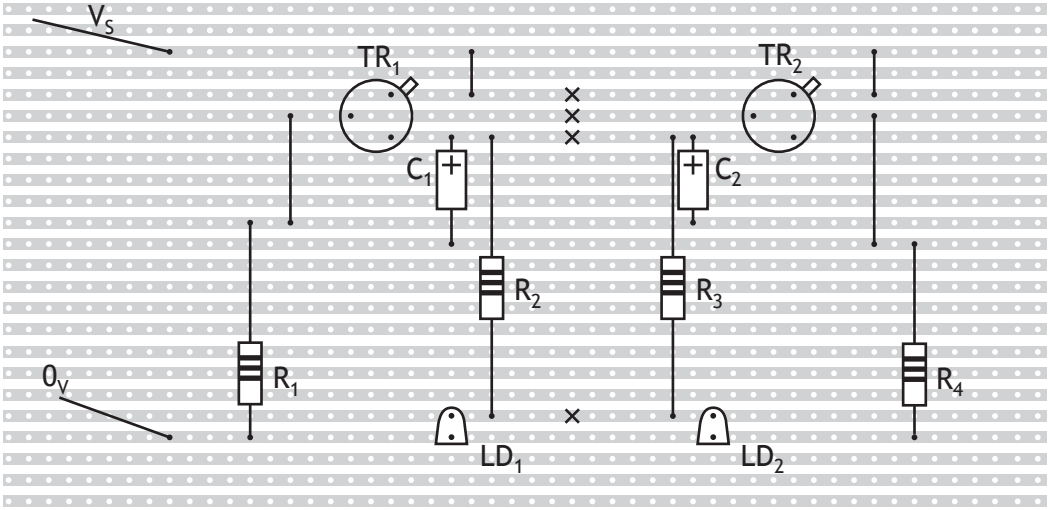
[Turn over

11.			**1 mark** for both power rails. **1 mark** for transistor oriented in a functional manner and uses correct symbol. **1 mark** for the four resistors (with correct symbol) relative to power rails and transistor. **1 mark** for C3 in parallel to R4 and uses acceptable symbol. **1 mark** for placement of input and C1(with correct symbol) relative to transistor. **1 mark** for placement of C2 (with correct symbol) and output relative to transistor.  	**6**	Component symbols must be used. Accept electrolytic capacitor symbols. Each component must be identifiable. Accept a mixture of component values and identifiers. (accept R notation) Node spot for input/outputs are not required. Power rails must have V+ or Vs and 0V with  end node.

[END OF MARKING INSTRUCTIONS]

MARGIN

10. The stripboard plan shows a **component (top) view** of a circuit with the following layout.



Component data

R_1 and R_4 – carbon film 2K2 0·25W

R_2 and R_3 – carbon film 390R 0·25W

C_1 – 100µF **electrolytic** capacitor

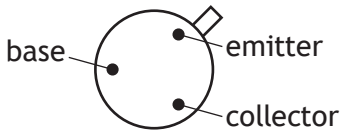
C_2 – 100µF **electrolytic** capacitor

LD_1 and LD_2 5mm standard LED

TR_1 and TR_2 – BC179 **pnp** bipolar transistors

X shows where a track has been cut

TR_1 and TR_2 pin connections



Draw a circuit diagram for this circuit.

Each component must be labelled. **6**

DO NOT WRITE IN THIS MARGIN

10. (continued)

[END OF QUESTION PAPER]

National Qualifications 2019

X860/75/11

Practical Electronics Data sheet

FRIDAY, 24 MAY

9:00 AM – 10:00 AM

SQA ©

Relationships required for National 5 Practical Electronics

$$V = IR$$

$$R_T = R_1 + R_2 + \ldots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$$

$$P = IV$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

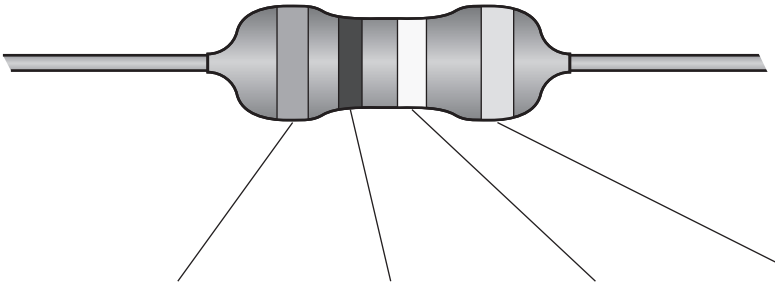
$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$$

$$f = \frac{1}{T}$$

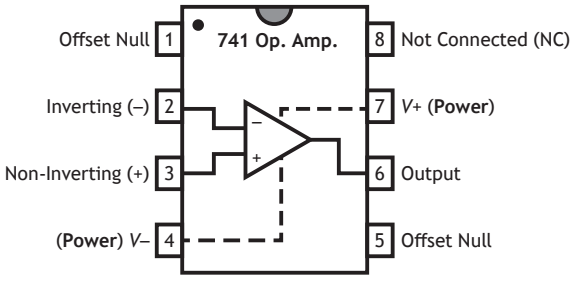
Resistor colour codes

4-band resistor

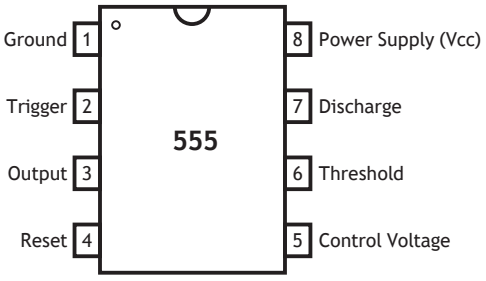


Colour	1st band value	2nd band value	Multiplier	Tolerances
Black	0	0	× 1	
Brown	1	1	× 10	±1%
Red	2	2	× 100	±2%
Orange	3	3	× 1 000	±3%
Yellow	4	4	× 10 000	±4%
Green	5	5	× 100 000	±0·5%
Blue	6	6	× 1 000 000	±0·25%
Violet	7	7	× 10 000 000	±0·10%
Grey	8	8	× 100 000 000	±0·05%
White	9	9	× 1 000 000 000	
Gold			× 0·1	±5%
Silver			× 0·01	±10%
No band				±20%

IC Pinout diagrams

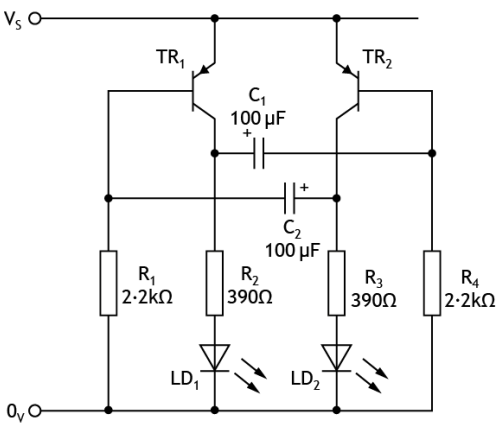


741 Op-amp



555 timer

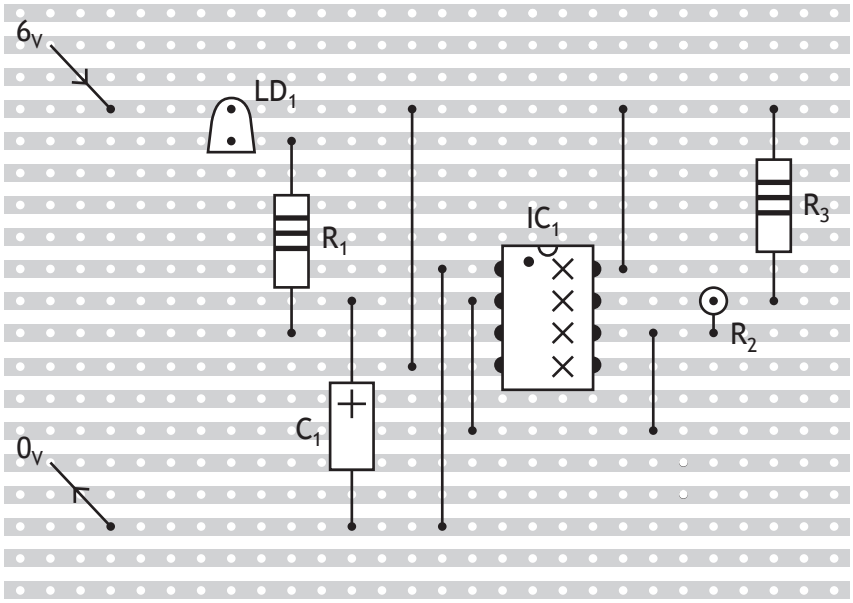
[Turn over

10.			Both power rails labelled with nodes **(1 mark)** Transistor positions and type (pnp used) **(1 mark)** Base to 0v via 2K2 resistors **(1 mark)** Collector to 0v via 390R resistor with LED correct orientation **(1 mark)** Both capacitor orientation and connections **(1 mark)** All components labelled (either values or identifiers) **(1 mark)**	**6**	 If npn used and other connections are in comparable places then treat as repeated error and deduct 1 mark for transistor position/type Accept a mixture of identifiers and values.

[END OF MARKING INSTRUCTIONS]

MARGIN

11. The stripboard plan shows a **component (top) view** of a circuit with the following layout.



Note R_2 vertically mounted.

Component data

IC_1 — NE555

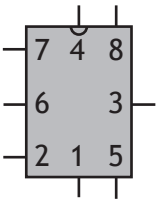
LD_1 — 5 mm standard led (red)

R_1 — carbon film 330 R 0·25 w

R_2 and R_3 — carbon film 10 K 0·25 w

C_1 — 100 µF 16 V **electrolytic** capacitor

Circuit symbol for IC_1



Draw a circuit diagram for this circuit.
Each component must be labelled. **6**

DO NOT WRITE IN THIS MARGIN

11. (continued)

[END OF QUESTION PAPER]

National Qualifications

X860/75/11

Practical Electronics Data sheet

Duration — 1 hour

©

Relationships required for National 5 Practical Electronics

$$V = IR$$

$$R_T = R_1 + R_2 + \ldots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$$

$$P = IV$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

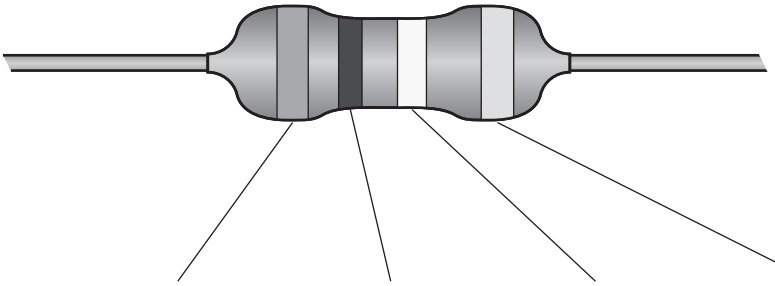
$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times V_S$$

$$f = \frac{1}{T}$$

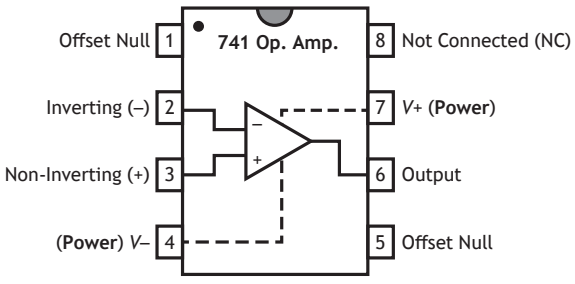
Resistor colour codes

4-band resistor

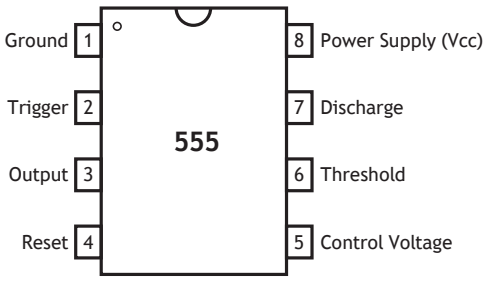


Colour	1st band value	2nd band value	Multiplier	Tolerances
Black	0	0	× 1	
Brown	1	1	× 10	±1%
Red	2	2	× 100	±2%
Orange	3	3	× 1 000	±3%
Yellow	4	4	× 10 000	±4%
Green	5	5	× 100 000	±0·5%
Blue	6	6	× 1 000 000	±0·25%
Violet	7	7	× 10 000 000	±0·10%
Grey	8	8	× 100 000 000	±0·05%
White	9	9	× 1 000 000 000	
Gold			× 0·1	±5%
Silver			× 0·01	±10%
No band				±20%

IC Pinout diagrams

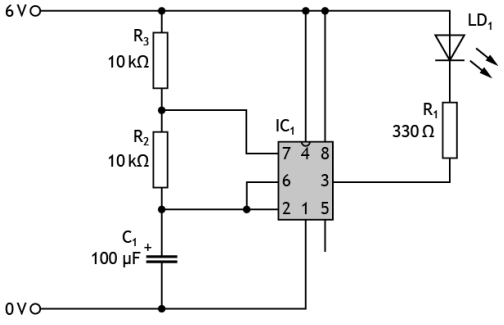


741 Op-amp



555 timer

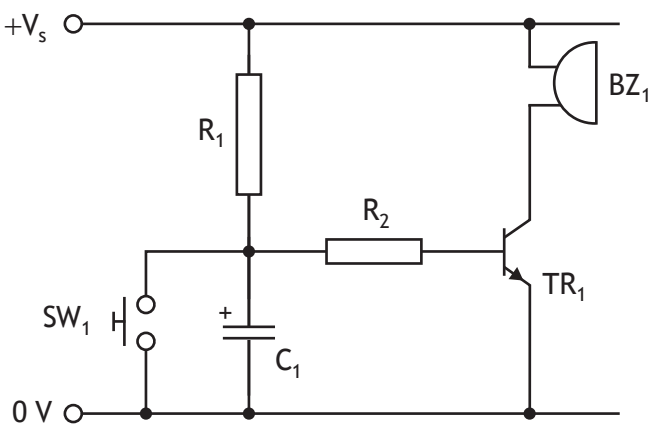
[Turn over

11.			Both power rails labelled with nodes **(1 mark)** Power rails to IC (4,8 to +6V and 1 to 0V) **(1 mark)** The "output" +6V to pin 3 via R_1 and correctly orientated LED **(1 mark)** R_2,R_3 and C_1 in series between power rails with C_1 **correctly orientated** **(1 mark)** For 6,7 and 2 connected, 5 not connected **(1 mark)** Component labelling **(1 mark)**	**6**	 Accept a mixture of identifiers and values Accept R notation

[END OF MARKING INSTRUCTIONS]

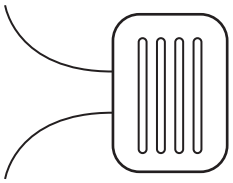
MARGIN

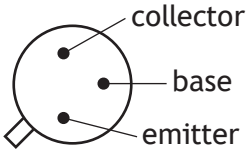
10. A time delay circuit is shown below.



Component data
R_1 — carbon film 10 kΩ 0.25 W
R_2 — carbon film 1 kΩ 0.25 W
C_1 — 2200 mF **electrolytic** capacitor
SW_1 — push-to-make switch
TR_1 — BC108 npn bipolar
BZ_1 — buzzer 6 V dc
TR_1 pin connections

buzzer with connecting wires





electrolytic capacitor



push-to-make switch



On the opposite page, a layout diagram includes the power rails and components TR_1 and R_2.

Complete the layout diagram with the remaining components.

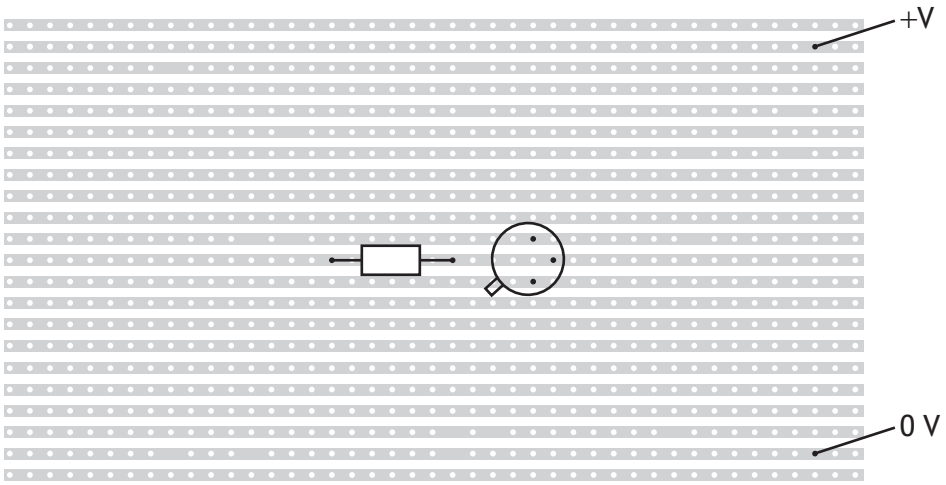
Each component must be identified by a label.

Mark on the layout diagram, with an X, where a track has been cut. **5**

DO NOT WRITE IN THIS MARGIN

10. (continued)

layout diagram

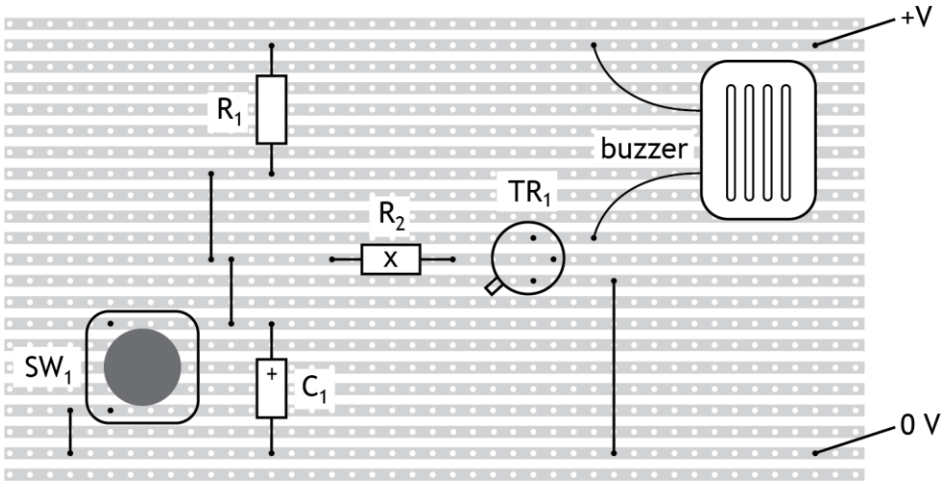


(An additional layout diagram, if required, can be found on *page 21*.)

[END OF QUESTION PAPER]

10.			**1 mark** for labels. **1 mark** for break(s): there should be a break across R_2. **1 mark** for SW_1, with solder points, connected across C_1 and connected to 0 V with appropriate links, breaks and connected to R_2. **1 mark** for branch containing R_1 and C_1 with appropriate links and breaks. **1 mark** for connecting TR_1 to 0 V and +V through buzzer, with appropriate links and breaks.	**5**	'Footprint' of components should be of an appropriate size. If candidate used circuit symbols, deduct label mark.

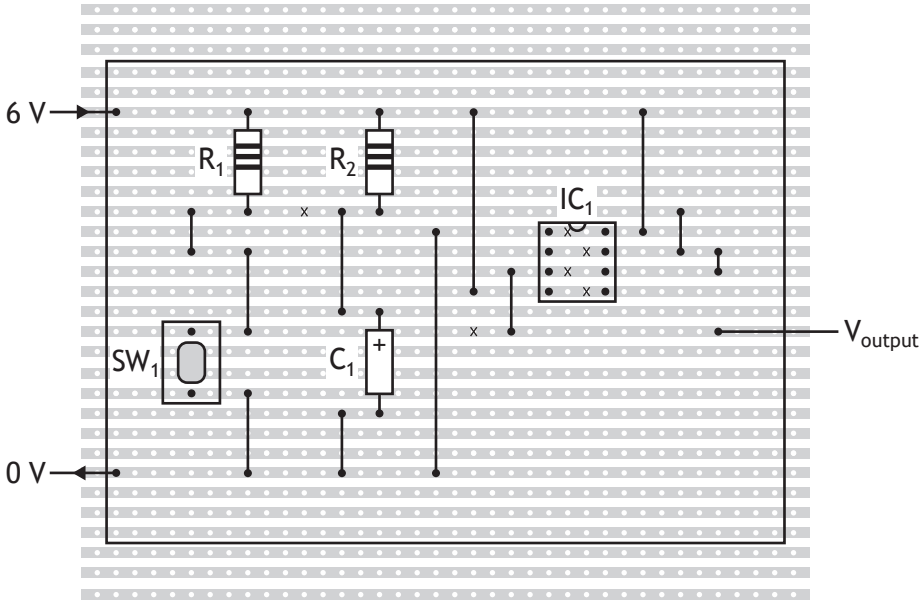
Layout diagram



[END OF MARKING INSTRUCTIONS]

MARGIN

11. The stripboard plan below shows a **component (top) view** of a circuit with the following layout.



Component data

IC_1 — NE555

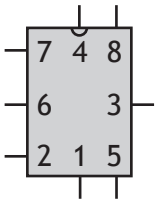
R_1 and R_2 — carbon film 10 KΩ 0.25 W

C_1 — 1000 µF 16 V **electrolytic** capacitor

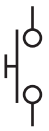
SW_1 — push-to-make switch

X shows where a track has been cut.

Circuit symbol for IC_1



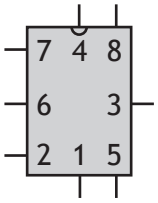
Circuit symbol for SW_1



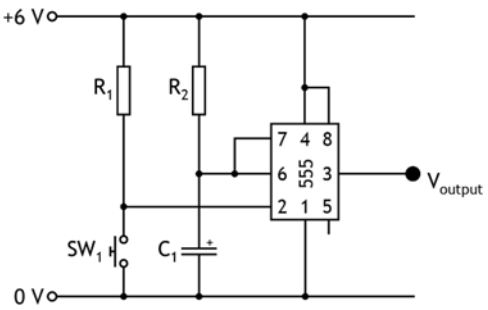
MARKS | DO NOT WRITE IN THIS MARGIN

11. (continued)

For the stripboard layout diagram, complete the circuit diagram below.
Each component must be labelled. **6**
(An additional circuit diagram template, if required, can be found on *page 21*.)



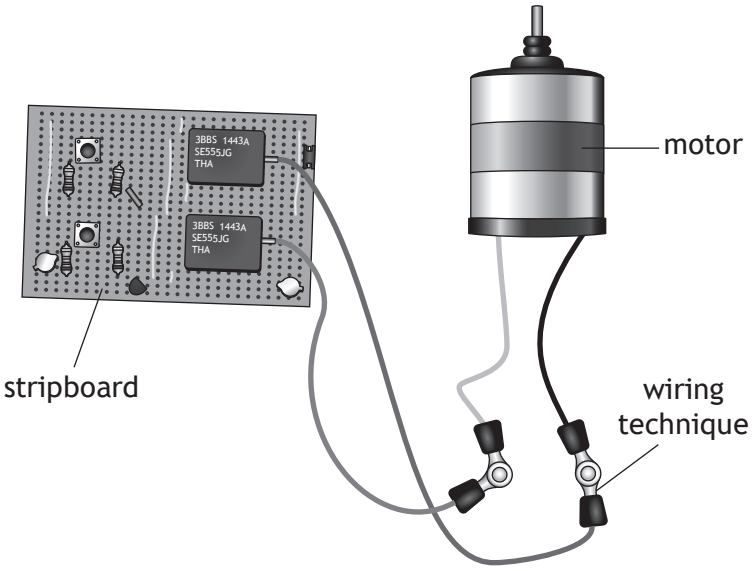
[END OF QUESTION PAPER]

11.			Both power rails labelled with nodes. **(1 mark)** Power rails to IC. (4,8 to +6V and 1 to 0V) **(1 mark)** SW_1 and R_1 in series between power rails, correctly positioned. **(1 mark)** R_2 and C_1 in series between power rails with C_1 correctly orientated. **(1 mark)** Pins 6, 7 and 2 connected, 5 not connected and correct connection of V_{output} to pin 3 of 555 timer. Node must be included. **(1 mark)** Each component must be identifiable and correctly used circuit symbols. (accept either values or identifiers). **(1 mark)**	**6**	 Accept IC1, 555 or NE555 for chip identifier.

[END OF MARKING INSTRUCTIONS]

MARGIN

6. The image below shows the wiring connections between a stripboard and a motor for a circuit.



The wiring technique used in the photo is crimping.

(a) State another wiring technique that could have been used to secure the wires together. **1**

(b) State one method that could be used to identify the correct wiring between the stripboard and motor. **1**

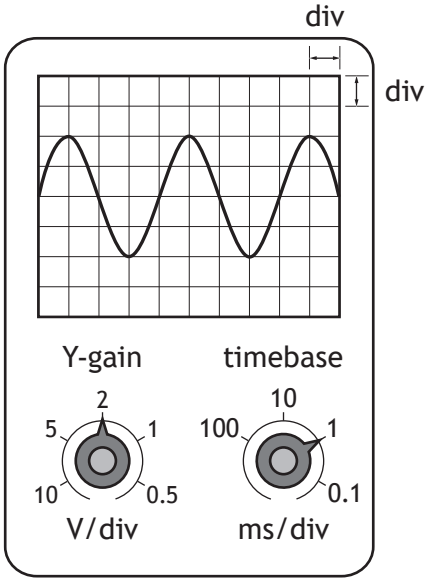
[Turn over

6.	(a)		Connector blocks **(1 mark)** **OR** Terminal blocks	**1**	Do not accept the following: Cable ties, heat shrink, velcro straps, spiral wrap or connector on its own. Accept Solder **and** heat shrink. Accept screw terminal.
	(b)		Colour coding **(1 mark)** **OR** Wire/cable numbering	**1**	Any other suitable response.

8. The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown below.



MARKS DO NOT WRITE IN THIS MARGIN

(a) State whether this signal is analogue or digital. **1**

(b) Determine the peak voltage of this signal. **1**

(c) Calculate the frequency of this signal. **3**

Space for working and answer

8.	(a)		Analogue (signal)	**1**	
	(b)		4 V	**1**	Must include unit.
	(c)		T = (4 × 1 ms) = 4 ms or 4 × 10^{-3} s or 0.004s **(1 mark)** $f = \frac{1}{T}$ **(1 mark)** f = 250 Hz **(1 mark)**	**3**	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula If period incorrect then award a maximum of 1 mark for the formula If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as *0·25* kHz, in which case 3 marks may be awarded Accept f=N/t

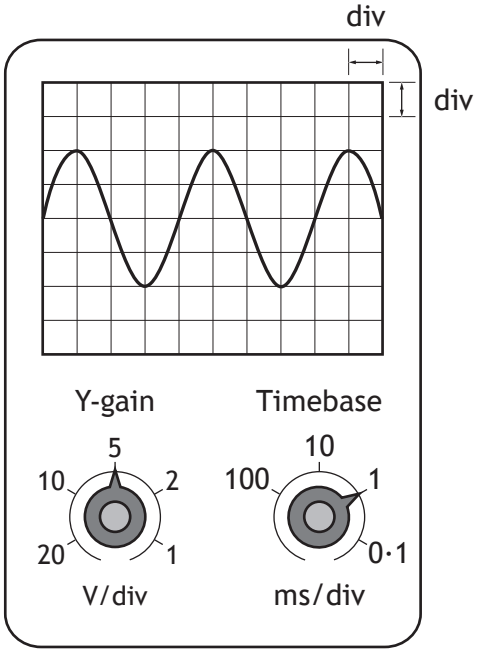
Question	Expected response	Max mark	Additional guidance

MARGIN

5. (a) The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown.



Determine the peak voltage of the signal. **1**

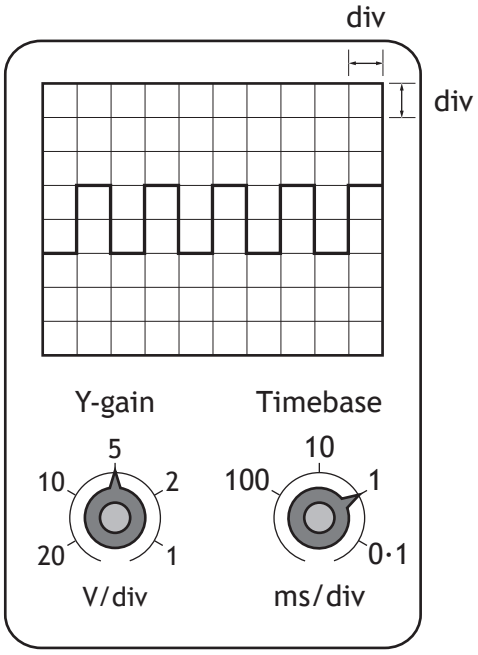
[Turn over

MARKS | DO NOT WRITE IN THIS MARGIN

5. (continued)

(b) The output from the signal generator was changed to produce the trace shown below.

The Y-gain and timebase settings are also shown.



(i) State whether the signal is analogue or digital. **1**

(ii) Determine the frequency of the signal. **3**

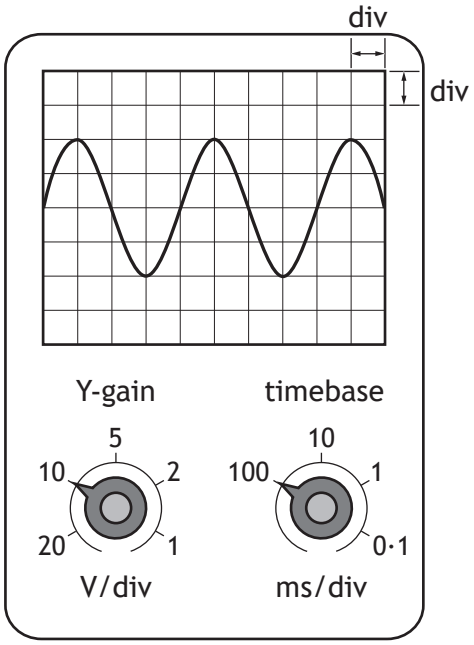
5.	(a)		10 V		**1**	
	(b)	(i)	digital		**1**	
		(ii)	T = (2 × 1 ms =) 2 ms $f = \frac{1}{T}$ f = 500 Hz	1 1 1	**3**	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0·5 kHz, in which case 3 marks may be awarded.

MARGIN

6. An oscilloscope is connected to show the output trace from a signal generator.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown.



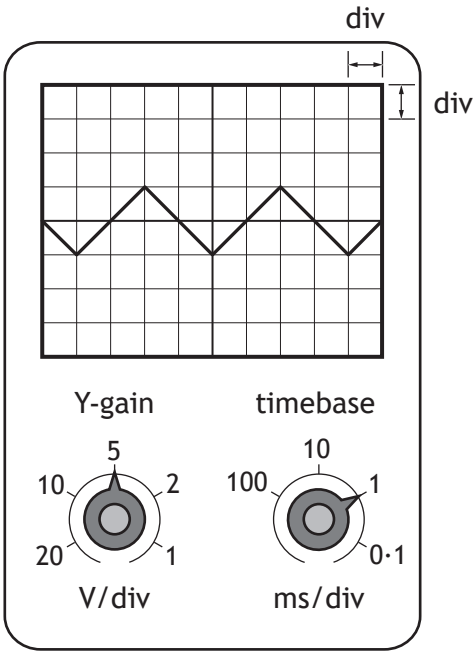
(a) Determine the frequency of the signal. **3**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

6. (continued)

The output from the signal generator was changed to produce the trace shown below.



(b) State whether the signal is analogue or digital. **1**

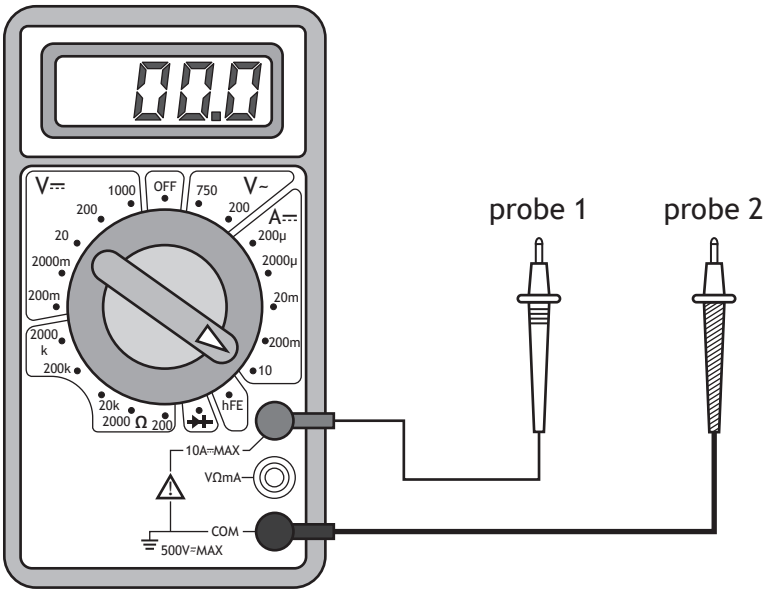
[Turn over

6.	(a)		Period t = 400ms or 0·4s **(1 mark)** $f = \frac{1}{T}$ **(1 mark)** $f = \frac{1}{0 \cdot 4}$ $f = 2 \cdot 5Hz$ **(1 mark)**	**3**	Must have unit If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0·0025 kHz, in which case 3 marks may be awarded.
	(b)		Analogue (signal)	**1**	

MARGIN

5. A student wants to measure the **voltage** across a component in a circuit and sets up the multimeter as shown below.

(a) Identify the two errors the student has made. **2**



Error 1:

Error 2:

MARKS | DO NOT WRITE IN THIS MARGIN

5. (continued)

(b) The student is given a resistor with no colour markings.

Describe how the student would set up and use a multimeter to accurately measure the resistance of the resistor. **3**

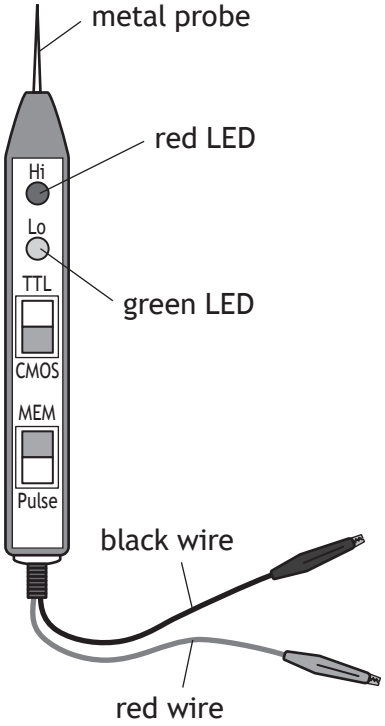
[Turn over

5.	(a)		Error: meter on incorrect scale/ range/setting **(1 mark)** Error: probe 1 in wrong socket **(1 mark)** **OR** probe 1 should be in VΩmA socket **OR** middle/centre socket	**2**	Accept Meter on ampere/current scale Multi-meter should be set to $V\overline{\overline{\cdots}}$ or V~ Probe must be identified as probe 1
	(b)		Connect probe 1 to middle socket/ VΩmA socket. (Probe 2 to the COM socket) **(1 mark)** Set scale to Ω/ohms/resistance **(1 mark)** Start at highest resistance range and work down until display reads 1 then go up one range. **(1 mark)**	**3**	Accept range/scale as interchangeable Accept: probe 1 => COM Probe 2 => VΩmA Or until measured resistance is bigger than the next smallest scale/range

MARGIN

5. A logic probe is used to test the inputs and outputs of a 7400 series logic chip.

The logic probe is set to TTL and pulse.



(a) State where the red and black wires should be connected. **1**

(b) Describe how a logic **0** would be detected. **2**

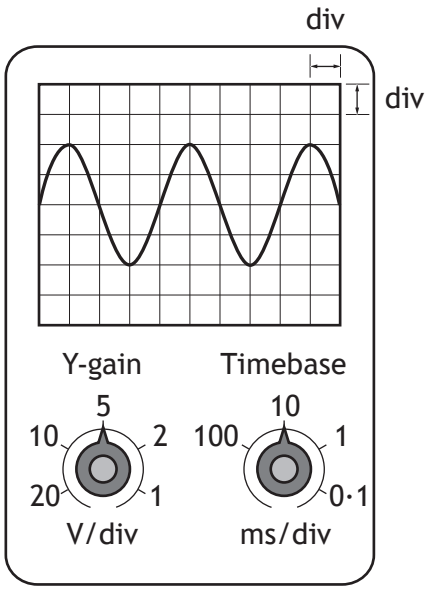
5.	**(a)**		Connect red wire of probe to V_{supply} and black wire to 0 V **(1 mark)**	**1**	Do not accept: connect the probe to the supply rails (on its own)
	(b)		Place the probe tip onto circuit at selected point **(1 mark)** Green LED lights **(1 mark)**	**2**	

MARGIN

6. (a) The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are also shown.



Determine the frequency of the signal. **3**

Space for working and answer

MARKS | DO NOT WRITE IN THIS MARGIN

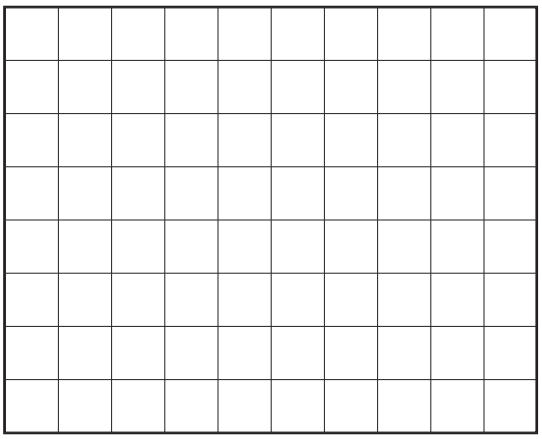
6. (continued)

(b) The peak voltage of the output signal from the signal generator is now **doubled**.

The frequency of the signal is unchanged.

The settings on the oscilloscope are unchanged.

Draw the new trace that would be shown on the screen. **1**



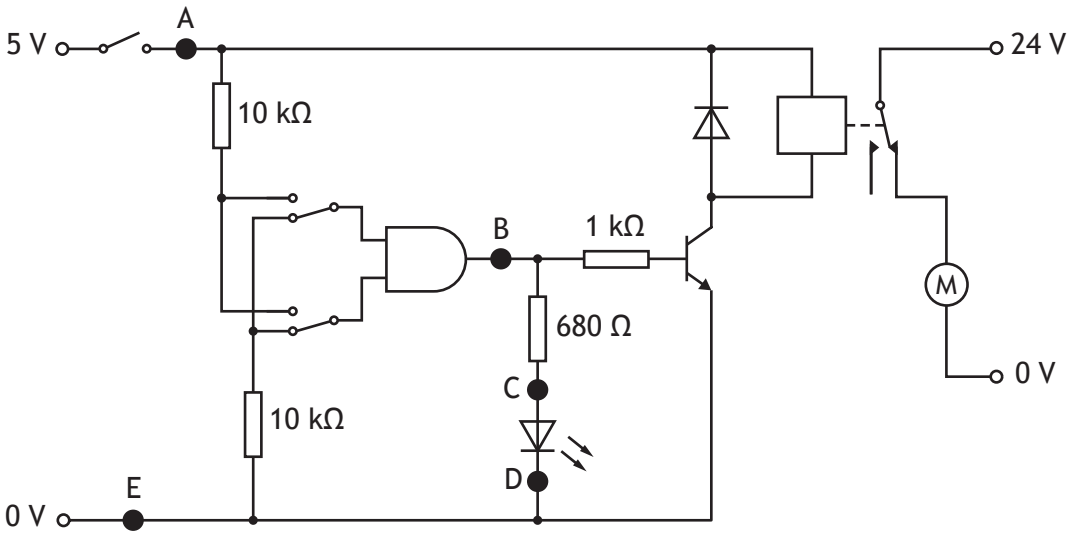
(An additional diagram, if required, can be found on *page 24*.)

[Turn over

6.	(a)		$T = (4 \times 10 \text{ ms}) = 40 \text{ ms}$ **(1 mark)** $f = \frac{1}{T}$ **(1 mark)** $f = 25 \text{ Hz}$ **(1 mark)**	**3**	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as 0·025 kHz, in which case 3 marks may be awarded.
	(b)		Trace should have double the amplitude (with 3 peaks and two troughs).	**1**	

MARGIN

7. The circuit diagram below has test points A, B, C, D and E as shown.



Complete the table below stating what the test points measure and the instrument used to carry out each test. **2**

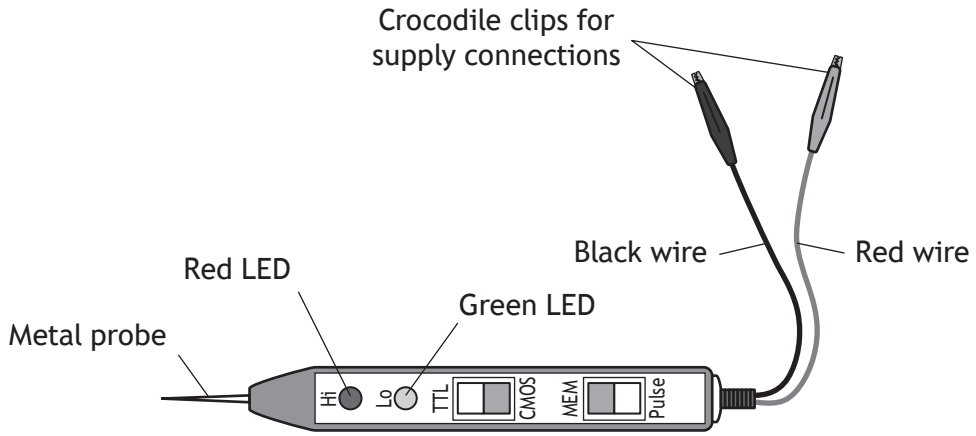
Test points	Measuring	Instrument
A and E	supply voltage	multimeter voltage setting
B		
C and D		

7.			<table><tr><th>Measuring</th><th>Instrument</th></tr><tr><td>supply voltage</td><td>multimeter voltage setting</td></tr><tr><td>Output logic level from gate</td><td>Logic probe</td></tr><tr><td>Voltage across LED</td><td>Multimeter voltage setting</td></tr></table>	**2**	1 mark for each row Accept logic level Accept voltmeter Do not accept multi-meter on its own

MARGIN

5. A logic probe is used to test the inputs and outputs of a 74 series logic chip.

The logic probe is set to TTL and pulse.



Describe how to connect and use the logic probe to test the inputs and outputs of the chip.

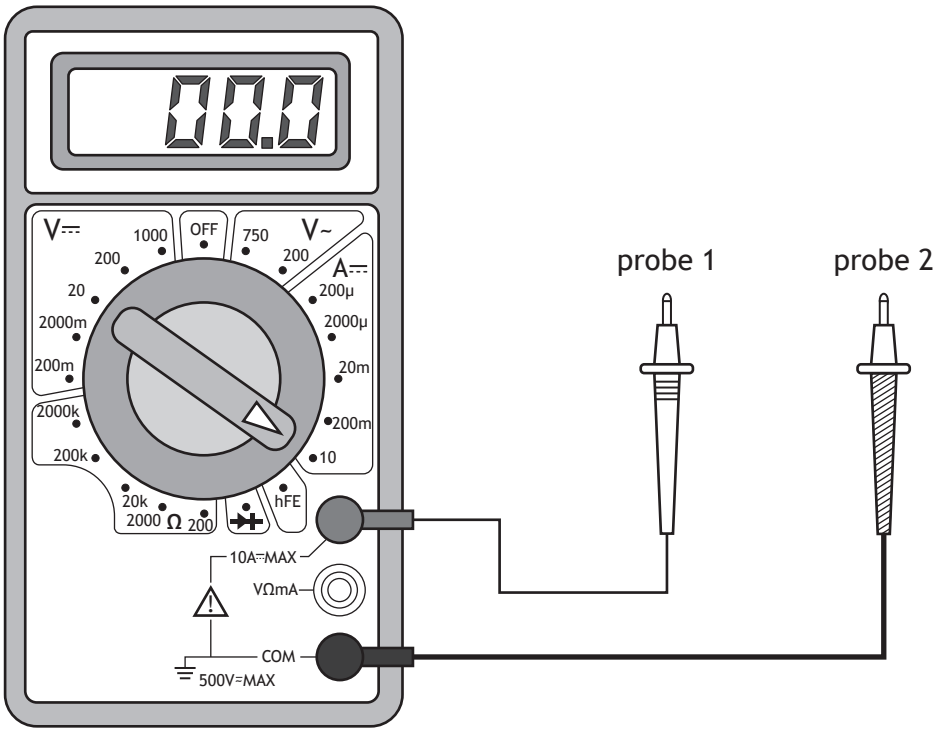
In your answer include how a logic 1 is detected. **3**

[Turn over

5.			Connect red wire of probe to V_{supply} and black wire to 0 V or equivalent. **(1 mark)** Place the probe tip onto circuit at selected point. **(1 mark)** Red LED lights. **(1 mark)**	**3**	Do not accept: connect the probe to the supply rails (on its own). 3 independent marks

MARKS DO NOT WRITE IN THIS MARGIN

7. A student wants to check that the current flowing through a component is **1.2 mA DC**.



(a) Identify **two** changes that must be made to the multimeter to get the most accurate measurement. **2**

Change 1:

Change 2:

(b) Draw the symbol for direct current that is shown on the meter. **1**

(c) State the most appropriate scale that should be used to check the resistance of a 15 000 Ω resistor. **1**

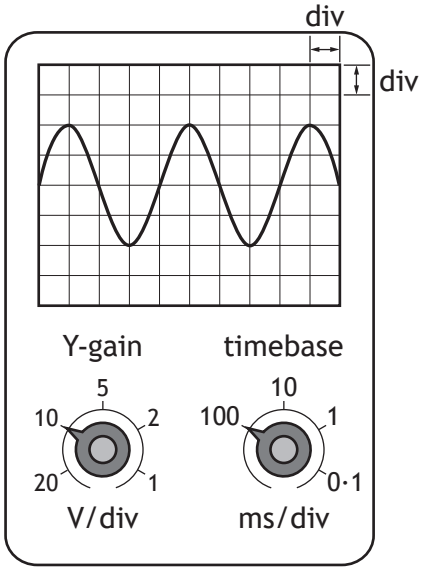
7.	(a)		Connect probe 1 to the VΩmA or middle socket of multimeter **(1 mark)** Select 2000 μ range **(1 mark)**	**2**	Probe must be identified as probe 1 Accept Red instead of Probe 1 Accept range/scale as interchangeable
	(b)			**1**	Also accept A and symbol but do not accept V and symbol.
	(c)		$20K$	**1**	Accept 20,000 no unit required

MARGIN

8. The output from a signal generator is connected to the input terminals of an oscilloscope.

The trace is shown on the screen.

The Y-gain and timebase settings are shown below.



(a) Determine the peak voltage of this signal. **1**

(b) Calculate the frequency of the signal. **3**

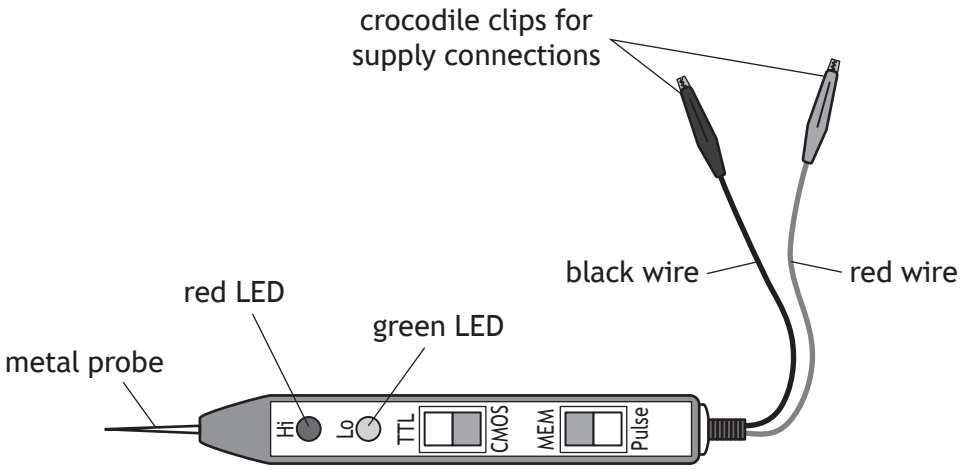
Space for working and answer

8.	(a)		20 V **(1 mark)**	**1**	Must include unit.
	(b)		T = (4 × 100 ms) = 400 ms or 0.4s **(1 mark)** $f = \frac{1}{T}$ **(1 mark)** f = 2.5 Hz **(1 mark)**	**3**	If incorrect setting chosen from oscilloscope award a maximum of 1 mark for the formula. If period incorrect then award a maximum of 1 mark for the formula. If milli omitted from the period of the wave, treat as a unit error and award a maximum of 2 marks, unless final answer is then quoted as *0·0025 kHz*, in which case 3 marks may be awarded. Do not accept the term “Secs” as unit for seconds.

MARGIN

7. A logic probe, shown below, is used to test the inputs and outputs of a 74 series logic chip.

The logic probe is set to TTL and pulse.



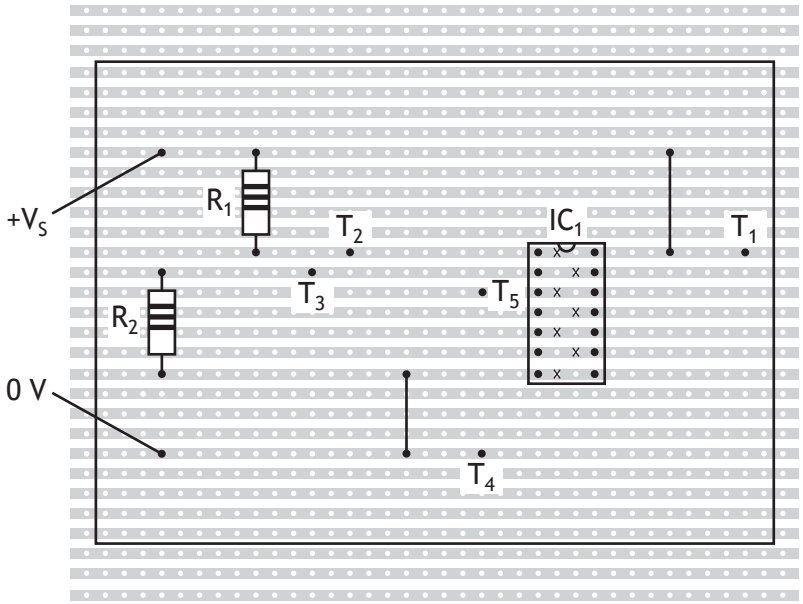
(a) (i) State where the red and black wires should be connected. **1**

(ii) Describe how a logic 1 would be detected. **2**

MARKS | DO NOT WRITE IN THIS MARGIN

7. (continued)

(b) A student sets up a circuit on stripboard as shown below. IC_1 is a 74 series logic chip.



Test points T_1 to T_5 are marked on the stripboard.

(i) Identify two test points that are used to measure the voltage across R_2. **1**

(ii) The logic probe is placed at pin 7 of the logic chip, IC_1.

Explain why the LED on the logic probe flashes green. **1**

[Turn over

7.	(a)	(i)	Connect red wire of probe to $+V_{supply}$ and black wire to 0v or equivalent. **(1 mark)**	**1**	Do not accept: connect the probe to the supply rails (on its own). Accept alternatives to V_{supply}, $+V_S$, +9V
		(ii)	Place the probe tip onto circuit at selected point. **(1 mark)** Red LED lights. **(1 mark)**	**2**	Independent marks.
	(b)	(i)	T3 and T4 **(1 mark)**	**1**	Answer must state test point numbers.
		(ii)	Pin 7 is connected to 0v or ground. **(1 mark)**	**1**	Accept: negative terminal (of the supply). Do not accept, "connected to the power supply" as it does not state high or low potential. Do not accept, "connected directly to the power supply". Do not accept "PIN 7 is logic 0 or negative" on its own.

2. State two safety precautions that must be taken when using a soldering iron. **2**

MARGIN

[Turn over

2.			**Any two from** • wear eye protection • use stand • use heat proof mat • avoid breathing in fumes • well-ventilated room • hold soldering iron correctly	**2**	Any acceptable alternate answer. Any mention of clothing needs a justification.

2. State **two** safety precautions that must be taken when using a soldering iron. **2**

MARGIN

2.			Any two from • Wear eye protection • Use stand • Use heat proof mat • Avoid breathing in fumes • Hold soldering iron correctly	**2**	Any acceptable alternate answer Do not accept “remove clothing” needs to justify why

Question	Expected response	Max mark	Additional guidance

End of the booklet

Finished every part? Go back to the questions where you lost the most marks. Do them again without looking at your first answer, then mark them again.

Sources and credits. Questions and marking instructions copied exactly from the real papers, © Qualifications Scotland (SQA), used for education: 2018 National 5 Practical Electronics Question paper; 2019 National 5 Practical Electronics Question paper; 2021 National 5 Practical Electronics Question paper; 2024 National 5 Practical Electronics Question paper; 2025 National 5 Practical Electronics Question paper; 2026 National 5 Practical Electronics Question paper; Specimen National 5 Practical Electronics Question paper. Every question in this booklet, with its paper and number, is listed on the part pages.

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Compiled by R Stewart. R Stewart - November 2026